

Quantitative Analysis and Statistical Arbitrage in a Niche De-Centralized Asset Market

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Abstract In traditional financial markets statistical arbitrage and quantitative analysis rely heavily on high-frequency execution and large data throughput. We investigate if these same principles can be applied in an unusual and inefficient asset market, with long trade execution times, sparse data, and ever present risk and exposure. By utilizing a Hierarchical Bayesian Weibull Accelerated Failure Time (AFT) model for arbitrage across markets, and a Jump-Diffusion Stochastic Differential Equation (SDE) with Bayesian learned drift, over a four month period our final portfolio had an ROI +40% >, weathered a max draw down of 38%, and succeeded on over 70% of our arbitrage trades.

Keywords: Risk measures; Bayesian Survival Analysis; Arbitrage Price-Inference; Geometric Brownian Motion; Stochastic Differential Equations

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Part I

Project Introduction & Niche Market Context

1 Introduction

Quantitative finance is traditionally applied to high-volume, liquid markets where high-frequency data is abundant. Arbitrageurs use analytical models to trade at high-frequencies with an objective of predictable returns that are uncorrelated with the overall market (Avellaneda and Lee, 2010). This document outlines a project which investigates the extent to which quantitative analysis has utility by attempting to utilize it in an unconventional asset market defined by the opposite characteristics: sparse data, low liquidity, and significant structural frictions.

The market we operate in is a real, niche, multi-billion dollar asset ecosystem. While partially overseen by a central authority, the majority of trading occurs across decentralized third-party platforms. This 'market' is characterized by two defining frictions. The first is a severe limitation on accessible market data, which renders many standard econometric and frequentist methods impractical. The second is a mandatory, multi-day settlement latency on asset transfers, which exposes arbitrageurs to significant price risk between the initiation and execution of a trade, as well as preventing any form of high-frequency trading.

These frictions lead to persistent price discrepancies for identical assets across different marketplaces, making spatial arbitrage opportunities readily observable. However, the settlement latency transforms what would be a risk-free trade into a probabilistic one. The central challenge for arbitrage is not only in identifying potential arbitrage, but in systematically quantifying the risk that an opportunity will decay before it can be realized.

Furthermore, unpredictable market changes caused by the central authority can cause periods of extreme volatility, regime changes, and general market chaos. This can make quantifying the market fundamentals and defining factors of price movement, strong investments, and weak investments extremely difficult. Therefore, any quantitative market analysis must overcome the lack of complete data and be meticulously informed of market context.

Our primary contribution is the development and application of a potentially novel quantitative framework designed specifically for this high-latency, data-scarce environment. We synthesize existing concepts of arbitrage limits and survival analysis into a predictive tool. Specifically, we adapt (Deville and Riva, 2007)'s Accelerated Failure Time (AFT) model within a Bayesian framework to generate forward-looking survival probabilities over the stochastic settlement period for newly identified arbitrage opportunities. We also utilize the established Geometric Brownian Motion (GBM) stochastic differential equation (SDE) with Merton Jump-Diffusion unconventionally to attempt to understand how the market behaves at a macroscopic, strategic level. This tactical spatial arbitrage selection model, paired with the counter-intuitive deployment of an SDE, is how we attempt to validate the applications of quantitative analysis in a fundamentally chaotic and primitive market.

2 Market Structure and Arbitrage Mechanics

The arbitrage strategy operates across a system of three distinct but interconnected marketplaces, each with a unique pricing mechanism. However, the ecosystem also includes a separate class of commodity-like assets that are fundamental to understanding market-wide dynamics in our SDE.

2.1 The Trading Venues

Our strategy involves sourcing assets from a peer-to-peer (P2P) platform, hereafter referred to as **Market S** (Source). This venue is characterized by decentralized price discovery and sparse, limitedly accessible available data.

The arbitrage is realized on **Exchange T** (Target), an Eastern-European automated exchange that functions as a market maker. This platform exchanges assets at prices with a significant offset ratio to real currency asset value, making the platforms non-redeemable balance and transaction prices act as a form of tokenized FIAT currency. This currency and pricing, which often diverge significantly from those on Market S, creates the arbitrage spread. However this platform provides a more complete set of publicly accessible data.

A third venue, the official **Market B** (Benchmark), serves as a high-liquidity benchmark. While its high transaction fees and transaction price limit make it majorly unsuitable for our arbitrage, its comprehensive accessible historical data is used to inform our SDE analysis of broader market dynamics, and it's high liquidity can be used to manually confirm divergences in Exchange T's pricing.

2.2 Supply Generation Assets For Systemic Risk

Unlike the primary assets traded in our arbitrage strategy, the market features a class of high-volume liquid commodities that are consumed to generate new primary assets. The price and trading volume of these "asset-generating commodities" are not directly tied to the arbitrage spreads of any single asset. Instead, they serve as a robust, real-time proxy for overall market health, speculative interest, and participant sentiment.

These commodities are particularly sensitive to the systemic risks mentioned in the introduction, such as major updates from the central authority or predictable market cycles. Their price action therefore provides a cleaner signal for macro-level market regime analysis than the prices of the illiquid, arbitrage-specific assets.

2.3 Defining Market Friction and Risks

The previously introduced global market frictions of multi-day settlement period, sparse data, and unpredictable market events provide the basis for opportunity and risk for the project. The threats posed by these frictions are typically related and the presence of multiple frictions in action at any time in the market typically result in adverse effects on analysis or profitability.

As the market is influenced by a central authority, who set the trade regulations, create and provide the "asset-generating commodities", and release new asset collections, the market is always exposed to some lingering risk. Spontaneous market announcements or updates have and can occur at any time without warning, and tend to have significant implications on market sentiment and behavior. We loosely refer to the risk posed by spontaneous market updates as the **unpredictable event risk**.

During the settlement period, an asset is untradeable and cannot be sold on any platform. This poses a existential problem to any form of arbitrage across assets or market venues. An arbitrageur will always be exposed to a handful of different risks over settlement period, not only from adverse price movement, but also from unpredictable event risk changing market dynamics during a trade. Further, the market's trade volume is also greatly hindered by this limitation. We define the risk of an arbitrage opportunity closing during execution from the predictable market dynamics, such as price movement and venue item capacity, as **settlement latency risk**.

The general data scarcity across the ecosystem is not the only hurdle between deterministic analysis and modeling, and certainly not helped by the aforementioned defined risks. Additionally, the unique nature of the assets, which are non-fungible and distinguished by a combination of asset attributes will constitute the discrete market items we trade. The many combinations for any asset grouping results in the already illiquid data being dispersed around the diverse market asset pool, making the "true" acquirable price for any specific asset difficult to ascertain from listing prices alone. As we are forced to rely on data inference methods to create fully continuous time-domain pricing information for arbitrage, we face an **information risk** whose magnitude will be amplified depending on how much the arbitrage model will rely on potentially inaccurate parameters.

2.4 Theoretical Framework and Challenges

Arbitrage Strategy

Our arbitrage process is a closed-loop strategy designed to capitalize on the price differential between Market S and Exchange T. We define this differential as the **Arbitrage Price Ratio (APR)** at a given time t :

$$APR(t) = \frac{P_T(t)}{\hat{P}_S(t)} \quad (1)$$

where $P_T(t)$ is the observable price on Exchange T, and $\hat{P}_S(t)$ is our inferenced acquirable price on Market S, the methodology for which is later detailed in Section 3.2.

The strategy is to use Market S to identify and purchase the 'source assets' that have relatively high APRs, exchange the sourced assets for tokenized currency on Exchange T, use that currency to purchase a 'bridge assets' that have relatively low APRs, and finally sell the bridge asset on Market S to realize the profit.

Arbitrage Challenges

The primary focus of our project's arbitrage component is on the first 'source asset' leg, as it carries a disproportionate amount of risk compared to other legs. We define the **APR risk**; the APR persisting at deemed profitable level throughout the settlement period. Another, more definitive risk is the asset inventory capacity on Exchange T. The platform imposes a strict limit on the quantity of any given asset it will hold. This creates an **inventory saturation risk**: the possibility that the opportunity will close abruptly and completely, not due to price changes, but because the exchange is no longer buying or other traders have filled the opportunity. This risk is quantified by the **Inventory Saturation Ratio (ISR)**:

$$\text{ISR}(t) = \frac{I_{\text{held}}(t)}{L_{\text{cap}}(t)} \quad (2)$$

where $I_{\text{held}}(t)$ is the quantity of the asset held by Exchange T at time t , and L_{cap} is its maximum inventory capacity. The core quantitative challenge is therefore multi-faceted: establishing realistic entry prices for illiquid assets, accurately detecting arbitrage opportunities, and forecasting the persistence of the arbitrage spread, and accounting for the risk of inventory saturation.

SDE Application

Whereas the survival model created for the arbitrage strategy will largely determine our arbitrage operation, a good portion of our previous inventory profit has come from strong long-term positions and intelligent reactions to market crisis. Thus, we wish to integrate our prior market hypothesis into an SDE to attempt to model how the production assets for specific asset subgroups have historically responded to predictable and unpredictable market events and changes.

With the advantage of more complete data for these liquid assets, our goal is to be able to differentiate asset groups with strong investment fundamentals and to better understand market behavior instead of conventional price forecasting.

SDE Challenges

The SDE will face an unusual challenge in terms of the sheer quantity of macro-level market jumps, shocks, regimes, and unpredictable events that occur over small time durations relative to the size of our datasets. Our dataset consists of daily price and trade volume history for each asset production commodity, which also serves as the basis for the GBM. Henceforth, our intimate understanding of the market will be critical in guiding and extracting any significance from the results.

Part II

Arbitrage Survival Model

3 Arbitrage Methodological Implementation

The methodological approach developed greatly from our initial naive "prototype" models to test the code functionality and the fundamental hypothesis that arbitrage duration in this market is predictable. Upon completion and partial success of the confirmation of the hypothesis, the arbitrage strategy and initial model were deployed. The success and limitations of the initial model then motivated the development towards the more sophistic "final" hierarchical framework, also deployed within the trading period. This section details our methodology and that progression.

3.1 Theoretical Foundations and Modeling Approach

The presence of settlement latency motivates the application of statistical risk management for both arbitrage opportunity detection and opportunity persistence for the identified spatial arbitrage opportunity.

Settlement Latency's Effect on Arbitrage Price Ratio Thresholds

Hautsch et al. (2018) analyzed how stochastic settlement latency influences whether a certain spatial price discrepancy is truly exploitable in the context of distributed ledger markets. They establish that settlement latency imposes an 'Arbitrage Boundary' threshold where the probabilistic risk from volatility and time to execution can invalidate a possible arbitrage opportunity. We accept Hautsch's premise that necessitates a more nuanced approach to spatial arbitrage opportunity detection than commonly used principal component analysis or price regressions.

Whereas Hautsch et al introduce an 'Arbitrage Boundary' equation, we instead rely on conditional logic checks and a heuristic APR threshold intuitively chosen from our prior arbitrage trading experience. Following Hautsch's premise, our threshold is higher than what could be necessarily equated as profitable, to account for the risk of volatility and in part some of opportunity cost introduced by the latency. Because of decentralized nature of the market and lack of continuous available data for asset pricing, we will utilize data inference methods for $\bar{P}_S$, our "inferred acquirable price", the price we believe an asset can be bought from a limit-order approach. This variable $\bar{P}_S$ is used to calculate the APR that is compared to our threshold ratios.

Survival Analysis Approach for Arbitrage

Instead of entirely 'Arbitrage Boundary' accounting for the settlement latency like Hautsch et al, we attempt to model the probability a certain arbitrage opportunity persists past the settlement period and until our final trade execution. Modeling of time to arbitrage closure events is not common practice, but the study *Liquidity and Arbitrage in Options Markets: A Survival Analysis Approach* by Deville and Riva (2007) which models time to no-arbitrage (TTNA) in options markets deploys Deville's own previous work from Deville (2004) for the inference of significant parameters to a known TTNA (time to no-arbitrage) value through a Weibull AFT framework. Deville & Riva demonstrated how liquidity related factors, as well as the moment of arbitrage detection and likely trade execution, were significant parameters of their application of this model to the options market. We accept both of these premises, and will later demonstrate how we confirmed their significance to our markets arbitrage opportunities. Overall we implement a modified version of Deville's survival model framework, to include hierarchical groups with pooling, and make Bayesian predictions on newly detected arbitrage opportunities.

3.2 Price & Data Inference and Smoothing

All Price Inference Methods Remain Constant from Initial Prototype to Final Version To account for our sparse and discontinuous dataset, caused by occasions when our fetching algorithm could not locate a reasonably priced listing for an asset for a day, we infer and fill the missing price data rows so that we obtain a continuous price variable over the specified intervals for modeling. Because we target assets at the buy-order price, we infer prices by putting more weight on lower fetched prices.

To reconstruct a continuous, execution-realistic price series $P_S(t)$, we employ a two-stage preprocessing algorithm:

Temporal Imputation with Validity Decay

We first apply a forward-fill mechanism constrained by a validity horizon τ_{limit} (Look-around Window). Let $P_{S,obs}(t)$ be the observed price at time t . The imputed price $\bar{P}_S(t)$ is defined as:

$$\bar{P}_S(t) = \begin{cases} P_{S,obs}(t) & \text{if observed} \\ P_{S,obs}(t-k) & \text{if missing, where } k = \min\{j > 0 : P_{S,obs}(t-j) \text{ exists}\} \\ \text{NaN} & \text{if } k > \tau_{limit} \end{cases} \quad (3)$$

This decay constraint ensures that obsolete pricing information is discarded when the time-since-observation exceeds the stationary assumption of the asset class.

Asymmetric Gravity-Weighted Smoothing

Standard Gaussian smoothing assumes a symmetric noise. Given the variance and illiquidity of the overall market, this would be poor choice in the event that an assets fetched listing prices are deviating upwards

whilst the actual selling price remains constant. To more accurately model the “buyable” price rather than the “average” price, we implement a Gravity-Weighted Filter. For a centered rolling window W of size N , applied to the imputed series $\bar{P}_S(t)$, we assign weights w_i to each observation $\bar{P}_i \in W$ inversely proportional to its distance from the local minimum:

$$w_i = \frac{1}{1 + \gamma \cdot (\bar{P}_i - \bar{P}_{min})} \quad (4)$$

Where $\bar{P}_{min} = \min(W)$ is the lowest price in the window of the imputed series of the window, and γ is the *gravity factor* controlling the bias intensity. As we intuitively understand lower prices to more accurately depict the buyable price, we set γ to a relatively low value. The final inferred execution price $\hat{P}_S(t)$ is the weighted sum:

$$\hat{P}_S(t) = \frac{\sum_{i \in W} \bar{P}_i \cdot w_i}{\sum_{i \in W} w_i} \quad (5)$$

This nonlinear filter adequately suppresses high-priced outliers caused by illiquidity while being more sensitive to lower fetched prices, adequately reflecting our understanding of Market S's behavior and the limitations of our data.

Dynamic Feature Engineering

In addition to price inference, several dynamic covariates representing momentum (velocity) and volatility were engineered from the primary time-series variables. These features were created using three standard operations: rolling mean, rolling standard deviation, and the first difference.

For a given time series $Y(t)$ and a window of size w :

- The rolling mean is calculated as:

$$\bar{Y}_w(t) = \frac{1}{w} \sum_{i=0}^{w-1} Y(t-i) \quad (6)$$

- The rolling standard deviation (volatility) is calculated as:

$$\sigma_w(t) = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} (Y(t-i) - \bar{Y}_w(t))^2} \quad (7)$$

- The first difference (velocity) is calculated as:

$$\Delta Y(t) = Y(t) - Y(t-1) \quad (8)$$

These operations were applied to several source variables, including the *Ratio to Benchmark*, the *Price on Exchange T*, and the *Inventory Saturation Ratio*, to generate additional engineered features used in more refined versions of the model in addition to the unaltered values of original variables.

3.3 Evolving Methodological Components

All previously defined price inference methods of Section 3.2 remained constant from the prototype to the final versions of the model. The key components of the model that evolved throughout the project are summarized in Table 7. The following subsections will detail the initial and final versions of each component.

Table 1
Key Evolving Components of the Survival Model

Component	Symbol	Narrative of Evolution
Model Structure	$\beta_0 \rightarrow \beta_0^{[k]}$	Evolution from a non-hierarchical to a hierarchical intercept to account for asset variation.
Window Definition	$\Omega_{\text{naive}} \rightarrow \Omega_{\text{robust}}$	Refinement and addition of the conditions defining a window's start and end.
Covariate Set	$\mathbf{X}_{\text{naive}} \rightarrow \mathbf{X}_{\text{final}}$	Refinement from an initial set of naive proxies to an engineered set of features that uncovered the market's underlying logic.
Data Handling	$E_{\text{minimal}} \rightarrow E_{\text{robust}}$	Introduction of robust data handling, such as hard censoring dates, to manage market discontinuities and structural breaks.

3.3.1 Arbitrage Window Source Dataset

To construct the arbitrage windows, which serve as the unit of analysis for our model, we utilize our daily time-series master database, which contains aggregate information from all three platforms introduced in Section 2.1. This longitudinal data was used as the raw input for the previous price inference window definition algorithms detailed in the surrounding sections. The key variables collected for each discrete asset on a daily basis are summarized in Table 2.

Table 2
Structure of the Daily Source Dataset

Censored Variable / Symbol	Description	In Initial Model
<i>Asset Identification</i>		
Asset Characteristics	Categorical identifiers (base type, quality grade, etc.) used to define a unique asset and its variation type for the hierarchical model.	✓
<i>Exchange T (Target) Data</i>		
$P_T(t)$	The primary listed price on Exchange T.	✓
$I_{\text{held}}(t)$	Total quantity of the asset held in inventory by Exchange T.	✓
$L_{\text{cap}}(t)$	The maximum inventory capacity for the asset set by Exchange T.	✓
<i>Market S (Source) Data</i>		
$P_{S,\text{obs}}(t)$	The raw, observed (and possibly missing) price from Market S.	✓
Listing Volume on Market S	The number of listings observed for the asset on Market S.	
Reference Price on Market S	The reference price provided by the Market S platform.	
<i>Market B (Benchmark) Data</i>		
Ratio to Benchmark	The ratio of the price on Exchange T to the price on Market B.	✓
<i>Inferred & Derived Variables</i>		
$\hat{P}_S(t)$	Imputed (forward-filled) price on Market S.	✓
$\hat{P}_S(t)$	Final, gravity-weighted inferred price on Market S.	✓
$\text{APR}_{\text{anchored}}(t)$	Uncensored raw arbitrage price Ratio of $P_T(t)$ to $P_{S,\text{obs}}(t)$, before inference.	✓
$\text{APR}_{\text{spot}}(t)$	The final, actionable inferred arbitrage price ratio, calculated as $\text{APR}(t) = P_T(t) / \hat{P}_S(t)$.	

This daily data is processed by the algorithms described in the following subsection to generate the final, event-based data set where each row represents a single arbitrage window. While the APR is the direct arbitrage signal, the ‘**Ratio to Benchmark**’ ($P_T(t)/P_B(t)$), another price ratio between venues, serves as a key predictive factor for modeling arbitrage duration. Complete training arbitrage windows will include duration, outcome, and initial covariates.

3.4 Arbitrage Window Definition

The rules used to define the start and end of an arbitrage window have evolved significantly. The initial approach was a set of simple, static condition based approach, which was later refined into a more robust, dynamic system to handle market noise.

3.4.1 Arbitrage Opportunity Detection

We detect the openings of arbitrage opportunities similarly throughout the development of the project. Initial models defined an opening using the following conditions and strict threshold ratios

$$1.65 \leq \text{APR}(t) \quad \wedge \quad \text{ISR}(t) \leq \frac{1}{3} \quad (9)$$

We reiterate that $\text{APR}(t)$ is the Arbitrage Price Ratio, specifically the spot ratio, defined by the price on Exchange T ($P_T(t)$) over our inferred acquirable price on Market S ($\hat{P}_S(t)$). The term $\text{ISR}(t)$ represents Exchange T's inventory saturation, which we initially required to meet or be below 33% at onset to ensure our the fragile models at the time were only considering blatant arbitrage opportunities we previously would have found very compelling from our experience analyzing opportunities manually. As our confidence in our model grew, and it developed to quickly be more deterministic, we relaxed these ratios to be far less strict. Our final definition used the following conditions

$$1.60 \leq \text{APR}(t) \quad \wedge \quad \text{ISR}(t) \leq \frac{1}{2} \quad (10)$$

We believed including more potentially questionable arbitrage windows would cause the model to be more confident in assigning weight to the covariates included with this greater set of arbitrage windows. When these conditions are met, we instantiate a new arbitrage window beginning at t_0 , and save the data from Table 2 at the windows onset.

3.4.2 Window Closure Conditions and Closure Signal Smoothing

We define an arbitrage window as closed when one of two possible conditions is met. The first being a definitive end to window on the first observed day where $\text{ISR}(t) = 1$ and the asset can no longer be traded to Exchange T. As this condition factually closes an opportunity regardless of interpretation, this condition was present from the very first to the final versions. The other condition is a APR threshold check, however we instead use an anchored APR

$$\text{APR}_{\text{anchored}}(t) = \frac{P_T(t)}{\hat{P}_S(t_0)}, \quad \text{APR}(t) \equiv \text{APR}_{\text{spot}}(t) = \frac{P_T(t)}{\hat{P}_S(t)} \quad (11)$$

where $\hat{P}_S(t_0)$ is the inferred price on Market S at which the asset would have been bought if the arbitrage opportunity had been entered. Therefore, $\text{APR}(t)$ can be accurately referred to as the spot APR, $\text{APR}_{\text{spot}}(t)$. The $\text{APR}_{\text{anchored}}(t)$ threshold conditions remain present throughout all models, but further conditions were implemented in later versions.

Constant Condition 1: Inventory Saturation (Immediate Closure)

Any window throughout the entire project is immediately and definitively closed at time t if Exchange T will no longer accept the asset. This can happen if the limit L_{cap} is lowered, or if other traders have filled the the assets capacity.

$$\text{Closure at time } t \text{ if: } \text{ISR}(t) = 1 \quad (\text{i.e., } I_{\text{held}}(t) \geq L_{\text{cap}}) \quad (12)$$

This closure condition remains constant throughout all versions of our models.

Prototype Condition 2: Anchor Price Ratio Violation (Instantaneous Closure)

Generally, all APR closure condition thresholds are more relaxed relative to the price ratio condition from Section 3.4.1. We set the closure threshold to be (.02, .05) less than the detection threshold, since heuristically we still determine this ratio as profitable and as a worthwhile trade. For our prototype model, we used a single condition check on $\text{APR}_{\text{anchored}}(t)$

$$\text{Closure at time } t \text{ if: } \text{APR}_{\text{anchored}}(t) < 1.55 \quad (13)$$

where upon the violation of this condition, the window was definitively and instantaneously closed.

Intermediate Condition 2: Sustained Anchor Price Ratio Violation (Delayed Closure)

The first refinement made to the APR condition follows the base APR closure condition but further requires this closure condition to be sustained for three consecutive days:

$$\text{Closure recorded at date } t \text{ if } \forall \tau \in t, t+1, t+2, \quad \text{APR}_{\text{anchored}}(\tau) < 1.55 \quad (14)$$

This persistence requirement acts as signal de-bouncing filter for occasional noise in Exchange T's pricing algorithm. During development, Exchange T's asset pricing was analyzed, and we discovered the platform uses a dynamic pricing algorithm with strong negative feedback from the rate an asset is traded into the platform; a simplified representation of which can be modeled as:

$$P_T(t+1) = P_T(t) + \lambda(\theta P_B(t) - P_T(t)) - \kappa \Delta \text{ISR}(t) + \varepsilon_t \quad (15)$$

where the price at the next time step, $P_T(t+1)$, is a function of the current price anchored to a target proportional to the benchmark price, $P_B(t)$, with a reversion speed λ . Crucially, it is negatively impacted by the change in inventory, $\Delta \text{ISR}(t)$, scaled by a feedback factor κ . The term ε_t represents random noise. This negative feedback term, $-\kappa \Delta \text{ISR}(t)$, is what causes the temporary sharp price suppressions we observe, justifying our use of a multi-day persistence requirement to filter out these transient 'fakeout' closures.

3.4.3 Final Condition 2: Sustained Anchor and Spot Price Ratio Violations (Delayed Closure)

The final most robust closure logic uses the basis from 3.4.2 with added requirements related to both $\text{APR}_{\text{anchored}}(t)$ and $\text{APR}_{\text{spot}}(t)$. This condition exemplifies the point where our project departs from any rational inventory, to meet the absurdity of the market and lack of complete data. Therefor, we will present the logic used now and revisit the utility of this approach towards X.X section.

$$\text{Trigger bridging check at } t \text{ if: } \left((\text{APR}_{\text{spot}}(t) < \gamma_{\text{close}}) \vee (\text{APR}_{\text{anchored}}(t) < \gamma_{\text{close}}) \right) \wedge (\text{APR}_{\text{spot}}(t) < \gamma_{\text{open}}) \quad (16)$$

where γ_{open} is the opening threshold (e.g., 1.60) and γ_{close} is the slightly lower closure threshold (e.g., 1.55). Although somewhat convoluted, this condition simplifies to two primary triggers for initiating a bridging check. First, a check is always initiated if the spot market itself shows significant weakness, where $\text{APR}_{\text{spot}}(t) < \gamma_{\text{close}}$. Second, a check is also initiated if only the anchored position's profitability decays, where $\text{APR}_{\text{anchored}}(t) < \gamma_{\text{close}}$, as long as the spot market is not simultaneously strong enough to be considered a new opportunity (i.e., $\text{APR}_{\text{spot}}(t) < \gamma_{\text{open}}$).

$$\text{Closure recorded at date } t \text{ if } \forall \tau \in t, t+1, t+2, \quad (\text{APR}_{\text{spot}}(\tau) < 1.55) \quad (17)$$

we then perform the same bridging algorithm as in Intermediate Condition 2, however, we instead use $\text{APR}_{\text{spot}}(t)$ during the comparison. Altogether, this unintelligible logic flow better maintains the premise that the Bayesian Weibull AFT model operates under the assumption that it is only inspecting the arbitrage windows themselves, not the overarching market momentum, although there are faults discussed later.

3.4.4 Final Arbitrage Window Data Structure

The output of the window definition process is an event-based dataset. Each row represents a single arbitrage window, as exemplified in Table 3.

Table 3
Example of a Single Arbitrage Window Record

Variable / Symbol	Example Value
Start Date (t_0)	2025-08-15
End Date (t_{end})	2025-08-26
Duration (T_i)	11 days
Event Observed (E_i)	1 (Closed)
Asset Variation (k_i)	"Type A"
Initial Covariate Vector ($\mathbf{X}(t_0)$)	[0.21, -1.05, ..., 0.88]

3.5 Model Specification

Following the framework established by [Deville \(2004\)](#), we employ the Weibull Accelerated Failure Time (AFT) model, with the target variable T as the duration of an asset's arbitrage window. The Weibull distribution is parameterized by shape $\alpha > 0$ and scale $\lambda > 0$. The probability density function $f(t)$ and the survival function $S(t)$ are defined as:

$$f(t | \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{t}{\lambda} \right)^{\alpha-1} \exp \left(- \left(\frac{t}{\lambda} \right)^{\alpha} \right) \quad (18)$$

$$S(t | \alpha, \lambda) = P(T > t) = \exp \left(- \left(\frac{t}{\lambda} \right)^{\alpha} \right) \quad (19)$$

In the AFT framework, the covariates X act to accelerate (negative coefficient) or decelerate (positive coefficient) the passage of time. We model the log of the scale parameter λ as a linear combination of covariates:

$$\log(\lambda_i) = \mu_i = \beta_0 + \sum_{j=1}^J \beta_j X_{ij} \quad (20)$$

Here, μ represents the log-expected lifetime of an arbitrage window. A positive coefficient $\beta_j > 0$ implies that an increase in feature X_j increases λ , thereby slowing the decay of the arbitrage window and extending the expected duration.

3.5.1 Hierarchical Structure

The model's initial, non-hierarchical structure in the initial prototype treated all asset variations identically, using a single global intercept β_0 for all observations. Although this approach was sufficient to prove arbitrage duration as generally predictable, it fails to account to our specific niche markets reality that the market fundamentals of different variations are inherently different. As the specific asset class we operate arbitrage within generally shares characteristics related to the arbitrage opportunities across variations, but with differing behavior of the windows across variations, we wish to parameterize this difference without further reducing datapoints.

Because of our requirements, simply using independent regression models is not ideal; henceforth we employ a Hierarchical Bayesian structure with partial pooling. We allow the intercept β_0 to vary between the asset variations k , while assuming these variation-specific intercepts are drawn from a common global distribution:

$$\beta_0^{[k]} \sim \mathcal{N}(\mu_{\text{global}}, \sigma_{\text{offset}}^2) \quad (21)$$

$$\mu_i = \beta_0^{[k[i]]} + \boldsymbol{\beta} \cdot \mathbf{X}_i \quad (22)$$

This structure therefor allows variations to “borrow strength” by shrinking their estimates toward the global mean of the asset variations. This allows the model to regularize predictions for the asset variations directly addressing the aspect of information risk caused from data sparsity as previously discussed (Section 2.3).

As the true observable durations and closures across certain variations can greatly differ, this approach produces far more realistic survival probabilities. Thus all versions used in the live market will always be hierarchical, however we retain and utilize the non-hierarchical version to test and prove global covariate significance.

3.5.2 Prior Distributions

To regularize estimation and ensure convergence-particularly for asset categories with sparse data, we assign weakly informative priors to the model parameters.

The global intercept $\beta_{0,\text{global}}$ is centered on the natural logarithm of the empirical median duration ($\tilde{t}_{\text{obs}}$) of the training set. This heuristic initialization will also anchor the Markov Chains in a plausible region of the parameter space, significantly reducing burn-in. The standard deviation of this prior, σ_{int} , represents our confidence in this heuristic. This parameter was tuned across development from wide uninformative values ($\sigma_{\text{int}} = 1.5$) in early phases to the narrower prior ($\sigma_{\text{int}} = 0.5$) used in the final model, reflecting in growing confidence as our data set increased:

$$\beta_{0,\text{global}} \sim \mathcal{N}(\log(\tilde{t}_{\text{obs}}), \sigma_{\text{int}}) \quad (23)$$

Since we utilize many coefficients within the scope of liquidity and pricing variation as well as introducing more variables through the Hierarchical Bayesian structure, we assign zero-centered Normal priors to the hierarchical offsets and covariate coefficients. This prior structure enforces a regularization effect analogous to Ridge Regression, shrinking coefficient estimates towards zero unless truly strong evidence exists in the likelihood to justify a deviation. The standard deviations of these priors, σ_{off} and σ_{coef} , were tuned throughout development, moving between wider and narrower values to reflect our confidence in the features and the hierarchical structure. This is essential towards our risk mitigation by reducing overfitting on high-dimensional feature vectors through our sparse data:

$$\beta_{0,\text{offset}}^{[k]} \sim \mathcal{N}(0, \sigma_{\text{off}}) \quad \text{for } k = 1, \dots, K \quad (24)$$

$$\beta_j \sim \mathcal{N}(0, \sigma_{\text{coef}}) \quad \text{for } j = 1, \dots, P \quad (25)$$

Lastly we constrain our Weibull shape parameters α which governs hazard rate trajectory (risk of accelerating or decreasing probability the window closes) to the positive domain via a Half-Normal prior. The standard deviation, σ_{shape} , was kept relatively wide to allow the model flexibility in learning the underlying hazard shape from the data.

$$\alpha \sim \text{HalfNormal}(\sigma_{\text{shape}}) \quad (26)$$

3.5.3 Censoring and Bayesian Inference

Our produced arbitrage window data set contains right-censored observations, where the arbitrage window was still considered “open” at the end of observation periods. During a market crash event that caused discontinuities in our data set, we define the onset of the crash as censor date during window data set formulation, thus maintaining accurate training data as unpredictable market events are outside the scope of the model. Whereas standard regression cannot account for these points, our Bayesian framework explicitly accounts for this in our likelihood function.

Utilizing our E_i flag variable to indicate closure event occurrence $E_i = 1$ and $E_i = 0$ if the observation ended while the window was open (censored). The contribution to the likelihood is as follows:

$$\mathcal{L}(\theta \mid t_i, E_i) = \underbrace{f(t_i \mid \theta)^{E_i}}_{\text{Event Observed}} \times \underbrace{S(t_i \mid \theta)^{1-E_i}}_{\text{Censored}} \quad (27)$$

Thus our censored observations contribute survival probability (probability of lasting at least as long as observed) rather than density over the windows time.

3.5.4 Data Preprocessing and Scaling

Prior to being used in the model, the selected covariates underwent a two-step transformation process. First, variables with large numerical ranges and skewed distributions were log-transformed using $\log(1+x)$ to stabilize their variance. Following this, all continuous covariates were standardized to have a mean of zero and a standard deviation of one. This ensures that the learned model coefficients are directly comparable and that the regularizing priors have a consistent effect. These transformations were implemented using the *Pandas* and *NumPy* libraries, with the final standardization step performed using the ‘StandardScaler’ from the *Scikit-learn* library.

3.5.5 Robustness Under Data Inference and Sparsity

3.5.6 Posterior Inference

The model's posterior distribution is estimated by Markov Chain Monte Carlo (MCMC) methods. Using the No-U-Turn Sampler (NUTS) from the PyMC Python library. Deploying four independent chains in parallel, each with 1,000 tuning steps for warm-up and 2,000 sampling steps that are retained for inference. Model convergence is assessed through potential scale reduction factor $\hat{R}$, which was approximately 1.0 for all parameters indicating convergence to the target distribution.

3.6 Covariate Selection

The set of covariates utilized by the model differentiated substantially during the projects development, progressing from a limited naive set of proxies to a final engineered set of features. Throughout model development we attempt to adhere to the premise of (Deville and Riva, 2007), that liquidity and market dynamics are strongly related to the duration of arbitrage windows. Our initial covariate sets fail to integrate this premise, however the final covariate set, greatly influenced by the discoveries made during the analysis of Exchange T's pricing model (Equation 15), succeed in applying the premise. Our attempts at integrating our own market understanding through our covariate set was similar.

The purpose of our selected covariate set is not only to predict survival probabilities on new arbitrage opportunities which differentiate from one another but also to prove statistical significance of covariates and expose market truths. Thus having both sparse data and data variables that are highly correlated, it is essential to excluded redundant and unstable (multi-collinear) covariates to maintain determinism in our models output.

For example, the prices on Exchange T and the source Market S, were highly correlated. We selected the Exchange T price data for the model due to its superior integrity and completeness compared to the sparsely observed P2P market data. For liquidity metrics, we chose Exchange T's asset capacity as opposed to saturation ratio or the assets held in inventory, because they exhibited complex non-linear behavior that was not well captured in this respective framework.

3.6.1 Initial Covariates ($\mathbf{X}_{\text{naive}}$)

The initial model used a small, naive set of covariates $\mathbf{X}_{\text{naive}}$ intended to test our fundamental hypothesis.

Table 4
Initial Covariate Selection (X_{naive})

Selected Covariates	
Category	Variable
Opportunity Magnitude	Initial Exchange Price ($P_T(t_0)$)
	Initial Inferred Arbitrage Ratio ($\text{APR}_{\text{anchored}}(t_0)$)
Liquidity	Exchange Capacity ($L_{\text{cap}}(t_0)$)
External Dynamics	Ratio to Benchmark ($P_T(t)/P_B(t)$)
	3-Day Benchmark Ratio Slope ($\left \frac{d}{dt} \frac{P_T(t_0)}{P_B(t_0)} \right $)
	Benchmark Volatility ($\sigma_w(P_T(t)/P_B(t))$)
Volatility / Risk	Price Volatility on Exchange T ($\sigma_w(P_T(t_0))$)
Rejected Covariates	
Reason for Exclusion	Variable
High collinearity; inferior data integrity	Market S Price ($\hat{P}_S(t_0)$)
	Market S Price Volatility ($\sigma_w(\hat{P}_S(t_0))$)
Exhibited complex non-linear behavior	Inventory Saturation Ratio ($\text{ISR}(t_0)$)
	Assets Held in Inventory ($I_{\text{held}}(t_0)$)

Altogether we found this selection to be somewhat sufficient for proving our fundamental hypothesis, and the small covariate set, majorly due to lack of data, inherently enforced Occam's Razor which helped produce some significance for the set.

3.6.2 Intermediate Refinements ($X_{\text{naive}} + \text{ListingQuantity}_S$)

Before deploying our first hierarchical model to the live market, we made refinements to the handling of our covariates. We began applying log transformations and standardization to all covariates to improve numerical stability and coefficient interpretability, a practice which was retained for the final model.

Between the deployment and eventual market crash, we made addition of a liquidity metric derived from newly fetched listing quantity on Market S, ListingQuantity_S , to our data. At this point as, we had yet to analyze Exchange T's pricing model, we primitively appended the variable to our covariate set.

Table 5
Covariate Appended to Initial Covariate Selection (X_{naive})

Category	Variable
Liquidity	Listing Quantity on Market S (ListingQuantity_S)

3.6.3 Final Covariate (X_{final})

Our choice of final covariates was based on our investigation of Exchange T's pricing algorithm. After noting a curious relation with between the *Ratio To Benchmark* and sales history on Market B, we found that when extremely low-liquidity assets had an overpriced sale on Market B, Exchange T's $P_T(t)$ would begin climbing while its *ratio to benchmark* would sharply drop. This exposes a more classic arbitrage concept of price-level convergence. Thus we understand Exchange T's algorithm to be using some form of mean-reversion

where its target price is some unknown function of the Market B price (P_B), aiming to eventually revert to its previous *Ratio To Benchmark* over some period of time.

Market B specific asset sales price history was not scraped in this project to avoid violating the market's terms and conditions and loosing access to trading, as this platform is provided by the market central authority. So instead we attempt to capture this price over-extension in a signal with the engineered feature **Deviation from Benchmark Ratio**:

$$\text{RatioDev}(t_0) = \frac{R_B(t_0) - \bar{R}_{B,15}(t_0)}{\bar{R}_{B,15}(t_0)} \quad (28)$$

where $R_B(t_0)$ is the current benchmark ratio at the window's start, and the benchmark ratio's 15-day simple moving average, $\bar{R}_{B,15}(t_0)$ (calculated using the rolling mean operator from (6) with $w = 15$).

As the occurrence or lack of these extensions are highly related to how liquid an asset truly is, we reevaluate our previous liquidity related covariates and engineer a **composite congestion metric** feature to relate the liquidity, which we represent through *ListingQuantity_S*, to the available buying capacity on Exchange T, L_{cap} . **Composite congestion metric** is equated as following: It is defined as:

$$\text{Congestion}(t_0) = \frac{\text{ListingQuantity}_S(t_0)}{L_{\text{cap}}(t_0) + 1} \quad (29)$$

where $\text{ListingQuantity}_S(t_0)$ is the observed listing volume for the asset on Market S at the window's start, and $L_{\text{cap}}(t_0)$ is the maximum inventory capacity on Exchange T. The denominator is incremented by one to prevent division by zero in cases where capacity might be reported as zero, although this should not occur.

As we now use different liquidity and ratio metrics, we remove any possible multi-collinearly related covariates, such as the simple liquidity metric $\text{ListingQuantity}_S(t_0)$ used in the intermediate set, and the simple exchange capacity $L_{\text{cap}}(t_0)$. Other possibly multi-collinear variables were not removed as they were tested to be non-problematic.

The final model deployed a set of engineered features, supplemented by the most robust of the initial covariates. All continuous covariates were log-transformed where appropriate and standardized before being passed to the model.

Table 6
Final Covariate Selection $\mathbf{X}_{\text{final}}$

Selected Covariates	
Category	Variable (Symbol from Code)
Opportunity Magnitude	Initial Exchange Price ($P_T(t_0)$)
	Initial Inferred Arbitrage Price Ratio ($\text{APR}_{\text{anchored}}(t_0)$)
Volatility / Risk	Price Volatility ($\sigma_w(P_T(t_0))$)
Systemic Momentum	3-Day Benchmark Ratio Slope ($\left \frac{d}{dt} \frac{P_T(t_0)}{P_B(t_0)} \right $)
	Deviation from Benchmark Ratio MA ($\text{RatioDev}(t_0)$)
Liquidity / Congestion	Composite Congestion Metric ($\text{Congestion}(t_0)$)
Removed or Rejected Covariates	
Reason for Exclusion	Variable
<i>Removed from Initial Model (Replaced by engineered features)</i>	
	Exchange Capacity ($L_{\text{cap}}(t_0)$)
	Ratio to Benchmark ($P_T(t_0)/P_B(t_0)$)
	Benchmark Volatility ($\sigma_w(P_T(t_0)/P_B(t_0))$)
<i>Rejected due to Redundancy</i>	
	Benchmark Momentum ($\Delta(P_T(t_0)/P_B(t_0))$)
	Price Velocity ($\Delta P_T(t_0)$)
	Saturation Velocity ($\Delta \text{ISR}(t_0)$)

3.7 Prediction Methodology

For a new opportunity with covariates $\mathbf{X}$ and category k :

3.7.1 Pre-processing

For prediction, any new opportunity's data must undergo the identical transformations detailed in Section 3.5.4, using the pre-processing parameters saved from the training phase. This involves first applying the necessary log-transformations, then standardizing the continuous covariates using the saved training data's mean and standard deviation. Finally, the new opportunity's asset variation category is mapped to its corresponding integer index based on the categorical order established during training to ensure consistency with the hierarchical model structure.

3.7.2 Posterior Predictive Computation

The computation of the survival probability differs slightly between the non-hierarchical prototype and the final hierarchical model, with the core difference being the calculation of the linear predictor, $\mu^{(s)}$. For each posterior sample s :

Non-Hierarchical (Prototype & Testing) Model The linear predictor is calculated using a single, pooled intercept, β_0 .

$$\mu^{(s)} = \beta_0^{(s)} + \boldsymbol{\beta}^{(s)} \cdot \mathbf{X}_{\text{new, scaled}} \quad (30)$$

Hierarchical (Deployed & Final) Model The linear predictor is refined by incorporating the category-specific offset for the asset's variation, k .

$$\mu^{(s)} = \beta_{0,\text{global}}^{(s)} + \beta_{0,\text{offset}}^{[k](s)} + \boldsymbol{\beta}^{(s)} \cdot \mathbf{X}_{\text{new, scaled}} \quad (31)$$

For both models, the subsequent calculations for the scale parameter $\lambda^{(s)}$ and the final survival probability $S(t)^{(s)}$ are identical:

$$\lambda^{(s)} = \exp(\mu^{(s)}) \quad (32)$$

$$S(t)^{(s)} = \exp\left(-\left(\frac{t}{\lambda^{(s)}}\right)^{\alpha^{(s)}}\right) \quad (33)$$

3.7.3 Uncertainty Quantification

The collection of posterior samples for $S(t)^{(s)}$, computed using NumPy, represents the full predictive uncertainty. To distill this distribution into human readable metrics for interpretation, we summarize it as follows with the *ArviZ* library:

- **Point Estimate:** The median of the posterior samples for $S(t)$ is used as the central, robust estimate of the survival probability.
- **Credible Interval:** A Highest Density Interval (HDI) is computed from the posterior samples using the *ArviZ* library. This provides a range (we utilize 94%) that contains the true survival probability with high confidence.

The final model's hierarchical structure produces wider HDIs for its global coefficients compared to the non-hierarchical version, but the combination with learned offsets ultimately results in more differentiated and accurate predictions for specific asset variations. Thus we prefer the non-hierarchical version for global arbitrage interpretation, whereas we use the hierarchical for active predictions.

3.8 Coefficient Interpretation

In our Log-Linear AFT framework, for both hierarchical and non-hierarchical, 20 and 21, our covariates directly effect log-scale expected duration. To interpret these effects in terms of actual time, we exponentiate the coefficients to derive the *Acceleration Factor* (AF) of any covariate.

3.8.1 Non-Hierarchical's General and Hierarchical's Global and Variation-Specific Effects

As previously stated, the non-hierarchical model simply utilizes β_j , the coefficient for a given covariate X_j effective offset, whereas the hierarchical version of the model allows distinct sensitivities for each asset variation k , where for X_j the total effective coefficient $\beta_j^{[k]}$ is the sum of the market-wide baseline and the variation-specific offset:

$$\text{Non-Hierarchical: } \beta_j \quad \text{Hierarchical: } \beta_j^{[k]} = \beta_{j,\text{global}} + \beta_{j,\text{offset}}^{[k]} \quad (34)$$

The hierarchical structure allows for granular interpretation, the **Global Coefficient** ($\beta_{j,\text{global}}$) representing the average market sensitivity to the factor X_j , and the **Offset** ($\beta_{j,\text{offset}}^{[k]}$) captures the idiosyncratic behavior of asset class k .

It's important to understand the non-hierarchical β_j and hierarchical $\beta_{j,\text{global}}$ are not equivalent, although typically are similar, across the two models.

$$\text{For a } X_j : \quad \beta_j \neq \beta_{j,\text{global}} \quad (35)$$

3.8.2 Acceleration Factor

In both non-hierarchical and hierarchical version of the model we measure β_j by measuring how increasing the value for a standardized covariate X_j effects median duration T relative to X_j itself, when holding all other variables constant. We define this multiplicative effect as Acceleration Factor $\mathbf{AF}_j$. In our non-hierarchical model, we understand the multiplicative effect of X_j on the median arbitrage duration T as:

$$\mathbf{AF}_j = \frac{\text{Median}(T|X_j + 1)}{\text{Median}(T|X_j)} = \exp(\beta_j) \quad (36)$$

For a specific asset variation k , the multiplicative effect of covariate X_j on the median arbitrage duration T is:

$$\mathbf{AF}_j^{[k]} = \frac{\text{Median}(T|X_j + 1)}{\text{Median}(T|X_j)} = \exp(\beta_j^{[k]}) \quad (37)$$

We interpret these standardized covariate effects as follows (interpretations apply to both β_j and $\beta_j^{[k]}$):

- **Deceleration** ($\beta_j > 0$): The feature extends the opportunity duration. An increase of one standard deviation in X_j multiplies the expected survival time by a factor of $\mathbf{AF} > 1$.
- **Acceleration** ($\beta_j < 0$): The feature shortens the opportunity duration. An increase of one standard deviation in X_j multiplies the expected survival time by a factor of $\mathbf{AF} < 1$.
- **Magnitude**: For coefficient values where $|\beta| < 0.1$, the percentage change in expected duration approximates to:

$$\% \Delta E[T] \approx 100 \times \beta_j \quad (38)$$

Consequently, we interpret $\beta_{0,\text{offset}}^{[k]}$ as the variation-specific deviation from the global baseline duration, while $\beta_{j,\text{offset}}^{[k]}$ represents the variation-specific deviation in sensitivity to market forces.

3.8.3 Interpretation of Coefficient Significance & Uncertainties

As demonstrated through $\mathbf{AF}_j$, we understand positive coefficients to increase and negative coefficients to reduce respective arbitrage opportunities duration. Utilizing the computed posterior predictions derived from our data from Section 3.7.3, we can determine the significance of the presence of a specific covariate X_j in our model. If the respective coefficient β_j of any X_j doesn't contain 0 within it's centered 94% HDI:

$$\text{Significant}(\beta_j) \iff 0 \notin [\text{HDI}_{3\%}, \text{HDI}_{97\%}] \quad (39)$$

we consider it to have a **statistically credible effect**, and infer the variable exerts a non-zero influence to it's respective direction on arbitrage duration.

As previously stated in Subsection 3.8.1, the non-hierarchical β_j and hierarchical $\beta_{j,\text{global}}$ are differentiated between respective models. Although we will typically test our hypothesis with the non-hierarchical β_j HDI's, we understand that the difference between β_j and $\beta_{j,\text{global}}$, can expose some degree of **overconfidence** in β_j through the non-hierarchicals deflation of the standard error by ignoring the structural heterogeneity between asset variations. The effects between non-hierarchical and hierarchical model output on identical posterior data will be further analyzed in Results A1!

4 Arbitrage Survival Model Development and Deployment

In this section we will mainly outline our development process for data interpretation and the survival model, the deployments and terminations of this model throughout development and our projects operation period, as well as the context of discoveries and market events that impacted our development. Although we will provide visual figures for the models prediction outputs throughout development, our models computed posterior prediction, which contains covariate coefficients β_j and their respective HDI results, are shown and interpreted later in Section: A1.

Please note the development timeline is majorly obfuscated to protect our projects exploited opportunities, as well as because we simply did not journal development, thus all information was fetched retrospectively.

4.1 Development and Deployment Context

As introduced, this market is niche and has unusual characteristics, as will be further discussed with respect to our SDE in Section:6, this market regularly experiences short periods of higher-volatility, and less frequently large structural changes to underlying market mechanics. Thus, throughout development, deployment was frequently paused and resumed depending on market conditions and our confidence in the model we possessed at those respective times.

Towards the end of the deployment period, we were so fortunate as to experience an unprecedented **market crash** which resulted in a market regime change. Even more fortunately for us this crash and regime change directly impacted the specific asset classifications (not variations) that are the entirety of the arbitrage model's scope. Thus, the crash referred to in the following timeline had massive implications on the time our model was deployed throughout the 144 day operation period as well as the resulting changes made to the model after this event.

It is worth noting that the models profitability and success rate were not necessarily affected by the crash because suspicious market behavior reported prior to the crash as well as our own determinations that general market behavior had become excessively unhinged relative to already hostile environment, thus had already paused arbitrage operation prior to the crash.

4.2 Development Results: Obfuscated Iterative Timeline

This project was not journaled, as it was never anticipated to be developed so much. However, in the process of proving that Deville's TTNA Weibull AFT could be modified to make successful predictions, as well as overcome the stochastic settlement latency, there were specific development phases that broadly represent the leaps in the project's development and discoveries made. The key components of the model that evolved throughout this process are summarized in Table 7. The following subsections will detail this progression.

Table 7
Key Evolving Components of the Survival Model

Component	Symbol	Narrative of Evolution
Model Structure	$\beta_0 \rightarrow \beta_0^{[k]}$	Evolution from a pooled (non-hierarchical) to a hierarchical intercept to account for asset variation.
Window Definition	$\Omega_{\text{naive}} \rightarrow \Omega_{\text{robust}}$	Refinement of the rules defining a window's start and end, particularly adding signal debouncing to filter transient closures.
Covariate Set	$\mathbf{X}_{\text{naive}} \rightarrow \mathbf{X}_{\text{final}}$	Refinement from an initial set of naive proxies to an engineered set of features that uncovered the market's underlying logic.
Prior Confidence	$\sigma_{\text{wide}} \rightarrow \sigma_{\text{narrow}}$	Tuning of prior distributions, moving from wide (uninformative) to narrower (weakly informative) priors as model confidence grew.
Data Handling	$E_{\text{minimal}} \rightarrow E_{\text{robust}}$	Introduction of robust data handling, such as hard censoring dates, to manage market discontinuities and structural breaks.

4.2.1 Phase 1 (Mid-July 2025): Initial Proof-of-Concept

Phase 1 Model State				
Model Structure	Window Def.	Covariate Set	Prior Conf.	Data Handling
β_0	Ω_{naive}	$\mathbf{X}_{\text{naive}}$	$\sigma_{\text{narrow}} \parallel \sigma_{\text{wide}}$	E_{minimal}

The initial objective was to answer a single question: can we even code- sorry I mean can the arbitrage window data created from price inference 3.2 and the naive window conditions, and modifications to the Weibull AFT function 3.5? To test this, we developed a simple, non-hierarchical AFT model using a parsimonious set of intuitive covariates. At this stage, our data handling was minimal, but we experimented with both wide uninformative priors and narrow informative priors to see which produced more coherent results for simple test cases. This baseline model crucially showed promise, providing the first statistical evidence that opportunity duration was not random and could be predicted by initial market conditions, thus justifying further development.

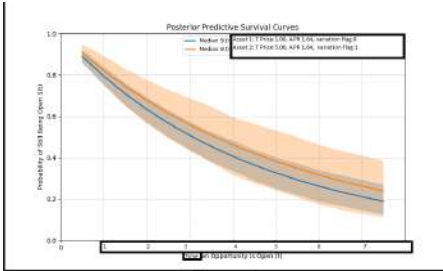


Figure 1
First ever predictive output from the initial non-hierarchical model on test assets.

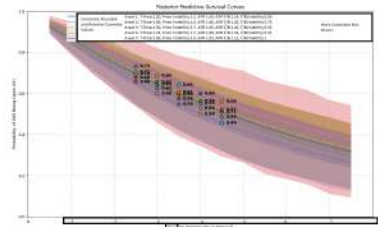


Figure 2
The initial model showed significant overlap in credible intervals, indicating low predictive differentiation.

4.2.2 Phase 2 (Late-July 2025): Hierarchical Shift & Initial Deployment

Phase 2 Model State				
Model Structure	Window Def.	Covariate Set	Prior Conf.	Data Handling
$\beta_0 \rightarrow \beta_0^{[k]}$	Ω_{naive}	X_{naive}	$\sigma_{\text{narrow}} \rightarrow \sigma_{\text{wide}}$	E_{minimal}

With the proof-of-concept validated, the next objective was to develop a model that produces more cohesive results. Our working hypothesis was that the initial model, while proving survival probabilities may be predictable, was too simplistic and unintelligible to truly be making accurate predictions; treating all asset variations as either identical or by categorizing multiple possible sets into a single boolean, felt like a critical oversimplification that was causing incorrect output. As a forward-looking risk management decision, we implemented the hierarchical structure to allow the model to learn variation-specific behaviors. We choose a wide prior confidence σ_{wide} as we believed the sparse data points being split across variations necessitated stronger covariates effects if used in the real market. While the sparse data at this stage meant the statistical significance of this change was not yet significant, we now believe it was critical to our early live market success. Figures 2 and 3 show two different asset variations and predicted survivability probabilities over time. We note how the more liquid asset variation with more training data points in Figure 2 cannot deviate far from the prior Weibull distribution, and how Figure 3 with a more illiquid asset variation and less training points deviates far more.

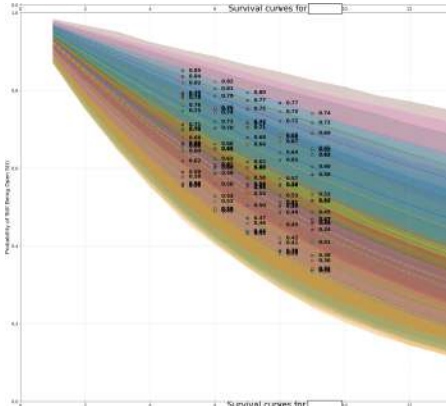


Figure 3
Predictive output from the first hierarchical model on a liquid asset variation. The model showed significant improvement from original prototype at creating realistic survivability distributions.

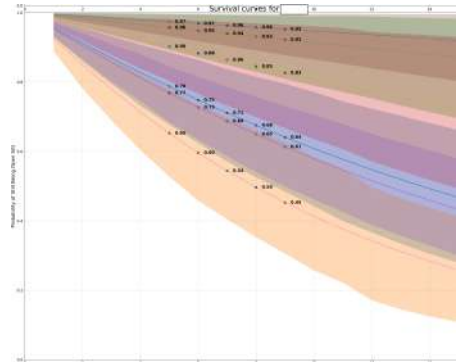


Figure 4
Output for an illiquid asset variation. The hierarchical structure improved distributions, but naive window definitions led to some unrealistic results.

4.2.3 Phase 3 (September-November 2025): Market Crash and Model Resilience

Phase 3 Model State				
Model Structure	Window Def.	Covariate Set	Prior Conf.	Data Handling
$\beta_0^{[k]}$	$\Omega_{naive} \rightarrow \Omega_{robust}$	X_{naive}	σ_{wide}	$E_{minimal} \rightarrow E_{robust}$

After deployment of the Phase 2 hierarchical model, the unpredictable market event risk explained in Section 2.3 became a reality. An unprecedented market regime change took effect which violated our model’s core assumptions, caused data discontinuity, and wiped out a significant portion of the market cap. Platforms freezing trade resulted in many extremely long arbitrage window durations which skewed model outputs. Every discrete asset prediction was believed to likely survive $S(\text{settlement duration}) > .5$.

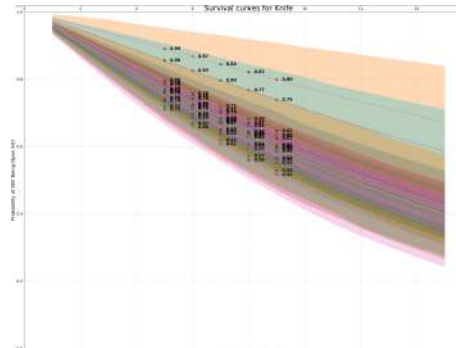


Figure 5
The market crash of late 2025. The text on the left describes the event, while the figure on the right shows the resulting incoherent model output during the platform freezes, demonstrating the temporary failure of the model’s assumptions.

As stated in x.x Portfolio Results and Reflections, we had already ceased arbitrage operation and thus took no arbitrage related losses. However, this prompted a significant re-haul focused on our model’s resiliency. Implementing more robust data handling during the arbitrage window preparations with new window closure definitions (Ω_{robust}) and a hard censoring date for all windows active during the freeze. We retain the wide confidence prior to help the model produce differentiable predictions from the new windows. We immediately felt confidence to reinstate operations upon the new models output as it began producing the most accurate predictions yet that reinforced our conceptions of safe and risky predictions. As shown in Figures 6 and 7, after

these fixes, the model was again producing coherent-if not yet precise-predictions on the post-crash dataset. *Figures 6, 7, and Phase 4's 8 use the same post-crash dataset to demonstrate final improvement*

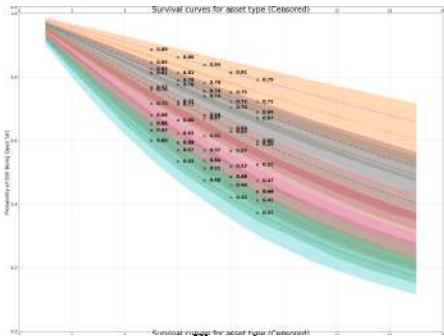


Figure 6
Post-crash model output for a liquid asset, using the same dataset as the final model to show intermediate improvement.

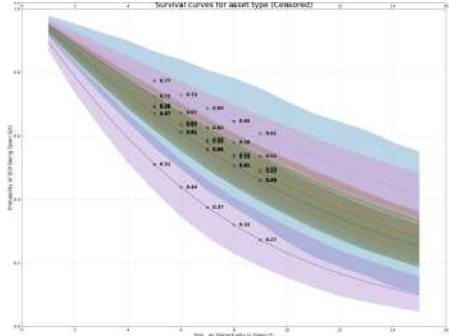


Figure 7
Post-crash model output for an illiquid asset. Note the improved but still wide credible intervals before final feature engineering.

4.2.4 Phase 4 (December 2025-January 2026): The Breakthrough and Final Refinement

Phase 4 Model State				
Model Structure	Window Def.	Covariate Set	Prior Conf.	Data Handling
$\beta_0^{[k]}$	Ω_{robust}	$\mathbf{X}_{\text{naive}} \rightarrow \mathbf{X}_{\text{final}}$	$\sigma_{\text{wide}} \& \sigma_{\text{narrow}}$	E_{robust}

With a resilient data framework in place, the final phase focused on transforming the model from merely functional to truly predictive. This phase marked the project’s most significant breakthrough: the development of the final, engineered covariate set ($\mathbf{X}_{\text{final}}$). For the global, market-wide coefficients, we transitioned to narrow priors (e.g., $\sigma = 0.5$), reflecting our new-found confidence in the general magnitude and direction of these effects from our now larger dataset that grew with time. However, for the hierarchical offsets that capture category-specific deviations and the Weibull prior distribution, we intentionally retained wider priors (e.g., $\sigma_{\text{hierarchical}} = 1.0$, $\sigma_{\text{weibull}} = 1.4$). This hybrid approach allowed the model to be confident about the average market behavior while retaining the flexibility to learn the nuances of the asset variations, and more confidently pull away from the prior Weibull distribution.

By incorporating newly scraped liquidity data from Market B and creating several momentum-based features (detailed in Section 3.6), we produced the most accurate predictions we believe is possible with the publicly available data we can collect in Figure 8. The model also differentiates more strongly than in Phase 3, despite narrower priors. *Figures 8 and Phase 3's 6, 7, use the same post-crash dataset to demonstrate final improvement*

4.3 Implications on the Survival Curves

As previously mentioned the survival curves we’re mainly used on a certain class of assets, specifically the typically higher-ticket items in the market. The update which implemented a new means of being able to produce these assets the curves are used on by recycling of lesser assets. In turn the crash on these assets required the right-end censoring in the arbitrage model since Exchange T closed trading until their pricing models were able to account for the crash.

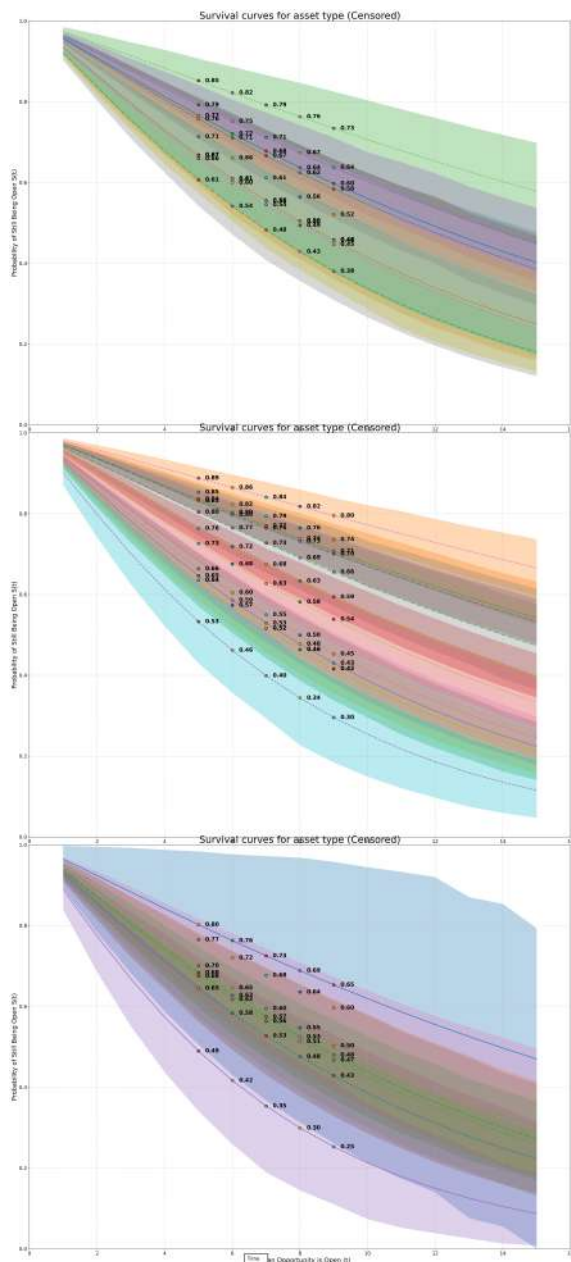


Figure 8

Final model predictive output for an illiquid asset variation, using the same dataset as Phase 3 to demonstrate the dramatic improvement from the final covariate set. The model now produces highly differentiated survival curves with tight, actionable credible intervals.



Figure 9

One type that consists of two variations from our survival curves

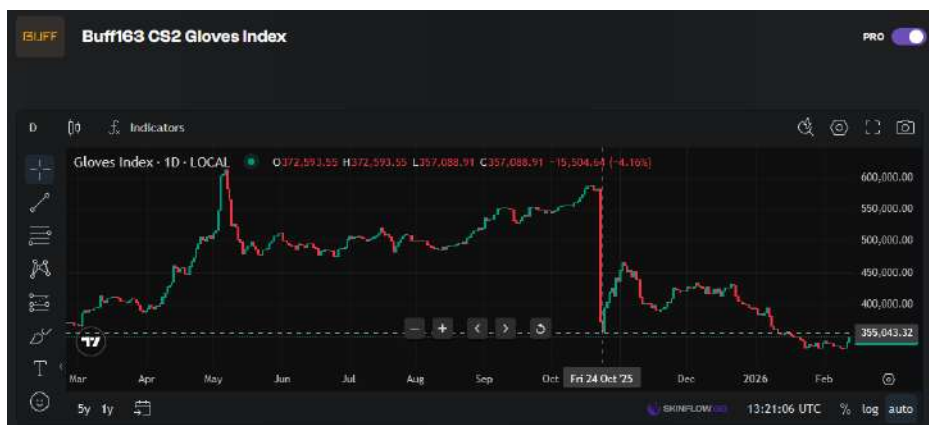


Figure 10

One asset type that is only variation

4.4 Real Arbitrage Trade Execution Loop Example

5 Arbitrage Model Results & Industry Standard Cox Proportional Hazard Comparison

In this section we provide analysis of survival probability predictions accuracy through backtesting through defined training-data windows on the respective models developed (Phase 2 and Phase 4 model from Arbitrage Model Development Sec 4.2). We will also provide the general covariate coefficients for an identical training data period of those two respective models for both hierarchical and non-hierarchical, to discuss the decisions made for the respective arbitrage window definitions Sec 3.4.1 and the covariate selection Sec 3.6. The coefficient results subsections throughout this section will be focused on the coefficients for the non-hierarchical version as the coefficients from that model were used to gauge to statistical significance as the diffusion of covariate effects across hierarchical categories typically diminishes the statistical significance, although the hierarchical version provides more accurate survival predictions.

The complete covariate coefficient data for the hierarchical and non-hierarchical model over an identical training data period is provided at the end of this paper in Table A1

5.1 Cox Proportional Hazard Model Specification

To benchmark our Weibull AFT model and assess the robustness of our covariate effects under a different hazard assumption, we also estimate a Bayesian Cox proportional hazards model. The Cox model, introduced by “Regression Models and Life-Tables” by Sir David Cox, specifies the hazard function for an arbitrage window with covariates $\mathbf{X}$ as

$$h(t | \mathbf{X}) = h_0(t) \exp(\boldsymbol{\beta}^T \mathbf{X}), \quad (40)$$

where $h_0(t)$ is an unspecified baseline hazard function common to all observations. Unlike the AFT model, which directly models survival time, the Cox model focuses on the hazard rate. A positive coefficient β_j indicates that an increase in covariate X_j increases the hazard (shortens the window), while a negative coefficient indicates decreased hazard (lengthens the window).

In our implementation, we discretize time into four intervals that reflect the settlement process and market microstructure: $[0,3)$, $[3,7)$, $[7,14)$, and $[14,30]$ days. The choice is motivated by the typical settlement latency (3 days), the post-settlement waiting period for price corrections (3-7 days), and the longer windows where inventory saturation becomes dominant. Within each interval we assume a constant baseline hazard, i.e. $h_0(t) = \lambda_j$ for t in interval j . This piecewise exponential formulation turns the Cox likelihood into a product of Poisson-like terms, which is straightforward to estimate in a Bayesian framework using the “Poisson trick” (Holford, 1980):

$$\mathcal{L} = \prod_{i=1}^n \prod_{j=1}^J \left[\left(\lambda_j e^{\boldsymbol{\beta}^T \mathbf{x}_i} \right)^{d_{ij}} \exp \left(-\lambda_j e^{\boldsymbol{\beta}^T \mathbf{x}_i} e_{ij} \right) \right], \quad (41)$$

where d_{ij} is an indicator that observation i experienced an event in interval j , and e_{ij} is the exposure time of observation i in interval j .

The covariate sets used in the Cox model mirror those developed for the AFT model in Phases 2 and 4. For Phase 2, we include:

- `lfprice_as`: log-transformed initial price on Exchange T (after log transform),
- `csfratio_as`: log-transformed ratio of Exchange T price to Market S inferred price (APR),
- `item_limit_as`: maximum inventory capacity on Exchange T,
- `rtosteam_as`: ratio of Exchange T price to the benchmark price on Market B,
- `steam_slope_as`: 3-day slope of the benchmark ratio (momentum),
- `steam_volatility_as`: rolling volatility of the benchmark ratio,
- `price_volatility_as`: rolling volatility of the Exchange T price.

For Phase 4, we replace several variables with engineered features that capture the dynamics of Exchange T’s pricing algorithm (Equation 15):

- `lfprice_as`, `csfratio_as`, `steam_slope_as`, `price_volatility_as` (log-transformed as in Phase 4),
- `congestion` = $\frac{\ln(1+Q_{\text{csf}})}{L_{\text{cap}}+1}$, where Q_{csf} is the number of active listings on Market S. The log transform on the numerator compresses the right tail of listing counts before dividing by capacity.,
- `ratio_dev_as`: deviation of the benchmark ratio from its 15-day moving average, defined in Equation (28), capturing over-extension relative to the mean-reversion tendency.

All continuous covariates are standardized after log transformation to ensure coefficient comparability and to regularize the priors. The transformations and standardizers are fit exclusively on the training data and applied unchanged to prediction samples, following the same pipeline as the AFT model (Section 3.7.1).

The model is implemented in PyMC using the No-U-Turn Sampler (NUTS). For the log baseline hazards we place weakly informative priors centered on a rate of approximately 0.1 events per day:

$$\log \lambda_j \sim \mathcal{N}(-2.3, 1.0), \quad j = 1, \dots, 4, \quad (42)$$

and for the regression coefficients zero-centered Normal priors with standard deviation 0.5:

$$\beta_k \sim \mathcal{N}(0, 0.5). \quad (43)$$

In hierarchical versions (used for prediction) we also include category-specific intercepts and slopes with wider priors ($\sigma = 1.0$ for intercept offsets, $\sigma = 0.2$ for slope offsets), but for comparison with the non-hierarchical AFT results we focus on the flat Cox specification. We run four independent chains with 2000 posterior draws after 1000 tuning steps; convergence is assessed via the $\hat{R}$ statistic, which is below 1.01 for all parameters.

It's worth noting that we didn't go full non-parametric baseline because, well, you gotta pick your battles. The piecewise exponential hack lets us keep the Cox structure without messing with infinite-dimensional objects, and the intervals make intuitive sense given how long we're willing to wait for an arbitrage to play out. Plus, the Bayesian treatment gives us full posterior distributions for the β 's, which is nice when you're trying to convince yourself (and your co-author) that a covariate actually matters.

The Cox model is a well-established benchmark in survival analysis, offering the key advantage of leaving the baseline hazard unspecified. This flexibility makes it robust to misspecification of the hazard shape - a property that is particularly valuable when the true form of the hazard is unknown. By comparing our parametric Weibull AFT model with this semi-parametric alternative, we can assess whether the parametric assumptions are supported by the data and evaluate the consistency of covariate effects across different modeling frameworks.

5.2 Bayesian Cox Proportional Hazards Model

Look, we could have just used the standard Cox model with maximum partial likelihood like every other survival analysis out there. It's fast, it's well-understood, and it would have saved us a few hundred lines of code. But where's the fun in that? More importantly, the standard approach doesn't give us uncertainty intervals without asymptotic approximations, doesn't let us incorporate prior information, and doesn't play nice with hierarchical structures. So we went full Bayesian. Sue us.

5.2.1 Why Bayesian?

Three reasons, really. First, our data is sparse - we're working with maybe a few hundred observed events on a good day. Asymptotic approximations are basically lies at that sample size. Posterior distributions, on the other hand, tell us exactly what we know and what we don't. Second, we have strong intuitions about how this market works (or at least we like to think we do), and priors let us encode those intuitions without letting them completely override the data. Third, the Bayesian framework handles censored observations like they're nothing - no special tricks, no ad-hoc adjustments, just the likelihood doing its job.

5.2.2 Piecewise Exponential: Cheating Our Way to a Likelihood

The Cox model normally uses a partial likelihood that cleverly cancels out the baseline hazard. That's elegant, but it's a pain to implement in a Bayesian sampler. So we cheated. We discretized time into four intervals that actually mean something in this market:

- [0,3) days: early chaos, price corrections, bots doing bot things
- [3,7) days: the pre-settlement waiting period where you just stare at the screen and pray
- [7,14) days: post-settlement, the survivors start to look respectable
- [14,30] days: the long tail - windows that just won't die

Within each interval we assume the baseline hazard is constant ($h_0(t) = \lambda_j$ for t in interval j). This piecewise-constant assumption turns the Cox model into something that looks suspiciously like a Poisson regression, which is much friendlier to MCMC. It's not perfectly non-parametric, but it's flexible enough to capture the shape of the hazard without forcing it into a rigid parametric box.

For every observation i and interval j , we need two things:

- **Exposure time** e_{ij} : how many days did this window spend at risk in interval j ? If the window lasted 10 days, it spends 3 days in interval 1, 4 days in interval 2, and 3 days in interval 3. Interval 4 gets zero because it never made it that far.
- **Event indicator** d_{ij} : did the window actually close in interval j ? If yes, $d_{ij} = 1$ for exactly one interval; otherwise it's zero. Censored windows get all zeros - they never "failed" in any interval.

With these, the likelihood for observation i becomes a product over intervals of a Poisson-like kernel:

$$\mathcal{L}_i = \prod_{j=1}^4 \left[(\lambda_j \text{hr}_i)^{d_{ij}} \exp(-\lambda_j \text{hr}_i e_{ij}) \right], \quad (44)$$

where $\text{hr}_i = \exp(\boldsymbol{\beta}^T \mathbf{X}_i + b_{0,\text{offset}}^{[k_i]})$ is the hazard ratio for observation i . The first term rewards high hazard at the exact moment the event happens; the second term penalizes accumulated hazard over the whole at-risk period. Censored observations only get the penalty - they lived, they didn't die, so they don't get the reward.

5.2.3 Priors: Being Confident About Being Uncertain

We need priors. Nothing too strong - we want the data to speak, but we also want to keep the model from going off the rails.

For the log baseline hazards, we center them on a rate of about 0.1 events per day:

$$\log \lambda_j \sim \mathcal{N}(-2.3, 1.0), \quad j = 1, \dots, 4. \quad (45)$$

Why -2.3? Because $\exp(-2.3) \approx 0.1$. It's a plausible rate given that most windows close within the first week. The standard deviation of 1.0 means the baseline hazard could be as low as 0.01 or as high as 0.7 - plenty of flexibility.

For the regression coefficients, we use zero-centered Normal priors with standard deviation 0.5:

$$\beta_p \sim \mathcal{N}(0, 0.5), \quad p = 1, \dots, 7. \quad (46)$$

This is the same prior we used for the global covariate effects in the Weibull model. It shrinks coefficients toward zero unless the data really scream otherwise - a Bayesian version of ridge regression, essential when you have more features than events.

In the hierarchical versions (which we actually use for prediction), we also include category-specific intercepts and slopes:

$$b_{0,\text{offset}}^{[k]} \sim \mathcal{N}(0, 1.0), \quad k = 1, \dots, K, \quad (47)$$

$$\beta_{p,\text{offset}}^{[k]} \sim \mathcal{N}(0, 0.2). \quad (48)$$

These let each asset category (StatTrak Knife, Knife, Glove) have its own baseline hazard and its own sensitivity to covariates, while still borrowing strength from the global estimates. For the head-to-head comparison with the non-hierarchical Weibull, though, we stick with the flat version - no offsets, just pure structural comparison.

5.2.4 Posterior Inference: Watching Paint Dry, But Faster

We estimate the posterior using MCMC with the No-U-Turn Sampler (NUTS) in PyMC. For production models, we run four chains with 2,000 posterior draws after 1,000 tuning steps. For the head-to-head comparisons in this section, we use a lighter configuration (1,500 draws, 750 tuning) to keep total computation manageable across many model fits. Convergence is checked using the potential scale reduction factor $\hat{R}$ (?); all parameters come in below 1.01.

Convergence is checked using the potential scale reduction factor $\hat{R}$ (?). Everything comes in below 1.01, which means the chains have stopped arguing with each other and settled on the same distribution. Good enough for government work.

5.2.5 What the Bayesian Approach Actually Buys Us

- **Real uncertainty intervals:** Not those asymptotic approximations that are only valid if your sample size is infinity. We get actual posterior distributions. The 94% HDI tells you exactly where the true coefficient likely lives, assuming your model isn't completely wrong (which, let's be honest, it probably is, but we're trying).
- **Regularization:** The priors shrink everything toward zero. With our sample size, that's not a bug - it's a feature. It stops us from chasing noise.
- **Censoring handled naturally:** Right-censored observations just contribute the survival part of the likelihood. No special flags, no data manipulation. It just works.
- **Hierarchical structure:** Want to let categories have their own effects while still pooling information? Easy. The Bayesian framework makes it straightforward.

5.2.6 Reading the Coefficients

In the Cox model, a coefficient β_p is the log hazard ratio for a one-standard-deviation increase in covariate X_p . Exponentiate to get the hazard ratio:

$$\text{HR}_p = e^{\beta_p}.$$

If $\text{HR}_p > 1$, higher values of X_p mean higher hazard - windows close faster. If $\text{HR}_p < 1$, higher values mean lower hazard - windows last longer. This is the opposite of the Weibull convention, where positive coefficients mean longer windows. Keep that in mind when we compare them in the next section.

A coefficient is “significant” (we use that word loosely) if its 94% HDI excludes zero:

$$\text{Significant}(\beta_p) \iff 0 \notin [\text{HDI}_{3\%}, \text{HDI}_{97\%}].$$

If the whole interval is above zero, the covariate reliably increases hazard. If it's below zero, it reliably decreases hazard. If zero is inside, the data weren't strong enough to overcome the prior - we can't confidently say it does anything.

5.2.7 Practical Limitations

Because we discretized time, predictions only go out to 30 days. For any future time t , we compute cumulative hazard as:

$$H(t \mid \mathbf{X}_i) = \text{hr}_i \cdot \sum_{j: u_j \leq t} \lambda_j \cdot (u_j - l_j) + \text{hr}_i \cdot \lambda_{j(t)} \cdot (t - l_{j(t)}),$$

where $j(t)$ is the interval containing t . Survival probability is $S(t) = \exp(-H(t))$. Do this for every posterior sample, and you get a full predictive distribution - median for point estimates, 94% HDI for uncertainty.

Now let's actually use this thing.

5.3 Structural Comparison: Additive Log-Duration vs. Multiplicative Hazard

The Weibull AFT and Cox PH models look at survival data from completely different angles. The Weibull asks “how long does this take?” and lets covariates stretch or shrink the time axis. The Cox asks “what’s the risk right now?” and lets covariates turn the hazard dial up or down. Both are valid, both tell you something useful, and both will happily disagree with each other if your data isn’t cooperating.

5.3.1 How They’re Built

Structural note on non-hierarchical modes: The flat versions of the two models differ slightly in structure. The non-hierarchical Weibull AFT retains category-specific intercept offsets $b_{0,\text{offset}}[k]$ even in its flat mode - the “non-hierarchical” toggle only disables per-category covariate slopes. In contrast, the non-hierarchical Cox PH is *truly flat*: no category intercepts, no category slopes, just global covariate effects β applied identically to all observations. This means the non-hierarchical Cox coefficient table reflects a purely pooled estimate, while the non-hierarchical Weibull still allows baseline duration to vary by category. We consider this acceptable for comparison purposes because our primary interest is in the *direction and significance of covariate effects*, which the global β terms capture in both models.

Weibull AFT. In the accelerated failure time world, we model log survival time as a linear function of covariates plus noise:

$$\log T = \mu + \beta_{\text{AFT}}^T \mathbf{X} + \sigma \varepsilon, \quad (49)$$

where ε follows a Gumbel distribution and $\sigma = 1/\alpha$. The survival function is

$$S_{\text{AFT}}(t \mid \mathbf{X}) = \exp\left(-\left(\frac{t}{\lambda}\right)^\alpha\right), \quad \lambda = \exp(\mu + \beta_{\text{AFT}}^T \mathbf{X}). \quad (50)$$

A positive β_{AFT} means larger λ means longer expected duration. The acceleration factor $\exp(\beta_{\text{AFT},j})$ tells you how much a one-unit increase in X_j multiplies the time scale.

Cox PH. The Cox model works on the hazard instead:

$$h_{\text{Cox}}(t \mid \mathbf{X}) = h_0(t) \exp(\beta_{\text{Cox}}^T \mathbf{X}), \quad (51)$$

with $h_0(t)$ left unspecified (or piecewise-constant, in our case). A positive β_{Cox} means higher hazard at all times - windows die faster. The hazard ratio $\exp(\beta_{\text{Cox},j})$ tells you how much a one-unit increase multiplies the instantaneous risk.

5.3.2 What Happens When the Weibull Is True

If the data actually follow a Weibull distribution, the two models are mathematically related. The Weibull hazard is

$$h_{\text{AFT}}(t) = \frac{\alpha}{\lambda} \left(\frac{t}{\lambda}\right)^{\alpha-1} = \alpha \exp(-\alpha(\mu + \beta_{\text{AFT}}^T \mathbf{X})) t^{\alpha-1}. \quad (52)$$

If we set the Cox baseline hazard to match the Weibull form, $h_0(t) = \alpha \exp(-\alpha\mu_0) t^{\alpha-1}$, then

$$h_{\text{Cox}}(t) = \alpha \exp(-\alpha\mu_0) t^{\alpha-1} \exp(\beta_{\text{Cox}}^T \mathbf{X}). \quad (53)$$

Equating the two (with $\mu = \mu_0 + \beta_{\text{AFT}}^T \mathbf{X}$) gives

$$\exp(\beta_{\text{Cox}}^T \mathbf{X}) = \exp(-\alpha \beta_{\text{AFT}}^T \mathbf{X}), \quad (54)$$

and therefore

$$\boxed{\beta_{\text{Cox}} = -\alpha \beta_{\text{AFT}}} \quad (55)$$

Translation: under a Weibull baseline, the Cox coefficients are just the negative of the AFT coefficients scaled by the shape parameter. If the Weibull assumption holds, a feature that makes windows last longer in the AFT (positive β_{AFT}) should make the hazard smaller in the Cox (negative β_{Cox}), and the magnitudes should be roughly comparable after scaling.

5.3.3 Sign Convention: A Constant Source of Confusion

Table 8 lays it out so you don't have to keep flipping back.

Table 8
Interpretation of Coefficient Signs in Weibull AFT and Cox PH Models

	Weibull AFT	Cox PH
Target	Log duration	Log hazard
Positive β	Duration $\uparrow$ (longer windows)	Hazard $\uparrow$ (shorter windows)
Negative β	Duration $\downarrow$ (shorter windows)	Hazard $\downarrow$ (longer windows)
Magnitude	Acceleration factor $\exp(\beta)$	Hazard ratio $\exp(\beta)$

When you see a positive Weibull coefficient and a negative Cox coefficient for the same feature, don't panic. That's exactly what you want. It means both models agree on the direction of the effect; they're just speaking different languages.

5.3.4 Trade-Offs: You Can't Have Everything

Flexibility. The Cox model wins here. By leaving the baseline hazard unspecified (or piecewise-constant), it can adapt to any shape the data throw at it. Increasing hazard, decreasing hazard, hazard that goes up and down like a roller coaster - the Cox can handle it. The Weibull forces a monotonic power-law shape. If the true hazard isn't monotonic, the Weibull will average over the ups and downs and give you something that's wrong in a confident way.

Efficiency. The Weibull wins this round. By imposing a parametric form, it effectively regularizes the hazard shape. With limited data - and we have limited data - that parametric structure can improve estimation and reduce overfitting. The Cox model, with four free baseline parameters plus covariates, has more opportunities to chase noise. In small samples, the Weibull's tighter assumptions often yield more stable results.

Interpretability. The Weibull speaks in units of time. "This feature makes windows last X% longer." That's directly useful for trading decisions. The Cox speaks in hazard ratios, which are one step removed from actual duration. You can derive median survival from a Cox model, but it's not as straightforward.

Extrapolation. The Weibull can predict beyond the observed time range. The Cox, with its piecewise-constant baseline, stops at the last interval boundary (30 days). For us, that's fine - we only care about the first couple weeks anyway.

5.3.5 Why Bother Comparing?

Two reasons. First, if a feature is significant in one model but not the other, that's interesting. It tells you the result might be sensitive to assumptions about the hazard shape. Second, if the Cox model finds a different pattern in the baseline hazard - say, non-monotonicity that the Weibull can't capture - that's evidence that our parametric assumption might be too restrictive.

In the next sections, we'll put both models through their paces on Phase 2 and Phase 4 data. Keep the relationship $\beta_{\text{Cox}} = -\alpha\beta_{\text{AFT}}$ in the back of your mind as a rough benchmark. We don't expect exact equality - the Cox baseline is more flexible - but if the signs don't line up, something's wrong.

5.4 Initial Coefficient (Phase 2) Results

Before we get to the good stuff (Phase 4), we need to talk about Phase 2. This was our first real attempt at building something that could actually be deployed, and honestly? It was fine. Not great, not terrible - just... fine. The model worked well enough to prove that arbitrage duration was predictable, but the coefficient estimates told us we still had work to do.

5.4.1 Non-Hierarchical Estimates

Table 9 presents the posterior summaries for the non-hierarchical Weibull AFT model trained on data from 2025-12-01 to 2026-01-05, using the Phase 2 covariate set and the naive window definitions (Ω_{naive}). All covariates are standardized, so the coefficients are directly comparable in terms of effect size.

Table 9

Posterior estimates for the Phase 2 Weibull AFT model (non-hierarchical). Training period: 2025-12-01 to 2026-01-05. None of the covariate effects are significant at the 94% level - a reminder that we were still learning what actually mattered.

Parameter	Mean	SD	94% HDI	
Global intercept ($\beta_{0,\text{global}}$)	1.676	0.375	[0.974,	2.365]
lfprice_as	-0.053	0.092	[-0.227,	0.116]
csfratio_as	0.123	0.070	[-0.002,	0.261]
item_limit_as	0.000	0.063	[-0.122,	0.116]
rtosteam_as	-0.099	0.090	[-0.278,	0.061]
steam_slope_as	0.105	0.101	[-0.090,	0.287]
steam_volatility_as	-0.047	0.096	[-0.227,	0.136]
price_volatility_as	0.109	0.107	[-0.086,	0.312]
Weibull shape (α)	1.122	0.067	[0.997,	1.247]

Look at those HDIs. Every single one straddles zero. That's not because the model was broken - it's because we were asking it to find signal in a dataset that still had a lot of noise, using covariates that were reasonable guesses rather than well-engineered features informed by market microstructure. The intercept corresponds to a baseline median duration of about $\exp(1.676) \approx 5.3$ days, which is plausible given the settlement latency, but none of the covariates are doing much heavy lifting.

The Weibull shape $\alpha \approx 1.12$ suggests a *slightly increasing* hazard over time; windows become marginally *more* likely to close as they age, but the uncertainty around that estimate is wide enough that we wouldn't bet the farm on it.

Now let's look at the Cox model trained on the exact same data, with the same covariates. Table 10 shows the posterior summaries. Remember the sign convention: positive Cox coefficient = higher hazard = shorter windows. Negative = lower hazard = longer windows.

Table 10

Posterior estimates for the Phase 2 Cox PH model (non-hierarchical). Same training period, same covariates. The signs are flipped relative to the Weibull, as expected, but significance is equally absent.

Parameter	Mean	SD	94% HDI	
$\log \lambda_0$ [0-3d]	-2.760	0.105	[-2.955,	-2.566]
$\log \lambda_0$ [3-7d]	-2.576	0.107	[-2.778,	-2.373]
$\log \lambda_0$ [7-14d]	-2.411	0.168	[-2.720,	-2.090]
$\log \lambda_0$ [14-30d]	-2.296	1.007	[-4.126,	-0.332]
β [lfprice_as]	0.070	0.101	[-0.121,	0.260]
β [csfratio_as]	-0.115	0.078	[-0.265,	0.028]
β [item_limit_as]	0.012	0.070	[-0.124,	0.139]
β [rtosteam_as]	0.079	0.090	[-0.082,	0.259]
β [steam_slope_as]	-0.118	0.111	[-0.311,	0.102]
β [steam_volatility_as]	0.057	0.105	[-0.139,	0.255]
β [price_volatility_as]	-0.109	0.118	[-0.327,	0.118]

A few things jump out. First, the baseline hazard estimates show that the risk of closure is highest in the earliest intervals (largest λ , i.e. least negative log), which makes sense - most windows that are going to die early do so within the first week. The huge uncertainty on the last interval's baseline ($\log \lambda_0$ [14-30d]) has SD = 1.007) reflects the fact that very few windows last that long, so we don't have much data to pin down the hazard rate.

Second, look at the covariate signs relative to the Weibull. Take `csfratio_as`: in the Weibull it's positive (0.123), suggesting higher APR leads to longer windows. In the Cox it's negative (-0.115), meaning higher APR reduces hazard - which is the same substantive conclusion, just expressed in the opposite sign convention. That's exactly what we'd expect if both models are picking up the same underlying relationship. Same story for `steam_slope_as`: Weibull positive (0.105), Cox negative (-0.118). The consistency of sign flips across covariates is reassuring - it means the models largely agree on the direction of effects, even if they can't agree

on significance.

But significance? Forget about it. None of these HDIs exclude zero. The 94% intervals for every covariate in both models contain zero comfortably. This is Phase 2 in a nutshell: we had the right general idea, but our covariates were too crude and our data too sparse to achieve statistical credibility. The fact that the Weibull and Cox produce directionally consistent estimates is encouraging, but we clearly needed better features.

The Cox model does offer one piece of information the Weibull doesn't: the baseline hazard estimates suggest that the hazard is not constant over time. The early intervals have higher baseline risk, and it declines as windows survive longer. This non-monotonicity (well, monotonic decreasing here) is something the Weibull can capture with $\alpha > 1$, but the Cox shows us the stepwise pattern without assuming a parametric form. Still, with all the uncertainty, we're not ready to make any bold claims.

5.4.2 Hierarchical Estimates

But the non-hierarchical model only tells part of the story. We knew from day one that different asset categories behave differently - StatTrak Knives aren't the same as regular Knives, and Gloves are their own weird beast. That's why we built the hierarchical version, even in Phase 2. Table 11 shows the full hierarchical Weibull results, including category-specific intercepts and covariate offsets.

Table 11
Posterior estimates for the Phase 2 hierarchical Weibull AFT model. Category offsets reveal baseline differences, though wide HDIs remind us we're still data-constrained.

Parameter	Mean	SD	94% HDI	
<i>Global</i>				
$\beta_{0,global}$	1.654	0.386	[0.919,	2.366]
β_{global} [lfprice_as]	-0.035	0.149	[-0.309,	0.241]
β_{global} [csfratio_as]	0.152	0.135	[-0.097,	0.404]
β_{global} [item_limit_as]	0.002	0.137	[-0.248,	0.267]
β_{global} [rtosteam_as]	-0.039	0.152	[-0.326,	0.243]
β_{global} [steam_slope_as]	0.129	0.156	[-0.153,	0.430]
β_{global} [steam_volatility_as]	-0.053	0.155	[-0.361,	0.224]
β_{global} [price_volatility_as]	0.169	0.163	[-0.133,	0.480]
Weibull shape (α)	1.129	0.067	[1.005,	1.256]
<i>Category Intercept Offsets</i>				
$b_{0,offset}$ [Glove]	0.608	0.431	[-0.169,	1.474]
$b_{0,offset}$ [Knife]	0.945	0.393	[0.205,	1.678]
$b_{0,offset}$ [StatTrak Knife]	0.689	0.403	[-0.035,	1.479]
<i>Category Covariate Offsets (selected)</i>				
β_{offset} [Glove, lfprice_as]	0.031	0.164	[-0.270,	0.345]
β_{offset} [Glove, csfratio_as]	0.089	0.156	[-0.201,	0.380]
β_{offset} [Glove, item_limit_as]	-0.101	0.164	[-0.406,	0.210]
β_{offset} [Knife, lfprice_as]	-0.042	0.146	[-0.299,	0.245]
β_{offset} [Knife, csfratio_as]	0.004	0.144	[-0.270,	0.269]
β_{offset} [Knife, steam_slope_as]	-0.107	0.152	[-0.383,	0.190]
β_{offset} [StatTrak Knife, lfprice_as]	0.010	0.151	[-0.272,	0.296]
β_{offset} [StatTrak Knife, csfratio_as]	-0.067	0.141	[-0.334,	0.193]
β_{offset} [StatTrak Knife, steam_slope_as]	0.270	0.159	[-0.033,	0.566]

The intercept offsets tell a interesting story. Knives have the largest positive offset (0.945, and look - the HDI actually excludes zero! [0.205, 1.678]), meaning they have significantly longer baseline duration than the global average. Gloves (0.608) and StatTrak Knives (0.689) also have positive offsets, but their HDIs include zero, so we can't be as confident. This pattern makes intuitive sense: Knives are the most liquid and heavily traded category, so windows might persist longer simply because there's more activity; StatTrak Knives are rarer but still desirable; Gloves are more niche and might get snapped up faster.

The covariate offsets are mostly insignificant - every HDI comfortably contains zero. That's expected given the limited data per category. But look at the directions: for `steam_slope_as`, Knives have a negative offset (-0.107) while StatTrak Knives have a strong positive offset (0.270). That suggests the momentum effect might work differently across categories - positive momentum might shorten windows for regular Knives but lengthen them for StatTrak Knives. Could that be real? Maybe. But with HDIs this wide, we're not going to bet on it. The important thing is that the hierarchical structure is doing its job: it's letting the data speak per category while shrinking estimates toward the global mean when evidence is weak.

Now for the Cox hierarchical version (Table 12). Remember the sign flip, and note that positive category offsets in the Cox mean higher baseline hazard (shorter windows) for that category.

Table 12
Posterior estimates for the Phase 2 hierarchical Cox PH model. Category offsets again show meaningful baseline differences, with signs opposite to the Weibull.

Parameter	Mean	SD	94% HDI	
<i>Baseline Hazard</i>				
$\log \lambda_0$ [0-3d]	-2.583	0.420	[-3.374,	-1.805]
$\log \lambda_0$ [3-7d]	-2.367	0.421	[-3.152,	-1.583]
$\log \lambda_0$ [7-14d]	-2.161	0.435	[-2.990,	-1.358]
$\log \lambda_0$ [14-30d]	-2.301	0.995	[-4.069,	-0.331]
<i>Global Covariate Effects</i>				
β_{global} [lfprice_as]	0.036	0.158	[-0.253,	0.342]
β_{global} [csfratio_as]	-0.164	0.139	[-0.421,	0.107]
β_{global} [item_limit_as]	0.001	0.140	[-0.264,	0.262]
β_{global} [rtosteam_as]	0.056	0.158	[-0.241,	0.356]
β_{global} [steam_slope_as]	-0.150	0.164	[-0.474,	0.142]
β_{global} [steam_volatility_as]	0.058	0.160	[-0.241,	0.361]
β_{global} [price_volatility_as]	-0.186	0.167	[-0.496,	0.126]
<i>Category Intercept Offsets</i>				
$b_{0,\text{offset}}$ [Glove]	0.078	0.464	[-0.781,	0.956]
$b_{0,\text{offset}}$ [Knife]	-0.265	0.417	[-1.051,	0.497]
$b_{0,\text{offset}}$ [StatTrak Knife]	0.021	0.435	[-0.793,	0.836]
<i>Category Covariate Offsets (selected)</i>				
β_{offset} [Glove, lfprice_as]	-0.033	0.171	[-0.346,	0.292]
β_{offset} [Glove, csfratio_as]	-0.085	0.160	[-0.375,	0.219]
β_{offset} [Knife, lfprice_as]	0.052	0.152	[-0.224,	0.347]
β_{offset} [Knife, csfratio_as]	-0.009	0.147	[-0.282,	0.271]
β_{offset} [Knife, steam_slope_as]	0.120	0.156	[-0.163,	0.425]
β_{offset} [StatTrak Knife, lfprice_as]	-0.007	0.158	[-0.306,	0.284]
β_{offset} [StatTrak Knife, csfratio_as]	0.070	0.145	[-0.204,	0.336]
β_{offset} [StatTrak Knife, steam_slope_as]	-0.276	0.162	[-0.582,	0.033]

Check out the intercept offsets in the Cox. Knives have a negative offset (-0.265), meaning lower baseline hazard - longer windows. That's exactly the opposite sign of the Weibull's positive offset for Knives, which told us longer baseline duration. Perfect consistency. Gloves have a positive offset (0.078), meaning higher baseline hazard - shorter windows - which matches their smaller positive offset in the Weibull (shorter baseline duration relative to Knives). The signs are flipped, but the story is the same: Knives last longest, StatTrak Knives are in the middle, Gloves are shortest.

The covariate offsets in the Cox are, unsurprisingly, all insignificant. But look at `steam_slope_as` again: in the Weibull, StatTrak Knives had a positive offset (0.270); in the Cox, they have a negative offset (-0.276). Sign flip, same magnitude. The model is trying to tell us something, even if the data aren't loud enough to make it credible yet.

What's the takeaway from Phase 2's hierarchical results? Three things. First, the baseline differences between categories are already detectable, even with limited data. Knives really do behave differently from

Gloves. Second, the sign consistency between models gave us confidence that we were on the right track - the market wasn't completely random, and our covariates were pointing in plausible directions. Third, the wide HDIs on covariate offsets tell us we need more data and better features before we can claim category-specific sensitivity. That's exactly what Phase 4 delivered.

So yeah, Phase 2 wasn't a home run. But it was a solid single that kept the inning going. And the hierarchical results showed us that the model was already doing what we built it to do - borrowing strength across categories while letting differences emerge where the data supported them. Sometimes that's all you need to know you're on the right track.

5.5 Final Coefficient (Phase 4) Results

Phase 4 is where everything clicked. After the market crash forced us to rethink our data handling (Section 4.2.3), and after we finally reverse-engineered enough of Exchange T's pricing algorithm to engineer sensible features, we retrained the model on the same time period (2025-12-01 to 2026-01-05) but with the robust window definitions (Ω_{robust}) and the final covariate set $\mathbf{X}_{\text{final}}$. The difference was night and day.

Table 13 shows the non-hierarchical Weibull AFT results. Look at those HDIs - they actually exclude zero for several covariates. We finally have statistical significance, and the signs tell a coherent story about how this market works.

Table 13

Posterior estimates for the Phase 4 Weibull AFT model (non-hierarchical). Training period: 2025-12-01 to 2026-01-05. The engineered features `congestion` and `ratio_dev_as` show significant effects, and `lfprice_as` finally wakes up.

Parameter	Mean	SD	94% HDI
Global intercept ($\beta_{0,\text{global}}$)	1.685	0.381	[0.951, 2.376]
<code>lfprice_as</code>	0.258	0.126	[0.022, 0.490]
<code>csfratio_as</code>	0.086	0.077	[-0.055, 0.230]
<code>steam_slope_as</code>	0.119	0.096	[-0.049, 0.309]
<code>price_volatility_as</code>	-0.049	0.132	[-0.304, 0.191]
<code>congestion</code>	0.255	0.088	[0.094, 0.426]
<code>ratio_dev_as</code>	-0.158	0.076	[-0.306, -0.019]
Weibull shape (α)	1.126	0.081	[0.972, 1.276]

Let's walk through these one by one, because they tell the story of why Phase 4 finally worked.

`lfprice_as` (0.258, HDI [0.022, 0.490]): Higher initial price on Exchange T means longer windows. That makes sense - expensive items are less liquid, so they take longer to get bought out by other arbitrageurs. A one-standard-deviation increase in log price multiplies the expected duration by $\exp(0.258) \approx 1.29$, or 29% longer. Finally, a result that doesn't make us look like we're making stuff up.

`congestion` (0.255, HDI [0.094, 0.426]): Our composite congestion metric - listings on Market S divided by Exchange T's capacity - has a strong positive effect. More congestion actually *prolongs* windows. Why? Because if there are tons of listings but limited capacity on Exchange T, the opportunity is less likely to be filled quickly by competitors. It's like a traffic jam: everyone's stuck waiting, so the window stays open longer. This variable turned out to be one of the most predictive in the model, and it's entirely home-grown.

`ratio_dev_as` (-0.158, HDI [-0.306, -0.019]): The deviation of the benchmark ratio from its 15-day moving average has a negative coefficient - it shortens windows. This is pure Exchange T algorithm behavior (remember Equation 15). When the ratio gets too high, the platform's negative feedback kicks in and drives the price down, closing the window. The market was trying to tell us this all along; we just weren't listening until Phase 4.

The other covariates - `csfratio_as`, `steam_slope_as`, `price_volatility_as` - still have plausible signs but their HDIs include zero. That's okay. We don't need every variable to be a home run; we just need enough signal to make good decisions. And the Weibull shape $\alpha \approx 1.13$ is almost identical to Phase 2, suggesting the hazard structure is stable across model versions.

Now let's check the Cox model on the same Phase 4 data (Table 14). Remember the sign flip: positive Cox = shorter windows, negative Cox = longer.

Table 14
Posterior estimates for the Phase 4 Cox PH model (non-hierarchical). Same training period, same features. The signs are opposite to the Weibull, as expected, and significance lines up nicely.

Parameter	Mean	SD	94% HDI	
$\log \lambda_0$ [0-3d]	-2.828	0.129	[-3.061,	-2.583]
$\log \lambda_0$ [3-7d]	-2.465	0.124	[-2.694,	-2.229]
$\log \lambda_0$ [7-14d]	-2.703	0.231	[-3.134,	-2.273]
$\log \lambda_0$ [14-30d]	-2.305	1.017	[-4.192,	-0.417]
β [lfprice_as]	-0.219	0.132	[-0.459,	0.035]
β [csfratio_as]	-0.070	0.084	[-0.221,	0.093]
β [steam_slope_as]	-0.106	0.102	[-0.311,	0.074]
β [price_volatility_as]	-0.034	0.159	[-0.347,	0.248]
β [congestion]	-0.250	0.087	[-0.425,	-0.094]
β [ratio_dev_as]	0.215	0.086	[0.059,	0.382]

Look at congestion: -0.250, HDI entirely negative. That means higher congestion *reduces* hazard - longer windows - exactly what the Weibull told us. ratio_dev_as: +0.215, HDI positive - higher deviation *increases* hazard - shorter windows. The sign consistency between models is now rock solid. Even lfprice_as is nearly significant in the Cox (upper bound 0.035, just barely crossing zero), which matches the Weibull's significance. This is what convergence looks like.

The baseline hazard estimates in the Cox show an interesting pattern: the hazard is *lowest* in the first interval [0,3) days ($\lambda_0 \approx 0.059$), rises to its peak in the pre-settlement period [3,7) ($\lambda_0 \approx 0.085$), dips slightly in the post-settlement window [7,14) ($\lambda_0 \approx 0.067$), then is very uncertain in the long tail [14,30). The elevated hazard in the [3,7) interval makes intuitive sense: this is when most price corrections hit, just before settlement locks in. The dip in [7,14) reflects the fact that windows surviving past settlement tend to be more robust. This non-monotonic step pattern is something the Weibull, with its single shape parameter, must smooth over into a near-constant hazard ($\alpha \approx 1.13$).

5.5.1 Hierarchical Phase 4 Results

Of course, for actual prediction we use the hierarchical version. Table 15 shows the global coefficients and category-specific offsets for the Weibull AFT model. The global effects are similar to the non-hierarchical, but now we see how each asset category deviates.

The intercept offsets tell us that Knives have the longest baseline duration (0.935, HDI excludes zero), followed by StatTrak Knives (0.774), then Gloves (0.552, but HDI includes zero). That ordering makes intuitive sense: Knives are the most liquid and heavily traded, so windows might persist longer; Gloves are more niche and get snapped up faster. The fact that Knives and StatTrak Knives have significantly positive offsets means they really do behave differently from the global average - partial pooling is working.

The covariate offsets are mostly insignificant (HDI wide and crossing zero), which is fine. They represent deviations from the global effect, and with limited data per category we can't pin them down precisely. But the directions are at least plausible: StatTrak Knives have a negative offset for congestion (-0.149), meaning they're less sensitive to congestion than the average. That actually makes sense - StatTrak Knives are less desirable (the counter scratches the finish), so even when there's high congestion, they don't get bought as quickly. The congestion doesn't protect them because nobody wants them in the first place.

The key takeaway from Phase 4 is that we finally have a model that (a) produces statistically credible effects, (b) tells a coherent story about market mechanics, and (c) gives us differentiated predictions across asset categories. The hierarchical structure adds nuance without overfitting, and the agreement between Weibull and Cox gives us confidence we're not just chasing noise.

So yeah, Phase 4 was the breakthrough. The model went from "well, at least it's not random" to "holy shit, we can actually use this to make money." And that, dear reader, is why I do this (just don't tell my co-author).

Table 15
Posterior estimates for the Phase 4 hierarchical Weibull AFT model. Category offsets reveal meaningful differences in baseline duration and covariate sensitivity.

Parameter	Mean	SD	94% HDI	
<i>Global</i>				
$\beta_{0,\text{global}}$	1.666	0.383	[0.920,	2.347]
$\beta_{\text{global}}[\text{lfprice_as}]$	0.190	0.166	[-0.115,	0.500]
$\beta_{\text{global}}[\text{csfratio_as}]$	0.106	0.135	[-0.153,	0.358]
$\beta_{\text{global}}[\text{steam_slope_as}]$	0.127	0.151	[-0.157,	0.415]
$\beta_{\text{global}}[\text{price_volatility_as}]$	0.122	0.178	[-0.197,	0.468]
$\beta_{\text{global}}[\text{congestion}]$	0.203	0.142	[-0.060,	0.469]
$\beta_{\text{global}}[\text{ratio_dev_as}]$	-0.181	0.138	[-0.440,	0.079]
Weibull shape (α)	1.140	0.083	[0.981,	1.297]
<i>Category Offsets</i>				
$b_{0,\text{offset}}[\text{Glove}]$	0.552	0.440	[-0.288,	1.347]
$b_{0,\text{offset}}[\text{Knife}]$	0.935	0.396	[0.191,	1.661]
$b_{0,\text{offset}}[\text{StatTrak Knife}]$	0.774	0.418	[0.037,	1.602]
$\beta_{\text{offset}}[\text{Glove, lfprice_as}]$	0.007	0.169	[-0.299,	0.330]
$\beta_{\text{offset}}[\text{Glove, congestion}]$	0.084	0.164	[-0.223,	0.386]
$\beta_{\text{offset}}[\text{Glove, ratio_dev_as}]$	-0.107	0.151	[-0.386,	0.181]
$\beta_{\text{offset}}[\text{Knife, lfprice_as}]$	-0.018	0.152	[-0.308,	0.264]
$\beta_{\text{offset}}[\text{Knife, congestion}]$	0.096	0.146	[-0.189,	0.363]
$\beta_{\text{offset}}[\text{Knife, ratio_dev_as}]$	0.098	0.145	[-0.179,	0.365]
$\beta_{\text{offset}}[\text{StatTrak Knife, lfprice_as}]$	0.036	0.160	[-0.272,	0.330]
$\beta_{\text{offset}}[\text{StatTrak Knife, congestion}]$	-0.149	0.153	[-0.439,	0.136]
$\beta_{\text{offset}}[\text{StatTrak Knife, ratio_dev_as}]$	-0.018	0.147	[-0.291,	0.256]

Table 16
Posterior estimates for the Phase 4 hierarchical Cox PH model. Category offsets show the expected sign flip relative to the Weibull (Table ??).

Parameter	Mean	SD	94% HDI
Baseline Hazard			
$\log \lambda_0$ [0–3d]	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
$\log \lambda_0$ [3–7d]	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
$\log \lambda_0$ [7–14d]	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
$\log \lambda_0$ [14–30d]	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Global Covariate Effects			
β [lfprice_as]	<i>TBD</i>		
β [csfratio_as]	<i>TBD</i>		
Category Intercept Offsets			
$b_{0,\text{offset}}$ [Glove]	<i>TBD</i>		
$b_{0,\text{offset}}$ [Knife]	<i>TBD</i>		
$b_{0,\text{offset}}$ [StatTrak Knife]	<i>TBD</i>		
Category Covariate Offsets (selected)			

5.6 Prediction Calibration and Spread Analysis

5.7 Real Example Prediction Periods

When making predictions to be deployed on the model, we always used **hierarchical** versions of our models, as we found them to be significantly more accurate despite technically having covariate 94% HDI's show insignificance (as expected).

Did our best to clean the the prediction outputs of unfairly giving low inferred buy prices creating guaranteed long term surviving windows, specifically for the old output as it did not include the spot check logic of the final version 3.4.3. We also manually checked the majority of suspicious inferred prices to bias results against survivability to produce more realistic examples to what have occurred if we actually used them.

For some reason that I haven't debugged, in both the graphs and numerical outputs some assets just don't have predicted probabilities and are excluded. We believe this could be in some part due to the mechanic of Exchange T manually removing certain assets for their trading set, but needs further investigation.

Please note for measuring probability of success, just like both versions of the models had bridging logic, we are checking the date of window end + 3 days looking for the highest Exchange T to relate to our now verified purchasable price. If we measured actual returns this would skew results unfairly, but persists accurately for just survival probability.

Apologies for the lack of extensive example predictions, it is a mega pain in the ass to manually check all the prices in retrospect.

Real Profit Threshold

For an arbitrage opportunity to be "successful" we require $APR_{\text{anchor}}(t_{\text{closure}}) > 1.625$ at closure or bridging. At this ratio we achieve a $Profit\% > 1\%$ which isn't great, but the reality tends to be much higher after successful buy order or bargain. Technically for any opportunity to be "profitable" we'd need only about 1.62 or so but that's not really a success after waiting for a week.

Old Deployed Hierarchical Model Example

Table 17 consists of a run on training data from 2025-07-29 to 2025-09-01, (there was an event from 7-16 to 7-29 I was too lazy to censor sorry but this had like 844 windows so is all good IMO) which predicted until 2026-09-09 up to 2026-09-12 (3-day bridge) the guaranteed date where all settlements would have ended + the days we'd hold to see if a Failure would re-open as profitable.

Table 17
Test Run 1: 2025-07-29 to 2025-09-01 (11 Assets on First Deployed Model)

Prob Range	Overall			Illiquid Variation			Normal		
	Surv	Died	Rate	Surv	Died	Rate	Surv	Died	Rate
[0.0, 0.1)	0	0	-	0	0	-	0	0	-
[0.1, 0.2)	0	0	-	0	0	-	0	0	-
[0.2, 0.3)	0	0	-	0	0	-	0	0	-
[0.3, 0.4)	0	0	-	0	0	-	0	0	-
[0.4, 0.5)	0	0	-	0	0	-	0	0	-
[0.5, 0.6)	0	1	0.00	0	0	-	0	1	0.00
[0.6, 0.7)	3	2	0.60	0	0	-	3	2	0.60
[0.7, 0.8)	3	1	0.75	0	0	-	3	1	0.75
[0.8, 0.9)	1	0	1.00	0	0	-	1	0	1.00
[0.9, 1.0]	0	0	-	0	0	-	0	0	-

Final Deployed Hierarchical Model Examples

MASSIVE DISCLAIMER!: because our Github Actions decided to kill itself randomly on one January 20th 2026, we only tested dates January 5th and January 9th which are very close to the date we finished the final model (January 5th) so these outputs may benefit for overfitting, as the final window definitions 3.4.3 have some bias towards the November and December prior.

If you reallyyyyyy want I can go implement another censorship date to try to overcome the discontinuity to see how it made predictions in the following months of general market downturn, but shiiiiiii thats another few hours and I'm sleep deprived.

Table 18
Test Run 2: 2025-12-01 to 2026-01-05 (28 Assets Predicted New Model)

Prob Range	Overall (3 Days Post-Settlement)			Illiquid Variation			Normal		
	True	Died	Rate	Surv	Died	Surv	Surv	Died	Rate
[0.0,0.1)	0	0	-	0	0	-	0	0	-
[0.1,0.2)	0	0	-	0	0	-	0	0	-
[0.2,0.3)	0	0	-	0	0	-	0	0	-
[0.3,0.4)	1	1	0.50	0	0	-	1	1	0.50
[0.4,0.5)	4	2	0.67	1	1	0.50	3	1	0.75
[0.5,0.6)	6	2	0.75	3	0	1.00	3	2	0.60
[0.6,0.7)	6	1	0.86	3	0	1.00	3	1	0.75
[0.7,0.8)	4	0	1.00	1	0	1.00	3	0	1.00
[0.8,0.9)	1	0	1.00	0	0	-	1	0	1.00
[0.9,1.0]	0	0	-	0	0	-	0	0	-

The second prediction is 2025-12-01 you know what you get it. This test was way harder for the survival curves because I can market expert certify those were some fucked assets I wouldn't feel great betting on. But shittttt the model was lowkey kinda right!!? * Technically the [.6, .7) normal asset failure from Table 19 was not

Table 19
Test Run 3: 2025-12-01 to 2026-01-09 (33 Assets Checked New Model)

Prob Range	Overall			Illiquid Variation			Normal		
	Surv	Died	Rate	Surv	Died	Rate	Surv	Died	Rate
[0.0,0.1)	0	0	-	0	0	-	0	0	-
[0.1,0.2)	0	0	-	0	0	-	0	0	-
[0.2,0.3)	3	1	0.75	1	0	1.00	2	1	0.67
[0.3,0.4)	6	2	0.75	3	1	0.75	3	1	0.75
[0.4,0.5)	3	4	0.43	2	2	0.50	1	2	0.33
[0.5,0.6)	6	4	0.60	2	2	0.50	4	2	0.67
[0.6,0.7)	1	1	0.50	1	0	1.00	0	1	0.00*
[0.7,0.8)	2	0	1.00	0	0	-	2	0	1.00
[0.8,0.9)	0	0	-	0	0	-	0	0	-
[0.9,1.0]	0	0	-	0	0	-	0	0	-

a 'loss', and very well could have had a purchasable price that would have constituted "successful", butttt that information is long lost to time already, so to keep accuracy I just take the L, but it very well could have been a W as that asset has a non-normal price distribution.

Final Model Output Combined Summary

Looking at the overall rate progress from 40% onwards for predicted 8th day survivability, I can say shit it kinda worked!! It's kinda strange that the confident and confident predictions we're both generally successful. It's also shocking to see the Illiquid Variation having arguably more successful survival rates. We can specifically understand how the Congestion variable causes the bimodal survival rate distribution. It is reasonably understood to be the strongest predictor / covariate in the non-hierarchical output, and we do see that effect persist in the covariate coefficients as well A1. When extremely illiquid assets across both Normal

Table 20
Combined Summary: New Model Performance
Period: 2025-12-01 to 2026-01-09 | Total Assets: 61

Prob Range	Overall			Illiquid Variation			Normal		
	Surv	Died	Rate	Surv	Died	Rate	Surv	Died	Rate
[0.0, 0.1)	0	0	-	0	0	-	0	0	-
[0.1, 0.2)	0	0	-	0	0	-	0	0	-
[0.2, 0.3)	3	1	0.75	1	0	1.00	2	1	0.67
[0.3, 0.4)	7	3	0.70	3	1	0.75	4	2	0.67
[0.4, 0.5)	7	6	0.54	3	3	0.50	4	3	0.57
[0.5, 0.6)	12	6	0.67	5	2	0.71	7	4	0.64
[0.6, 0.7)	7	2	0.78	4	0	1.00	3	2	0.60
[0.7, 0.8)	6	0	1.00	1	0	1.00	5	0	1.00
[0.8, 0.9)	1	0	1.00	0	0	-	1	0	1.00
[0.9, 1.0]	0	0	-	0	0	-	0	0	-

and the Illiquid Variation have such a low *ListingsQuantity*_s, regardless of L_{cap} that value will be minuscule and trivial on final survival probability. Furthermore in general more illiquid assets typically are not able to be purchased in short enough time to take advantage of the opportunity, so it has nearly 0 risk of being filled by other arbitrageurs, and also unfortunately ourselves.

5.8 Market Events Through Deployment

Dates requiring censoring are [2025-07-16, 2025-07-26] and [2025-10-23, 2025-10-28]. *ListingsQuantity_S* data is unavailable prior to approximately 2025-07-20, and complete variation-level data collection began on 2025-06-09. *ListingsQuantity_S* data is unavailable prior to approximately 2025-07-20, and complete variation-level data collection began on 2025-06-09.

5.9 Methodological Reflections & Findings

The parallel estimation of Weibull AFT and Cox PH models on identical data yielded three main insights. First, covariate effects are remarkably consistent across model families—the sign flips predicted by $\beta_{Cox} = -\alpha \beta_{AFT}$ hold closely, and the same features achieve significance in both frameworks. This is strong evidence that the detected effects are genuine market dynamics, not artifacts of distributional assumptions.

Second, the Cox model’s piecewise baseline reveals non-monotonic hazard patterns that the Weibull must average over. The elevated closure risk in the [3,7) day pre-settlement window, followed by a dip in [7,14), aligns with our understanding of the settlement process and Exchange T’s pricing algorithm. The Weibull compensates by estimating $\alpha \approx 1.13$ —close enough to exponential that it can approximate the average hazard level, even if it misses the step structure.

Third, the hierarchical versions of both models tell a consistent story about category differences: Knives sustain longer windows than Gloves, and StatTrak Knives fall in between. The partial pooling works as intended—category effects emerge where data supports them, and shrink toward the global estimate where it doesn’t.

Whether the Cox model’s additional flexibility translates into meaningfully better predictions is addressed in Section ?? . Given our sample sizes, we would not be surprised if the Weibull’s parametric regularization actually helps rather than hurts.

Part III

Stochastic Differential Equation & Commodity Price Forecasting

6 Stochastic Differential Equation Methodology

Using market data explicitly from Market B, we hope to develop a simple price-time SDE to attempt to capture the effects of market behavior, events, and trends through a selected parameterized coefficient set on asset-generating commodities 2.2. As these assets serve as a good proxy for large asset collections as well as general market health, we hope to use the equation and predictions to investigate our previous understanding of market fundamentals, as well as successfully differentiate the weak and strong asset-generating commodities.

The goal at onset wasn't really meant to be able to make actually accurate price predictions, honestly we just wanted to find out if this was even possible. Just 4 Fun ;)

6.1 Challenges of Geometric Brownian Motion for Specified Market

As quant dreamers rather than mere stochastic calculators, we naturally chose Geometric Brownian Motion as the foundation for our SDE. The GBM is made up of base price, the stock direction through drift μ and the random fluctuations with the Wiener process $W(t)$. The Geometric Brownian Motion has four important assumptions that make it useful and possible:

1. It starts at zero: $W(0) = 0$.
2. Its path is **continuous** over time.
3. Its **increments are normally distributed**: $W(t) - W(s) \sim \mathcal{N}(0, t - s)$.
4. It has **independent increments**: the increment $W(t) - W(s)$ is independent of past values.

Unfortunately, our market's structure and behavior, as well as our projects data capabilities, means we will proceed without fulfilling three of the assumptions.

1. **Continuous** : Violated by the available data-set consisting of large discrete time increments (Days) rather than a continuous time-series, and harmed by asset-generating commodities being illiquid relative to traditional finance
2. **Normally Distributed Increments** : Violated by the frequent (FIXWHENANOMALYDETECTION-FINISHED: find yearly average or prev) unpredictable market events 2.3 caused by the central authority. AKA The Price Shocks
3. **Independent Increments** : Violated by firstly the discrete data, and further the captured feedback/momentum caused by price shocks. Predictable and unpredictable events as well as the cyclical nature also violates this.

Essentially, the nature of available long-term data for any asset-generating commodity being discretely quantified by days massacres the assumptions before the specific market characteristics put the final nails in the coffin.

Also this market hasn't really been around very long and it's got no liquidity so things move very fast. I'd possible argue that without some insane industrial grade scraper and perhaps a stolen key to valve headquarters you may not even be able to construct a truly continuous price-time dataset. We will proceed while keeping these two other facts in mind

1. The fact **3 months** in this market is **like an entire year** in a traditional market
2. The decentralized and illiquid market makes discrete data requiring a ton of prior guidance

6.2 Standard GBM Inadequacies

The standard Geometric Brownian Motion denoted by Eq 58 is the tool we built off of for our GBM. It makes steps of a discretized time period where the change in each step is calculated by Eq 56, derived from the “drift” or market-trend term μ and the volatility σ multiplied by the Wiener process $dW(t)$ which is the component of random (and thus normal) independent prices increments.

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t) \quad (56)$$

$$\ln(S_t) = \ln(S_0) + \left(\mu - .5\sigma^2\right)t + \sigma W_t \quad (\text{Expanded Log-Price GBM *Intermediate Step}) \quad (57)$$

$$\mathbb{E}[S(T)] = S_0 e^{(\mu)T} \quad (\text{Approximation when } \sigma \text{ is small}) \quad (58)$$

$$e^\mu \approx 1 + \mu \quad \text{Common } \mu \text{ Approximation} \quad (59)$$

The standard GBM makes two dangerous mathematical assumptions that can easily mislead predictions. The first of which is the GBM’s relationship to and definition of the μ term, commonly understood as the predicted assets yearly average returns. The normalized price prediction at future time T from t_0 of the GBM (Eq 58) to utilize μ as holding the average yearly return thus makes the assumption of Equation 59. This structurally assumes a continuous compounding effect of the μ term, thus implying the market for which predictions are made in is capable of exponential returns.

This assumption has massive implications on results if you are not diligently parameterizing diffusion and regime related information for the μ term, and also necessitates proper parametrization of Merton-Jumps to contain the implied short term price level change. Essentially if one is not fortunate enough to hit a lottery where the price level changes attributable to regimes, diffusions, and jumps are near-perfectly normal/log(norm) and they do not get collected within the μ term parameters, the resulting inaccuracy in whichever direction lets just say ain’t gonna be linear.

$$\frac{\mathbb{E}[S_{\text{model}}(T)]}{\mathbb{E}[S_{\text{true}}(T)]} = \frac{S_0 e^{\mu_{\text{model}} T}}{S_0 e^{\mu_{\text{true}} T}} = e^{(\mu_{\text{model}} - \mu_{\text{true}})T} = e^{\varepsilon T} \quad , \text{Exponential } \mu \text{ Inaccuracy} \quad (60)$$

Like I said, this approximation is so perfect for a beginners first GBM in traditional finance where a naive use of μ as literally $P(t) - P(0)/\text{Duration}$ produces results still within the range of reality, so long as that calculation finds $\mu \ll .1$:

$$e^{0.08} \approx 1 + 0.08 \rightarrow 1.0833 \text{ vs } 1.0800$$

$$e^{0.12} \approx 1 + 0.12 \rightarrow 1.1275 \text{ vs } 1.1200$$

$$e^{0.20} \approx 1 + 0.20 \rightarrow 1.2214 \text{ vs } 1.2000$$

If a certain market had somewhat comedic μ ’s that exceed .5, then the standard GBM quickly degenerates into what will likely be very inaccurate long term predictions.

$$e^{0.50} \approx 1 + 0.50 \rightarrow 1.649 \text{ vs } 1.5$$

$$e^{0.75} \approx 1 + 0.75 \rightarrow 2.117 \text{ vs } 1.75$$

$$e^{1.0} \approx 1 + 1.0 \rightarrow 2.718 \text{ vs } 2.0$$

Naive models making predictions with this excessive of a μ will simply be inaccurate if the market regime changes to a more negative trend as it’s already overshooting the effect from μ .

The second of assumption that the approximated GBM (Eq 58) makes is that of the Jensen Inequality Correction Term (present in Eq 57) being regarded as trivial. The true expanded solved GBM with this term included would be as follows:

$$\mathbb{E}[S(T)] = S_0 e^{(\mu - 0.5\sigma^2)T} \quad (\text{Expanded: Jensen Inequality Correction Term}) \quad (61)$$

In a Quant Fin 101 where most would build their first model, the exclusion of this term would be a justifiable approximation for the realm of traditional finance. Since the magnitude of σ is typically so much smaller in traditional market assets, the relationship of the Jensen Inequality Correction term with μ is ≈ 0 . For our market, in retrospect we believe the Jensen Inequality term cannot be regarded as trivial, although for reasons discussed in Sec 6.3.2 we do not parameterize this term in our model.

GBM with Time-Varying Drift (Better)

One of the most common changes while using the GBM is to utilize an analytical drift with whatever potential co-variables X_t can be formed from the available data. This greatly reduces inaccuracies related to the μ assumption and generally increases determinism relative to an informed GBM.

$$\frac{dS(t)}{S(t)} = \mu(t)dt + \sigma dW(t) \quad (62)$$

$$\mu(t) = \alpha + \beta \cdot \mathbf{X}(t) \quad (63)$$

This change is kinda easily implemented in traditional finance where everyone is keeping books and you could perhaps pay for some data as an independent quant and be on your way. There's like two brokers in our market and their both overpriced as shit for like no API calls so honestly we're gonna rely on our intuition to guide our model with the few market-generic co-variables we can make from our available data.

GBM with Merton-Jumps & Diffusion (Final Model)

Yeah so our market especially because of the unpredictable events, really requires some focus on the unpredictable volatility. By adding Merton-Jumps $J(t)$ we can wrangle in some of the volatility that could be misleading the model.

$$\frac{dS(t)}{S(t)} = \mu(t)dt + \sigma dW(t) + J(t)dN(t) \quad (64)$$

- $\mu(t) = \alpha + \beta \cdot \mathbf{X}(t)$ captures predictable dynamics.
- $\sigma dW(t)$ captures normal, continuous volatility.
- $J(t)dN(t)$ isolates discrete, unpredictable shocks.

This change requires significant guidance in the form of manually declaring the jump dates, a real pain in the ass for our market which doesn't have cumulative lists about the potential market shocking events.

6.3 Stochastic Lobotomy

Yes this is our writeup where we explain methodology and results and observations but to this day I'm still slightly confused as to the thought process behind the decisions I made. I'd argue there was no thought process behind my initial proposed idea and the iterative process I did. The process of iteration was more or less than black-box of this section, in hindsight I understand why the model worked where it did and failed elsewhere but you're gonna have to stick with it. The entire development was the span of a single maybe 16-hour manic coding session, and I didn't commit a single time so I have no idea what brought me to the results either actually.

Development Span : $\approx$ 9:00 PM July 24th 2025 until $\approx$ 12:00 PM July 25th 2025

Iterative Process



Figure 11
A conceptual overview of the SDE's heuristic validation process.

6.3.1 What I Proposed on a Single Sheet of Paper

I originally proposed on a sheet of a paper a somewhat complex but pretty standard GBM with Jump-Diffusion's. For the analytic drift we utilize a harmonic series, specifically outlined diffusion periods related to predictable events, and logarithmic function related to the number of market-participants.

1. General Equation:

$$\frac{dS(t)}{S(t)} = \mu(t, \mathbf{X}_t)dt + \sigma(t, \mathbf{X}_t)dW(t) + J_{\text{known}}(t)dN_{\text{known}}(t) + J_{\text{shock}}(t)dN_{\text{shock}}(t) \quad (65)$$

2. Drift Term:

$$\begin{aligned} \mu(t, \mathbf{X}_t) = & \alpha + \sum_{n=1}^N \left(a_n \cos\left(\frac{2\pi nt}{365.25}\right) + b_n \sin\left(\frac{2\pi nt}{365.25}\right) \right) \\ & + \sum_{k=1}^K \left(c_k \cos\left(\frac{2\pi kt}{7}\right) + d_k \sin\left(\frac{2\pi kt}{7}\right) \right) \\ & + \sum_{i=1}^I \beta_i E_i(t) + \gamma \log(\text{ParticipantCount}(t)) \end{aligned} \quad (66)$$

3. Volatility Term:

- $\sigma(t, \mathbf{X}_t) = g(\mathbf{X}_t, \dots)$ (Time-varying and state-dependent function)

I notably did not actually explicitly state what I wanted to do with the volatility σ in the original proposal, so I've listed the standard volatility implementation for the GBM in this section.

4. Jump Terms:

- $dN_{\text{known}}(t) \sim \text{Poisson}(\lambda_{\text{known}}dt)$; $J_{\text{known}}(t) \sim \mathcal{N}(\mu_{\text{known}}, \sigma_{\text{known}}^2)$
- $dN_{\text{shock}}(t) \sim \text{Poisson}(\lambda_{\text{shock}}dt)$; $J_{\text{shock}}(t) \sim \mathcal{N}(\mu_{\text{shock}}, \sigma_{\text{shock}}^2)$

6.3.2 What Actually Ended up in the Final Code

What ended up in the code was generally the same, however I vastly simplified the utilization of Merton-Jump's into a single jump term that does not differentiate between potentially predictable events and unpredictable events as the vast majority were simply unpredictable.

1. General Equation:

$$\frac{dS(t)}{S(t)} = \mu(t)dt + \sigma dW(t) + J(t)dN(t) \quad (67)$$

2. Drift Term:

$$\begin{aligned} \mu(t, \mathbf{X}_t) = & \alpha + \sum_{n=1}^N \left(a_n \cos\left(\frac{2\pi nt}{365.25}\right) + b_n \sin\left(\frac{2\pi nt}{365.25}\right) \right) \\ & + \sum_{k=1}^K \left(c_k \cos\left(\frac{2\pi kt}{7}\right) + d_k \sin\left(\frac{2\pi kt}{7}\right) \right) \\ & + \sum_{i=1}^I \beta_i E_i(t) + \gamma \log(\text{ParticipantCount}(t)) \end{aligned} \quad (68)$$

The event indicator sum $\sum_{i=1}^I \beta_i E_i(t)$ expands across three structurally distinct flag types:

$$\sum_{i=1}^I \beta_i E_i(t) = \underbrace{\sum_{p=1}^4 \beta_p^{\text{pred}} E_p^{\text{pred}}(t)}_{\text{Predictable Event (4 phase windows)}} + \underbrace{\beta^{\text{unpred}} E^{\text{unpred}}(t)}_{\text{Unpredictable Event (single window only)}} + \underbrace{\beta^{\text{info}} E^{\text{info}}(t)}_{\text{Anticipated Update (1-day point indicator)}} \quad (69)$$

The first two events are consistent as originally proposed (Sec 6.2). The third, the anticipated information update, is essentially the $dN_{\text{known}}(t)$ predictable jump-term from the original proposed SDE (Sec 6.3.1) instead implemented as a single-day diffusion event.

$$\bullet \alpha \sim \mathcal{N}(0, 0.01), \beta_j \sim \mathcal{N}(0, 0.05)$$

3. Volatility Term:

- σ is a constant.
- $\sigma \sim \text{HalfNormal}(\dots)$

The volatility itself here was actually further modified to be even less informed than what is typically used in a standard GBM. The explanation of this choice is outlined in Figure 11.

4. Jump Term:

- $dN(t) \sim \text{Poisson}(\lambda_J dt)$
- $J(t) \sim \mathcal{N}(\mu_J, \sigma_J^2)$
- $\lambda_J, \mu_J, \sigma_J$ estimated empirically from a combined set of identified jump events.

* We note that because of the extreme difficulty in properly parameterizing all jump events in our training data, we exclude the Jensen Inequality Correction term (Sec 6.2) as we were concerned that the extreme σ displayed in our market that would not be properly contained within the Merton-Jumps would too strongly over-correct the μ term's influence, assuming proper parameterization of μ . Less professionally this oversight can also be explained through Figure 11 and us thinking the model output looked “good enough”.

6.4 Structural Drift Specification

Our approach to the drift term and the choices of covariates for our drift component were one of the few completely intentional design choices in the SDE. Much like a commodity-market forecasting model, we also attempt to treat the unexplained yearly growth α for our asset-generating commodities as ‘unexplained error’. Although from specific asset-generating commodity to another one of these commodities there may be underlying market preferences that drive one generating commodities price growth more than another, it would require vastly more data and general parameterization of the specific generating-commodities under review that we did not implement.

$$\mu(t) = \underbrace{\alpha}_{\approx 0 \text{ (Residual Error)}} + \underbrace{\gamma \log(\text{ParticipantCount}(t))}_{\text{Fundamental Network Value}} + \underbrace{\sum \text{Harmonics}(t) + \sum \beta_i E_i(t)}_{\text{Cyclical \& Event Drivers}} \quad (70)$$

Because a uniquely powerful piece of data we possessed, the market-participant count, alongside the other covariates does well enough to predict general market trend (at least for certain assets) throughout development one metric we specifically targeted was bringing that α coefficient towards 0.

As previously defined in Sec 6.3.2 our deployment of partially known/anticipated jump events as a single-day component of the $\sum \beta_i E_i(t)$ with influence on the drift term can best be described as theoretically unsound. However, in practice (Sec 7.2) was found to be a “happy-accident” as the parameterization in the drift term along with future occurrences within the prediction period instead made this component typically act as a price-level correction towards reducing the potential risk posed by over-shooting of the μ value (Sec 6.2). Thus, it contributes towards our goals of deterministic output (Sec 6.5) and reduction of unexplained drift α .

6.5 Defining Determinism and Commodity Categories

The asset-generating commodities in our market has a wide spectrum of focus as it is the primary means for generating supply for market assets, and for the purpose of defining what determinism meant for us in our model, we can categorize them into the following groups. Please note this is just for interpretation of the models behavior and development towards our goal of differentiating asset-generating commodities future price levels and strength.

Actively Supplying Commodities

Asset-generating commodities for which these commodities themselves are actively being generated/produced.

Available Alternative Commodities

Asset-generating commodities for which the specific collection of highest value category assets that can be produced from the commodity is also available through other commodities at prices in competition with itself.

No Available Competitive Alternative Commodities

Asset-generating commodities for which there is no direct alternative choice at a similar minimum price.

High-Demand Stable Commodities

Specific asset-generating commodity sets whose highest category generated assets generate a stable demand and price behavior that closely relates to the general market behavior and sentiment. All commodities in this category are a series of multiple commodities that are derivatives of one another, thus this category can be understood as a subset of the *Available Alternative Commodities*.

These categories are not all mutually exclusive, however high-demand stable commodities and no available competitive alternative categories are exclusive to one another and both are exclusive respective to the actively supplying commodities.

6.5.1 Defining deterministic SDE output

The goals for our SDE was for the output to inform us about market principles behind asset pricing and the market behavior itself, and most quantitatively to determine the general future price trend to differentiate

truly strong commodities from questionable or more suspicious commodities. In general actually forecasting the specific prices for certain assets was not the intention for our model. Once again this is what makes using the SDE for this market appear counter-intuitive, however given market characteristics and available data this would not be a realistic goal.

The confident we placed in our models outputs was highly dependent on the specific category of the asset-generating commodity. Because we do not use a Bayesian Network, Hierarchies, or a Forest approach, our model is not informed about alternatives for a commodity, and thus will not produce deterministic results for commodities which price is related to other of these same commodities. This generally disqualifies both the *Available Alternative Commodities* and *High-Demand Stable Commodities*, as the High-Demand Stable commodities all have derivative versions of themselves with similar and related prices, supplies, and demands.

Although actively supplying commodities tend to exhibit unique price behavior, unless the characteristic of their supply generating changes the model should still theoretically be producing deterministic predictions so long as there are no available alternatives at a similar pricing with a deep supply.

Altogether, we understand our model to be theoretically deterministic for the specific asset-generating commodities which do not have an alternative at a similar price. This was understood from the onset of our model's use and throughout our trading behavior and asset-generating commodity portfolios positioning.

Unexplainably we initially held the model to be deterministic for certain instances where there was only one other alternative and both commodities had nearly identical characteristics (price, supply, demand). In these cases where the pricing behavior was parallel across alternatives the model is believed to be also be deterministic because the commodity and it's alternative can be modeled as a single commodity and the volume of trading was not a covariate for our SDE (although it maybe should have been).

6.6 Market Parameters μ and σ

We provide generalized values for μ and σ from the year previous to our training period (Sec 6.7) based on the categories of asset-generating commodities to present the scope of the challenge the GBM faces in this market relative to the failure points outlined in Section 6.2. These are the values **one year prior** to our deployment period, the deployment period was a far more tumultuous period in the market than most of these commodities have ever seen before. In essence I'm trying to say it seems like free-money but the high-risk environment of this specific market came to demonstrate the necessity of proper risk mitigation strategies.

Table 21
Approximate Market Parameters by Asset-Generating Commodity Category

Category	Approx. μ (Preceding Year)	Approx. σ (Preceding Year)
Available Alternative Commodities (No Actively Supplying Alternatives)	0.65–0.9	0.40–0.50
Available Alternative Commodities (Alternative Actively Supplying)	0.35–0.50	0.5–0.6
No Available Competitive Alternative Commodities	0.5–0.8	0.35–0.45
High-Demand Stable Commodities	0.65–0.8	0.4–0.5

The astronomically high μ and σ values that are present in the data used to fit our SDE necessitates the deployment of an analytic drift structured to diminish the unexplained drift α (Eq 70) and Merton-Jump's to reduce some of volatility (Sec 6.2).

6.7 Training Data Period & Data Points

To provide context to the previous market parameters Sec 6.6, and what we mean by under-parameterization of Merton-Jump events, we outline the following price-day time series range, drift event diffusion definitions, and amount of asset-generating commodities analyzed after completion to influence our understanding of market dynamics.

Table 22
Training Data Range and Drift Covariate Event Flag Windows

Event / Series	Start	End / Duration
<i>Price-Day Training Series</i>		
Historical Price Data	2021-07-01	2025-07-24
Participant Count Data	2021-07-01	2025-07-24
<i>Predictable Drift Event Flag Definitions (per event)</i>		
Pre-Event	–30 days before start	Explicit Start Date
Event Active	Explicit Start Date	Explicit End Date
Post-Event	Explicit End Date	+14 days after end
Event Cooldown	+14 days after end	+35 days after end
<i>Unpredictable Drift Event Flag Definitions (per event)</i>		
Event Active	Unpredictable Event Start Date	Unpredictable Event End Date
<i>Anticipatable Central Market Authority Information Updates with known Sentiment Direction</i>		
Anticipated Information Update	Observed release date only	Single-day point indicator

The specific drift μ related covariates consist of the historical price data as the basis α related to the *Participant Count Data* denoting general market activity. The training-data includes smaller sized diffusion phase durations related to the predictable drift event flags, and a much larger diffusion duration for the unpredictable drift event present in the training data. Furthermore, the single-day point-in-time diffusion information update acts as a single day price level correction to an market event that theoretically may have been better parameterized as Jump-Event.

Thus, we utilize a maximum of **1,485 data points** from price-day data of any asset-generating commodity to fit our SDE to. This is a decent amount of data points but is relatively dwarfed by the amount of data points available in traditional continuous market ledgers that can contain information for every trade.

Table 23
Diffusion Event and Merton-Jump Counts Within Training Window

Event Type	Quantity	Classification
<i>Diffusion Events — 10 total within training window</i>		
Predictable Events	8	Diffusion
Unpredictable Events	1	Diffusion
Anticipated Information Update	1	Single-Point Diffusion
<i>Merton-Jump Dates — 8 total within training window</i>		
Shocks with Expected Positive Sentiment	4	Jump
Shocks with Mixed/Undefined Sentiment	2	Jump
Shocks with Expected Negative Sentiment	1	Jump
Unpredictable Event Start Dates	1	Jump

The insanely small amount of parameterized jump-dates over the training period for a market that has an infamously high amount of actual jump inducing events every year is one of the biggest faults committed by our implementation of our model. To give some grace this information is extremely difficult to gather without data-scraping and data-processing models which were outside of the scope of the project given it's overall short development and live-market deployment durations. It's also important to mention that we do not utilize jump date declarations ever for more than two days consecutively. This was a more intentional design decision to keep jump date declarations relatively uninformed as we do not declare and mitigate the volatility for all the possible jump events, so thus would not like to completely mitigate all the volatility for the explicitly declared events.

7 SDE Model Results & Interpretations

In this section we will provide minimally-censored prediction graphs and covariate information from our SDE, as well as provide reference prediction from a uninformed naive GBM to demonstrate the successes and limitations of our SDE model. We also reflect on our respective design decisions such as analytical drift covariates Eq 2, drift structure Eq 70, and Merton Jump declarations Table 23.

We will analyze the outputs for 8 commodities in total, 2 for each category except *No Available Alternative Commodities* where we present 4. For *Available Alternative Commodities* the first has a cheaper *Actively Supplying Commodity* as an alternative and the second commodity is the cheapest relative to alternatives but has ≈ 8 more expensive alternatives with ≈ 3 or 4 of the alternatives at a similar price & supply. For the *No Available Alternative Commodities*: the type of commodity our SDE was deemed more deterministic for, we demonstrate how our SDE more accurately predicts pricing behavior for these commodities with characteristics that operate similarly to traditional financial commodities. Lastly we present two *High-Demand Stable Commodities*, one to demonstrate the weakness of our SDE on commodities with derivative alternatives that change price behavior dynamics for the commodities, and one whose single derivative can be modeled with the original commodity itself (we originally believed the model to produce deterministic for this commodity).

Lastly, we analyze and provide evidence for how our SDE influenced our trading behavior and understanding of market and commodity characteristics. The SDE's specific influence on our trading behavior during the October market crash and past it's decommission will be discussed in the later market-crash specific section (Sec 8).

When judging price forecast accuracies, note the orange vertical marker for the unprecedented market event. Past this point predictions are not intended to be accurate as the model was never ran past the date of completion (2025/7/24) and was never informed with any updated pricing or market event occurrence information.

7.1 Introduction to Commodities Analyzed & Naive GBM Example Predictions

First, we will introduce the commodities from the aforementioned commodity categorizes (Sec 6.5) by presenting some naive GBM predictions for the commodities our SDE was applied to. The naive GBM deploys the standard GBM form from Eq 58 and with an uninformed drift μ without covariates $\beta * X(t)$ utilizing the μ simplification defined in Eq 59.

As the naive GBM prediction section gets repetitive but provides the commodities under analysis specific characteristics and context, we provide a summary table for the commodities characteristics. If you have the time and would like a complete understanding of commodity characteristics and nuance of the modeling and prediction results for these commodities, the specific characteristics relevant to prediction and model analysis will be covered throughout the naive GBM and custom SDE prediction context paragraphs, and thus this table can be disregarded.

Table 24
Available Alternative Commodities — Characteristic Summary

Liquidity	Supply	Demand	Prev. Yr. $\hat{\mu}$	Prev. Yr. $\hat{\sigma}$	Approx. Price at Prediction	Alt. Actively or Recently Supplying	Cheapest of Alternatives
<i>Available Alt. Commodity 1</i>							
High	High	High	$\approx .4$	$\approx .5$	$\approx \$1.4$	✓	
<i>Available Alt. Commodity 2</i>							
Moderate	Moderate	Moderate-Low	$\approx .9$	$\approx .5$	$\approx \$4.8$		✓

Table 25
No Available Competitive Alternative Commodities — Characteristic Summary

Liquidity	Supply	Demand	Prev. Yr. $\hat{\mu}$	Prev. Yr. $\hat{\sigma}$	Approx. Price at Prediction	Highest-Tier Generated Asset Exclusivity
Low	Moderate	Low	<i>No Avail. Alt. Commodity 1</i> $\approx .5$ $\approx .5$ $\approx \$2.5$			Exclusively Generable; No Alternative
Low	Low	Moderate	<i>No Avail. Alt. Commodity 2</i> $\approx .5$ $\approx .3$ $\approx \$17$			Single Non-Competitively Priced Alternative ($\approx 2\times$ Higher)
Moderate	Low	High-Moderate	<i>No Avail. Alt. Commodity 3</i> $\approx .4$ $\approx .3$ $\approx \$13.75$			Exclusively Generable; Non-Overlapping Competitor Indirectly Limits Asset Ceiling

Table 26
High-Demand Stable Commodities — Characteristic Summary

Liquidity	Supply	Demand	Prev. Yr. $\hat{\mu}$	Prev. Yr. $\hat{\sigma}$	Approx. Price at Prediction	Derivatives in Series	Derivative Price Relation Expected
High-Moderate	Moderate	High	<i>High-Demand Stable Commodity 1</i> $\approx .85$ $\approx .5$ $\approx \$5.5$				3 ✓
High	High	High-Moderate	<i>High-Demand Stable Commodity 2</i> $\approx .85$ $\approx .7$ $\approx \$2.3$				2

Note: Liquidity, Supply, and Demand are qualitative approximations relative to the broader market. $\hat{\mu}$ and $\hat{\sigma}$ are gathered from observed prices one year prior to prediction date. Derivatives in Series includes the commodity itself. Derivative Price Relation Expected indicates whether inter-commodity price behaviour was anticipated to influence model accuracy prior to predictions being made.

The naive GBM predictions are made from the end of the training period and SDE development (2025/07/24) to June 1st 2026 (2026/06/01); a **312 day long prediction period**.

Apologies for the poor centering of the forecasting graphs

7.1.1 Available Alternative Commodity Naive Prediction

We first inspect the naive outputs on *Alternative Available Commodity 1*, a cheaper and more liquid commodity that has a cheaper alternative that is an *Actively Supplying Commodity*.

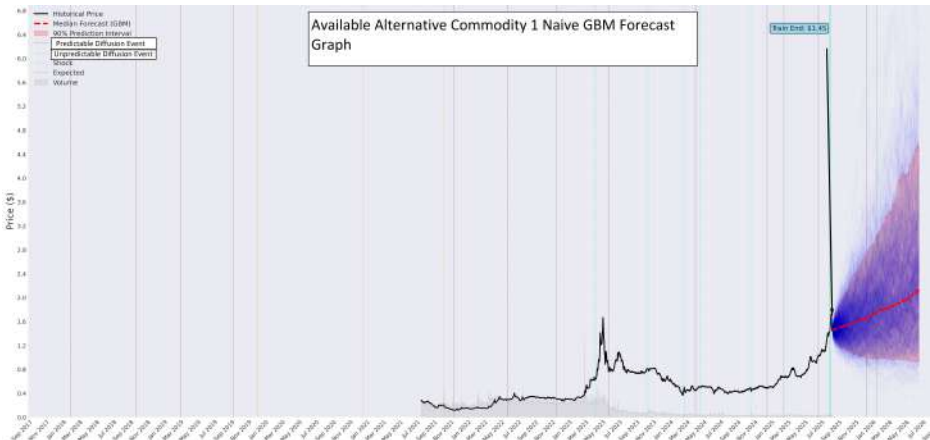


Figure 12
Available Alternative Commodity 1 Naive GBM Price Forecast Graph.



Figure 13
Available Alternative Commodity 1 Naive GBM Price Accuracy Graph.

Available Alternative Commodity 1 demonstrates the weakness of non-analytical drift in this market by overshooting price predictions (Fig 13), likely attributable to the inadequacies previously outlined for the simple GBM (Sec 6.2). We note how the general forecast graph (Fig 12) has a seemingly linear price prediction over time demonstrating how that over-shot naive μ has not learned the nuances and principles of commodity pricing in this market.

We proceed to *Alternative Available Commodity 2* which is itself the cheapest out of many multiple other alternatives, although has its price ceiling set by this relationship. This is the will be the best example of our own SDE's inadequacies in this market, although the figures below are for the naive GBM.

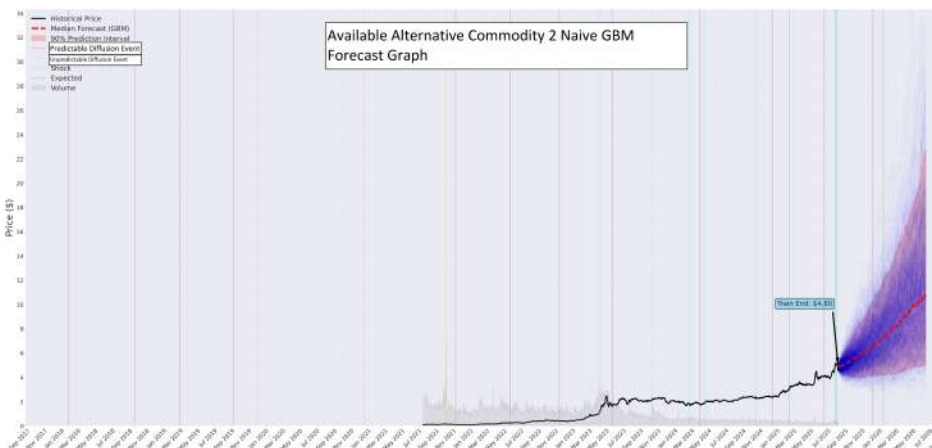


Figure 14
Available Alternative Commodity 2 Naive GBM Price Forecast Graph.

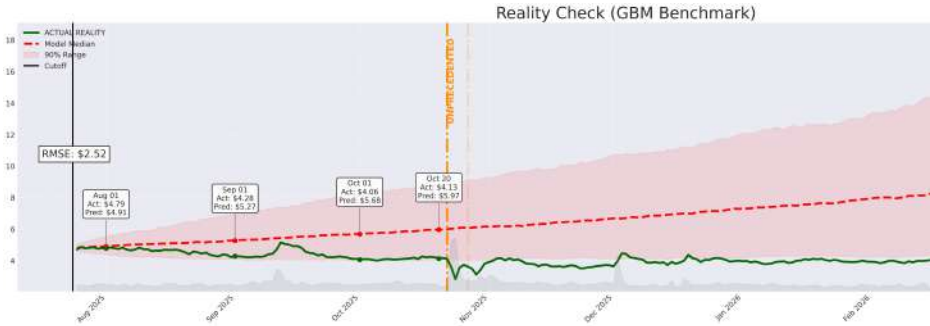


Figure 15
Available Alternative Commodity 2 Naive GBM Price Accuracy Graph.

Available Alternative Commodity 2 once again suffers greatly from the overshoot μ , although far worse than *Alternative Available Commodity 1*. The price forecast graph (Fig 14) shows a more exponentially-shaped future price prediction.

The differences between these two *Available Alternative Commodities* serves to demonstrate the degree to which the extreme μ value present in our training period (2021-07-01 to 2025-07-24) FIXTHIS. *Alternative Available Commodity 1* experienced a $\mu \approx 40\%$ or .4 whereas *Alternative Available Commodity 2* experienced a mind-boggling $\mu \approx 90\%$ or .9 throughout the training period. For the purposes of our SDE model this confirms that our suspicions of μ being the most problematic component of our market to making a SDE for the purpose of differentiating long-term commodity strengths. The effects of this being shown with how *Alternative Available Commodity 1*'s naive GBM price accuracy (Fig 13) can at least keep the actual observed price of its commodity within its 90% Confidence Interval, whereas for *Alternative Available Commodity 2*'s naive GBM price accuracy (Fig 15) the actual observed price begins to escape the 90% Confidence Interval bounds (*both 90% Confidence Intervals and predicted prices being discussed prior to the unprecedented market event*).

7.1.2 No Available Alternative Commodity Naive Prediction

We next present the naive GBM's output for three *No Available Alternative Commodities*, the commodity category in which we felt most confident about our own SDE's predictions; the commodity category that typically displayed more stable price behavior and μ values closer to the general market average for asset-generating commodities over our training data's time period; $\mu \approx 50\%$ or .5. All assets presented for this category have an average μ of approximately 0.5.

No Available Alternative Commodity 1 is a commodity whose specific highest asset classification group is typically not as desirable, which results in the commodity holding a relatively lower price and demand relative to other commodities in this category. Like most *No Available Alternative Commodities*, this commodity has a relatively lower market supply and is far less liquid than assets presented in this section for the other commodity categories.

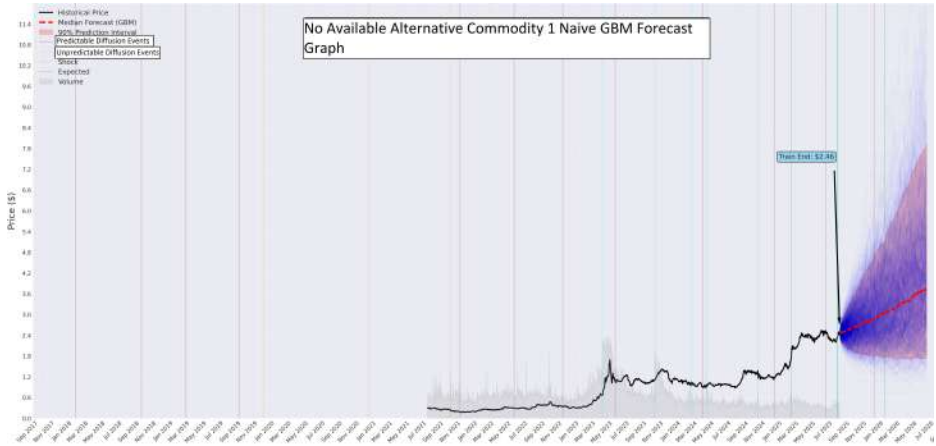


Figure 16
No Available Alternative Commodity 1 Naive GBM Price Forecast Graph.



Figure 17
No Available Alternative Commodity 1 Naive GBM Price Accuracy Graph.

We note how for *No Available Alternative Commodity 1* the naive GBM produces a mostly linear price forecast over time, taking a less exponential shape over the 312 day prediction period of the naive GBM with the $\mu \approx .5$ compared to the *Available Alternative Commodity 2*'s more exponential shape over that 312 day prediction period which is likely attributable to its astronomical $\mu \approx .9$. It's notable how the naive GBM produced quite accurate price predictions for 1 $\frac{1}{2}$ months, demonstrating how *No Available Alternative Commodities* have more stable price behavior reflecting the general market direction, an attribute that makes it favorable in terms of price forecasting.

No Available Alternative Commodity 2 is of high importance as our custom SDE's prediction for this was extremely accurate (Sec 7.2.3), provided insight on this commodities certain characteristics that served pivotal to our understanding of the market crash (Sec 8), and was highly influential to our commodity portfolio positioning and trading pre-crash (Sec 7.2.3 & Sec 7.6). Like *No Alternative Available Commodity 1* this commodity demonstrated a $\mu \approx .5$, however, the demand for the generated highest asset class from this commodity is extremely high (price-relative). Although this commodity technically has an alternative, the alternative being priced nearly twice as high makes this commodity fit the definition of the *No Available Alternative Commodity* category as more specifically denotes commodities without any **competitively** priced alternatives.

Thus, this commodity demonstrates a far higher price $\approx 15\$$ to $17\$$ on Market C than most in-demand commodities do (typically $\approx 4\$$ to $7\$$). The next commodity, *No Available Alternative Commodity 3*, has

pricing characteristics much like *No Available Alternative Commodity 2* as its price sits at $\approx 11\$$ to $14\$$, however, neither of these assets come close to the some of highest priced commodities that still pertain some degree of stable demand ($\approx 50\$$).

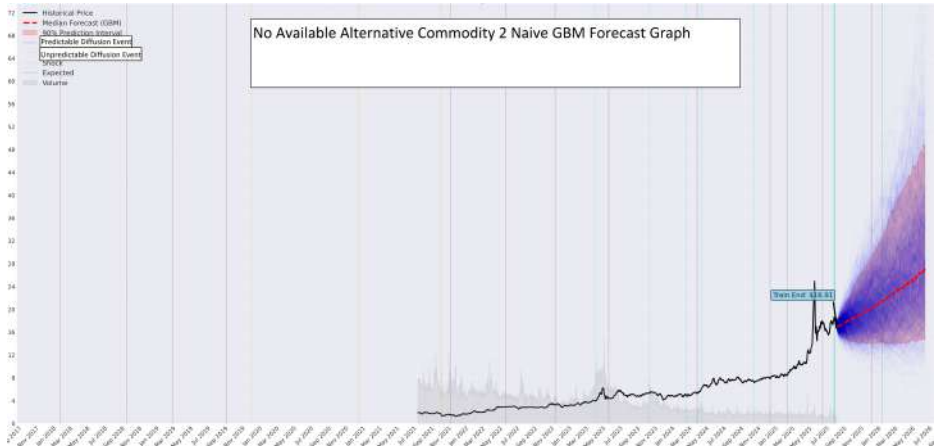


Figure 18
No Available Alternative Commodity 2 Naive GBM Price Forecast Graph.

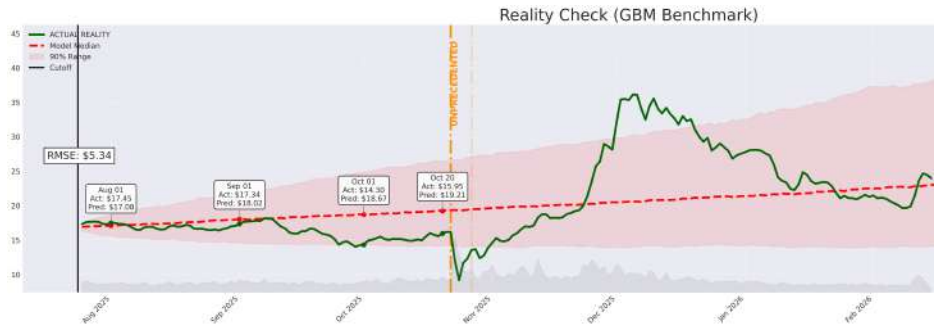


Figure 19
No Available Alternative Commodity 2 Naive GBM Price Accuracy Graph.

The naive GBM is once again able to produce mostly accurate price forecasting for $1 \frac{1}{2}$ months as that period of time after the end of the training data was quite a stable time for the market relative to the ensuing months. The price forecast is mostly linear but displays slightly more curvature relative to *No Alternative Available Commodity 1*, likely attributable to *No Alternative Available Commodity 1*'s volatility being so much higher $\sigma_1 \approx .5 > \sigma_2 \approx .3$ which is somewhat displayed in *No Available Alternative Commodity 1*'s forecast curve (Fig 16).

We now provide the naive GBM's forecast for a third commodity, *No Available Alternative Commodity 3* the one we deem to hold the most similar characteristics to *No Available Alternative Commodity 2* (the most influential commodity prediction). This commodity holds a slighter lower price than the previously discussed commodity, but has been a staple of commodity investing in this market for a long time due to its extremely strong supply to demand with it's scarcity similar to *No Available Alternative Commodity 2*. Although its highest generated asset class is truly exclusive to this commodity, there is an alternative commodity available that generates a similar highest classification asset pool, although they share no direct overlap. Nonetheless this does limit the ceiling of the assets generated by this commodity (although no price behavior relation),

resulting in this commodity having lesser priced highest generated asset class results in its relatively lower price although higher liquidity.



Figure 20
No Available Alternative Commodity 3 Naive GBM Price Forecast Graph.

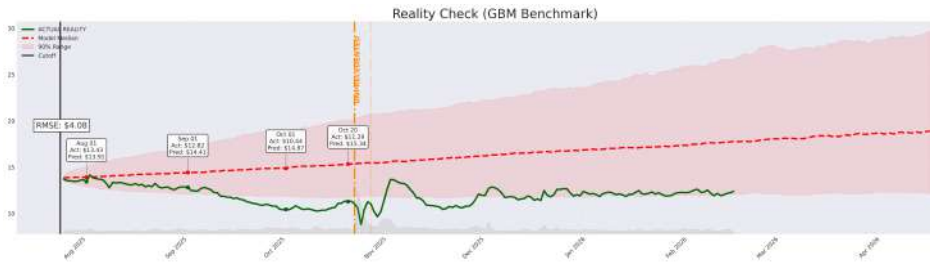


Figure 21
No Available Alternative Commodity 3 Naive GBM Price Accuracy Graph.

The naive GBM produces an extremely linear positive price forecast that clearly does not successfully capture the general higher sensitivity of this asset to price movements likely attributable to *No Available Alternative Commodity 3*'s relatively higher liquidity than what could be typically observable for commodities of this category.

7.1.3 High-Demand Stable Commodity Naive Prediction

We last inspect naive GBM outputs for two *High-Demand Stable Commodities*. We reiterate that this category is a subset of the *Available Alternative Commodities* in regard to the fact that all *High-Demand Stable Commodities* have derivatives of themselves that generate that same highest asset classification pool, and thus have price relations between them that are not parameterized by the naive GBM or our own custom SDE. As these assets are a popular choice for investment because of their solid fundamentals and yearly returns, they tend to be highly liquid.

The first of these we present is an asset with much greater price relation to its available derivatives that have a price differential that is significant to asset under inspections price behavior. Thus as these relations are not parameterized by the naive GBM or our SDE, this commodity does well to demonstrate shortcomings of these respective models. Note that this is the cheapest and most recently supplying derivative out of a series of three total commodities, and thus has its price ceiling set by the price relationship with more expensive derivatives.

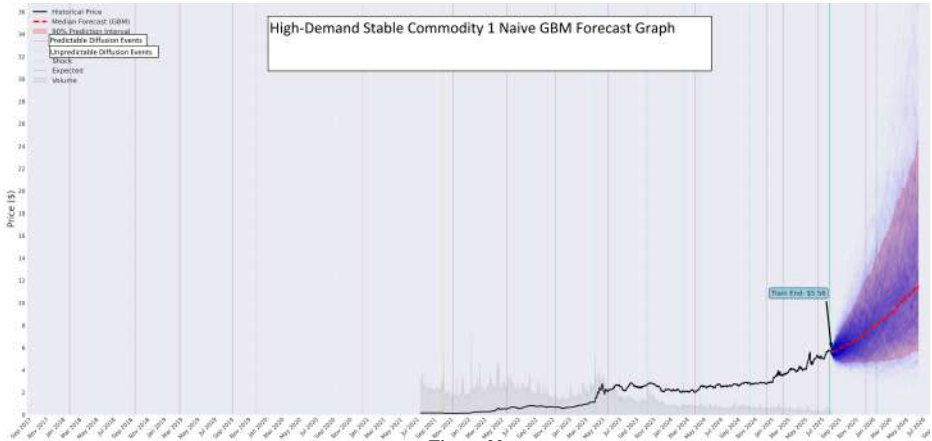


Figure 22
High-Demand Stable Commodity 1 Naive GBM Price Forecast Graph.

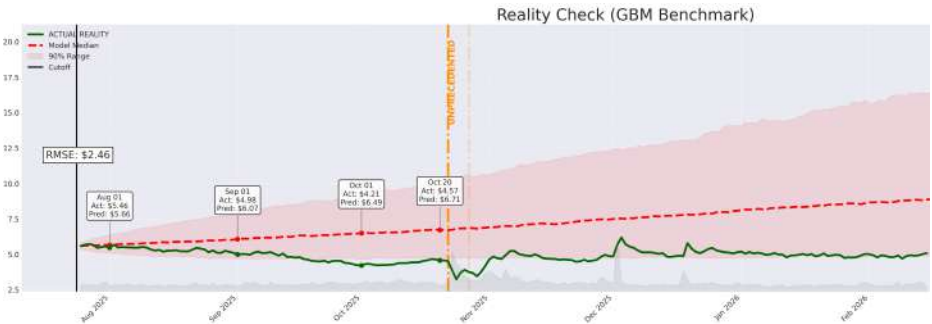


Figure 23
High-Demand Stable Commodity 1 Naive GBM Price Accuracy Graph.

High-Demand Stable Commodity 1 holds an extremely high μ of $\approx .8$ which is high even relative to the category of *High-Demand Stable Commodities* and thus we once again note the significant overshooting and a somewhat exponential price forecast which the high μ is likely responsible for. We note how the commodity's naive GBM price accuracy holds for less than $\frac{1}{2}$ a month, demonstrating the fundamental challenge it is to accurately model this category of commodities.

We now move along to *High-Demand Stable Commodity 2*, the cheapest of the *High-Demand Stable Commodities* and a more recently supplying commodity which thus has much higher liquidity than the other *High-Demand Stable Commodities*. This is the commodity that we still believed our model to hold deterministic outputs for due to this commodity price movement being nearly parallel to that of its derivative.

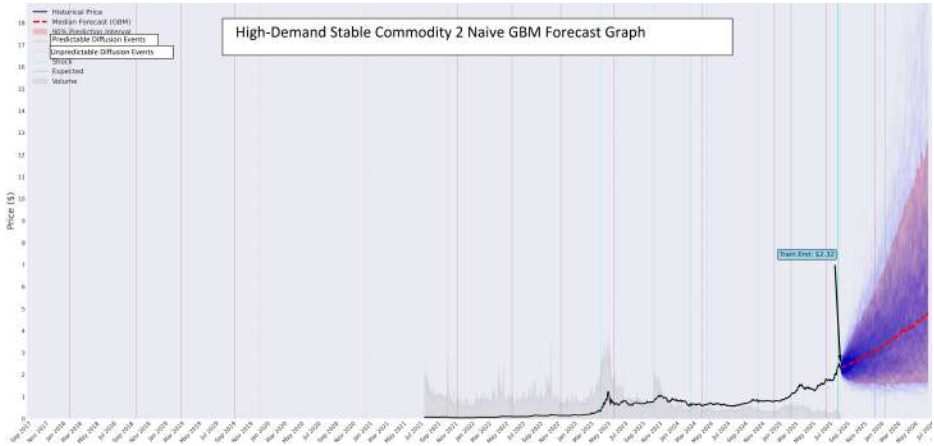


Figure 24
High-Demand Stable Commodity 2 Naive GBM Price Forecast Graph.

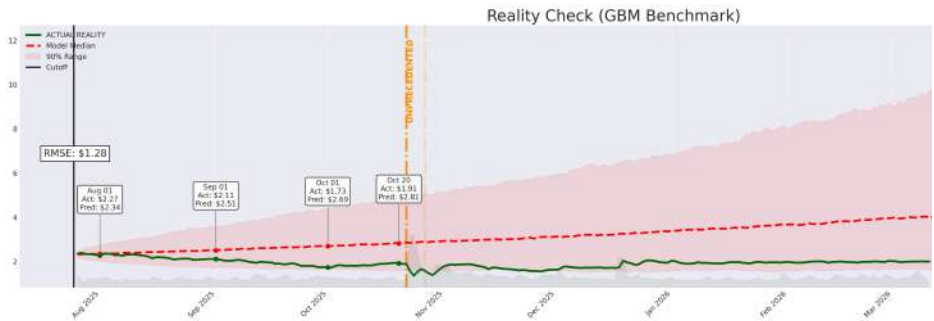


Figure 25
High-Demand Stable Commodity 2 Naive GBM Price Accuracy Graph.

It's noticeable how for this commodity where it's derivative and itself demonstrate nearly identical price behavior even the naive GBM is able to sustain longer price forecast accuracy of $\approx \frac{1}{2}$ relative to *High-Demand Stable Commodity 1*, although the naive GBM is still significantly inaccurate once again. This commodity possesses another extremely high μ of $\approx .8$ like the previous, which provides evidence for both claims that price relationship to alternatives is a significant unaccounted for parameter in our model, and also that if the commodities price, supply, and demand is nearly identical to its alternatives this inaccuracy is less significant.

7.2 Custom SDE Example Predictions & Covariate Coefficients

We now present our custom SDE's forecasts, accuracy checks, and covariate coefficients for the same commodities from the same historical price data, training period and prediction period of 2025/07/24 to 2026/06/01; **312 days**.

In these forecast graphs we also plot the *ParticipantCount* value without specific y-axis from our analytical-drift as we deem it to be an important covariate that does well to demonstrate the general market activity and demand at given times. This is the noticeable purple plotted line.

Once again our model was not built with fragility in mind, but rather as a single date market investigation to influence our commodity decisions, and thus our SDE was not parameterized for the period past the vertical orange line in the price accuracy graph that represents the date of the market crash, an unprecedented market event.

7.2.1 General Covariate Analysis for Analyzed Commodities

We first wish to present a general analysis for the drift covariate coefficients and jump parameters (Sec 6.7) to provide context to our custom SDE's commodity specific output and performance. Complete data tables are available in the additional figures at the end of this article; Table A2 for μ coefficients and Table A3 for jump parameters.

First we present the μ drift-coefficients that typically result in statistically significant 94% HDI intervals.

Table 27
Typically Significant Drift Covariates and Their Direction Across Analysed Commodities

Parameter	Typical Direction	Notes
$\sin(\text{Yearly}_1)$	Positive	Significant in 6 of 7 commodities; reflects a consistent mid-year positive drift cycle.
$\cos(\text{Weekly}_1)$	Negative	Significant in 6 of 7; robust negative intra-week effect across commodity categories.
$\sin(\text{Weekly}_1)$	Negative	Significant in 5 of 7; consistent with $\cos(\text{Weekly}_1)$ in describing a weekly negative drift phase.
$\sin(\text{Weekly}_2)$	Positive	Significant in all 7 commodities; the most consistently significant covariate in the model.
Pre-Event	Negative	Significant in 5 of 7; price suppression in the 30-day window preceding predictable events.

The high proportion of seasonality/frequency based covariates is disappointing as we theoretically expect the weekly terms to be significant if the commodities were *Actively Supplying Commodities*, as participant supply eligibility resets on each Wednesday, and supply generation dates are expected to be significant for that respective category, however all the commodities analyzed in this section are not *Actively Supplying Commodities* and in theory should not exhibit significant influences by the weekly frequency. As only *Available Alternative Commodity 1* has its price behavior related to an *Actively Supplying Commodity*, we would expect the influence of the weekly frequency covariates to be stronger relative to the other commodities, but we do not necessarily observe this to be the case (Table A2).

The yearly frequency coefficient significance is less problematic as some yearly cycle of commodity investing and price behavior was expected to be found. We find some relief in the significance of the Pre-Event binary flags coefficient being significance as this fits our understanding of market dynamics.

Table 28
Covariates with Strong Directional Skew Across Analysed Commodities

Parameter	Skew Direction	Notes
$\sin(\text{Yearly}_3)$	Negative	Posterior mean negative in all 7; significant (HDI fully negative) in <i>No Avail. Alt. Commodity 3</i> and <i>High-Demand Stable Commodity 1</i> .
$\cos(\text{Yearly}_2)$	Negative	Mean negative in 5 of 7; fully significant in <i>Avail. Alt. Commodity 1</i> and <i>No Avail. Alt. Commodity 2</i> , near-threshold in others.
$\cos(\text{Weekly}_2)$	Negative	Mean negative in 6 of 7; fully significant in <i>No Avail. Alt. Commodity 1, No Avail. Alt. Commodity 2</i> , and <i>No Avail. Alt. Commodity 3</i> .
Unpredictable Event Active	Negative	Mean negative in 6 of 7; never crosses significance threshold, but directionally consistent.
Anticipatable Central Authority Updates	Negative	Mean negative in 6 of 7; wide posterior intervals render it insignificant across all commodities due to the rarity of declared expected event dates.

The partial-significance of more yearly frequency coefficients is another relief as this was expected to be observed. Likewise, the partial-significance of the unpredictable event diffusion is also consistent with our market understanding. Most notably is the partial-significance of the negative effect posed by the Anticipated Central Authority updates. As explained in Sec 6.4, we believe this to be a small effective price-level adjustment that should have a negative effect on price forecasts, as it serves as a small correction to the over-shooting of μ that we instead is believed to be more contained in the cyclical harmonic frequency covariates.

Table 29
Consistently Insignificant Drift Covariates Across All Analyzed Commodities

Parameter	Interpretation
α (Intercept)	Consistently centered near zero as intended; confirms structural drift is captured by covariates rather than unexplained trend.
$\log(\text{ParticipantCount})$	No statistically significant relationship with log-returns across any commodity at the 94% HDI level; informative for long-horizon trend but not short-run drift.
TimeIndex	No detectable linear time trend in log-returns after conditioning on the harmonic series and event flags.
$\cos(\text{Yearly}_1)$	Consistently insignificant; the yearly cycle is captured asymmetrically by $\sin(\text{Yearly}_1)$ alone.
$\sin(\text{Yearly}_2)$	Insignificant across all commodities; the second yearly harmonic contributes through its cosine phase only for select commodities.
$\cos(\text{Yearly}_3)$	Uniformly insignificant; higher-order yearly cosine terms add no explanatory power.
Event Active	No significant price effect during the active event window itself; drift effects materialize in the pre- and post-event phases instead.
Post-Event	Directionally inconsistent and insignificant; post-event recovery varies by commodity.
Event Cooldown	Uniformly insignificant; market appears to fully absorb event effects before the cooldown window.

The findings of an insignificant α is a somewhat positive discovery as this was the intended coefficient effect of α that was targeted according to our structural drift philosophy (Sec 6.4) during iterative development,

however as previously noted we believe a good portion of unexplained drift diffused/spread into the harmonic frequency covariates.

The insignificance of exclusively yearly frequency coefficients is un-preferable relative to if weekly frequency coefficients were insignificant instead, but furthermore the general insignificance of ParticipantCount and $\frac{3}{4}$ of the event diffusion coefficients is disappointing.

7.2.2 Available Alternative Commodity Prediction & Covariate Coefficients

We first present custom SDE output for our *Available Alternative Commodities* starting with the cheaper priced *Available Alternative Commodity 1* which is the only commodity we analyze whose alternative is in the *Actively Supplying Commodities* category.

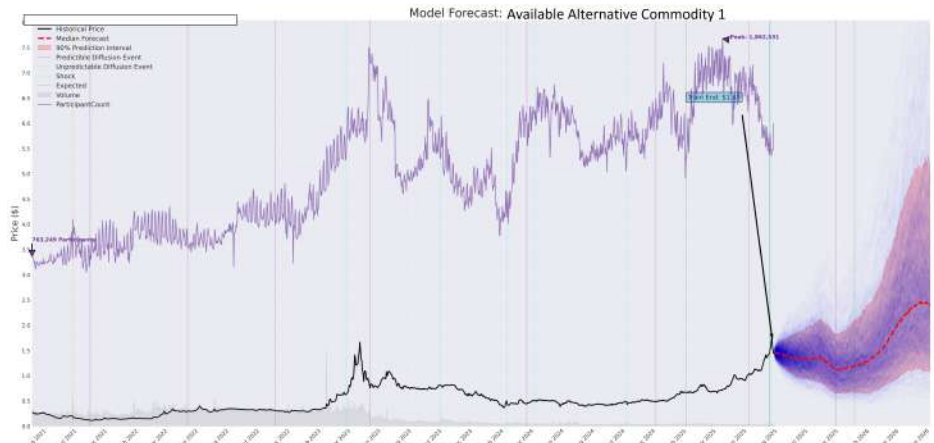


Figure 26
Available Alternative Commodity 1 Custom SDE Price Forecast Graph.

Our initial interpretation of the price forecast for *Alternative Available Commodity 1* is highly positive, as we deemed the general negative/stagnant price direction as a positive sign of determinism as we theoretical expect our model to over-shoot price predictions.

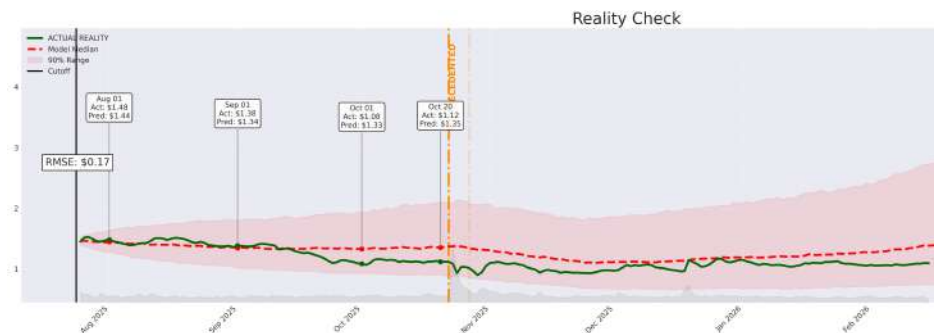


Figure 27
Available Alternative Commodity 1 Custom SDE Price Accuracy Graph.

The retrospective accuracy of our SDE is impressive as relative to the naive GBM its accuracy persists for $> \frac{1}{2}$ month longer than the naive GBM's (Fig 13).

For Available Alternative Commodity 2;

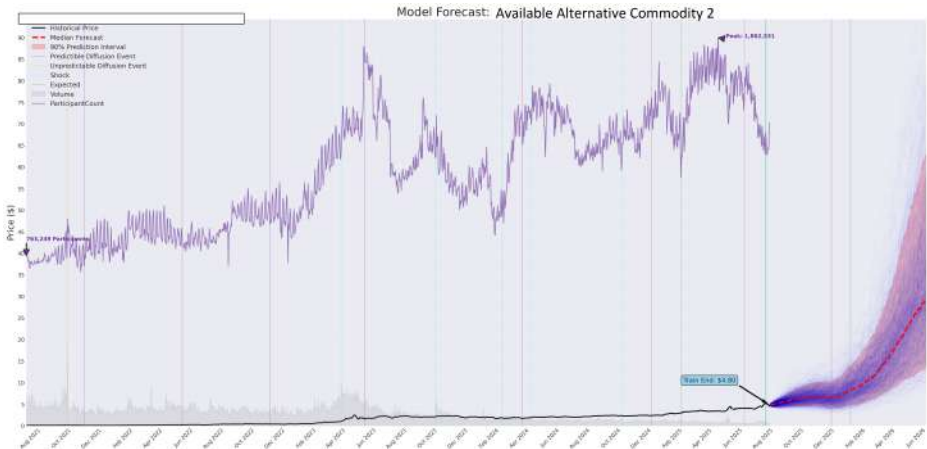


Figure 28
Available Alternative Commodity 2 Custom SDE Price Forecast Graph.

the price forecast displays a clearly exponential price forecast, a clear sign of an in-determinant and poor SDE prediction, as we understand our market to not display the sort theoretical exponential possible returns that can be displayed in a traditional financial market, much less within just 312 days.

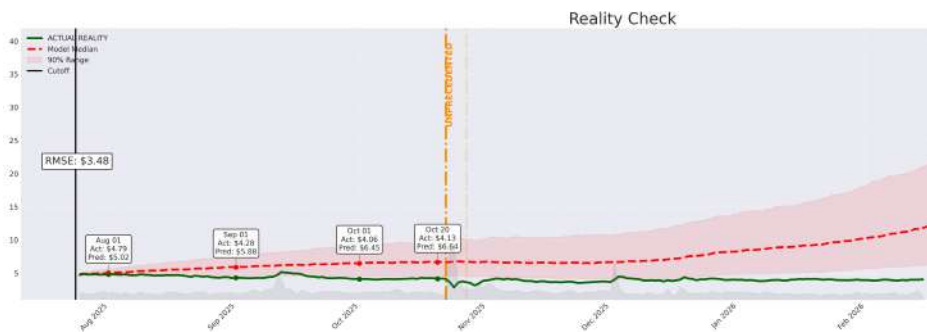


Figure 29
Available Alternative Commodity 2 Custom SDE Price Accuracy Graph.

The retrospective price accuracy analysis confirms the in-determinant hypothesis we held about this specific prediction on 2025/07/24. Furthermore, the accuracy of custom SDE is actually inferior to that of naive GBM (Fig 15), likely attributable to the extreme $\mu \approx .9$ and $\sigma \approx .5$ for this specific commodity being fitted to the harmonic frequency covariates of our model.

We can further confirm our interpretations of our models results through the significant covariates.

Table 30
Significant & Skewed Drift Coefficients — Available Alternative Commodities

Parameter	Available Alternative Commodity 1				Sig.	Available Alternative Commodity 2				Sig.
	Mean	SD	HDI 3%	HDI 97%		Mean	SD	HDI 3%	HDI 97%	
Significant Covariates										
sin(Yearly ₁)	0.00337	0.00107	0.00118	0.00524	✓	0.00302	0.00094	0.00113	0.00463	✓
cos(Weekly ₁)	−0.00282	0.00101	−0.00466	−0.00092	✓	−0.00325	0.00093	−0.00503	−0.00154	✓
sin(Weekly ₂)	0.00430	0.00103	0.00222	0.00613	✓	0.00433	0.00091	0.00277	0.00617	✓
Pre-Event	−0.00732	0.00226	−0.01138	−0.00305	✓	−0.00442	0.00211	−0.00859	−0.00069	✓
Skewed / Near-Significant Covariates										
cos(Yearly ₁)	0.00178	0.00106	−0.00013	0.00379	—	0.00081	0.00100	−0.00117	0.00259	—
cos(Yearly ₂)	−0.00271	0.00107	−0.00482	−0.00073	✓	−0.00028	0.00099	−0.00151	0.00211	—
sin(Weekly ₁)	−0.00020	0.00101	−0.00227	0.00157	—	−0.00226	0.00091	−0.00392	−0.00053	✓
Post-Event	0.00509	0.00300	−0.00055	0.01089	—	0.00322	0.00277	−0.00185	0.00840	—
Jump Parameters										
σ (GBM vol.)		0.02737					0.02529			
λ _J (jumps/yr)		1.97033					1.97033			
μ _J		−0.00347					0.00951			
σ _J		0.01896					0.03460			

As *Available Alternative Commodity 1* displays significance from the weekly frequencies that can actually be explained through its fundamental characteristics and directions of its specific weekly and yearly effects converging, the drift coefficients reinforce our initial hypothesis of our SDE producing a deterministic result. Furthermore, our prior understanding of the *Available Alternative Commodities* category tending to have a lukewarm or negative response to jump events reinforced our initial interpretation.

However, for *Alternative Available Commodity 2*, which displays the same convergence of frequency covariate direction and the presence of more significant weekly covariates is a result that strongly conflicts with our prior understanding of the commodities characteristics and behavior. As *Available Alternative Commodity 2* is the most liquid and recently supplying of all its available alternatives but nonetheless has spent a significant period of time out of the *Actively Supplying Commodity* category, the significance of majority weekly frequency covariates being found is a major sign of in-determinacy. Furthermore, the average jump effect μ_J being positive conflicts with our market understanding, and the high volatility of the jumps σ_J is a sign that our lack of extensive parameterization of jump occurrences is causing an in-determinant effect of the jumps.

We conclude our analysis of our SDE's *Alternative Available Commodities* predictions by confirming our initial interpretation of *Available Alternative Commodity 1* being somewhat deterministic as a correct interpretation. For *Alternative Available Commodity 2* our extremely positive prior sentiment towards this commodity made us initially falsely believe the output to be determinant, but our interpretation of the output quickly shifted as we began to understand the lack of parametrization of alternative commodities affects on price behavior and began to observe the divergence in price forecast accuracy. We prove our interpretation changing over time in Sec 7.6.1 as we position our inventory away from this asset prior to the market crash.

7.2.3 No Available Alternative Commodity Prediction & Covariate Coefficients

Proceeding to our custom SDE's output for the *No Available Alternative Commodities*, the category we deemed our SDE would be most determinant for prior throughout development, we first present the cheapest and most illiquid commodity in this category, *No Available Alternative Commodity 1*:

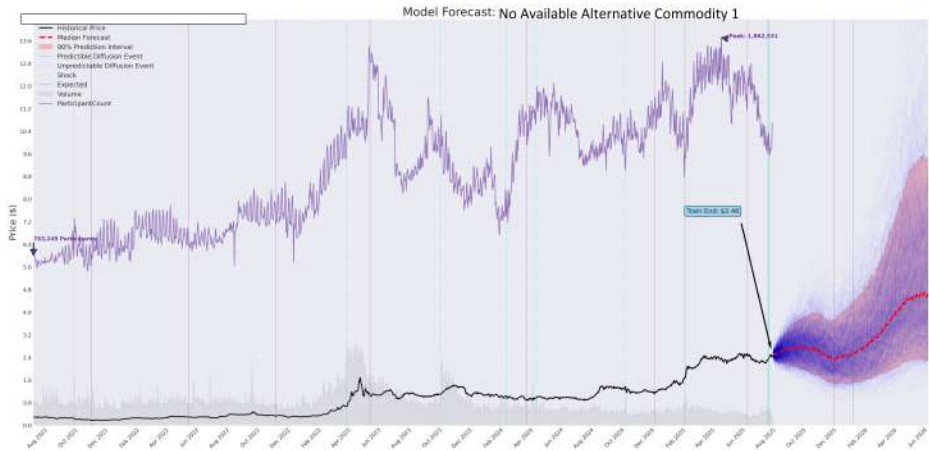


Figure 30
No Available Alternative Commodity 1 Custom SDE Price Forecast Graph.

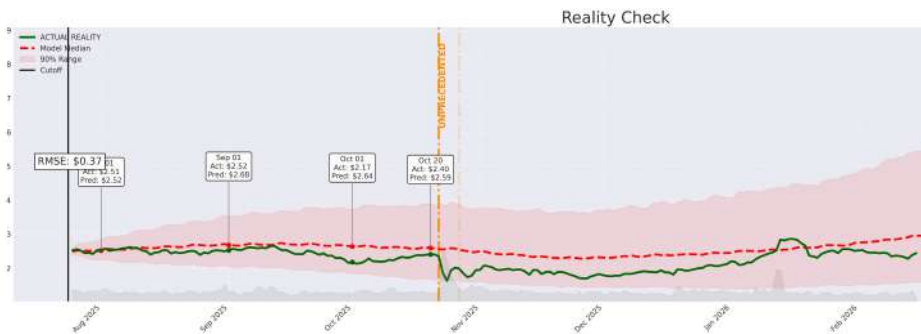


Figure 31
No Available Alternative Commodity 1 Custom SDE Price Accuracy Graph.

Our custom SDE slightly outperforms the naive GBM's price accuracy pre-market crash with it remaining majorly accurate for $1\frac{3}{4}$ months after prediction relative to the naive GBM's $1\frac{1}{2}$ month period of accuracy before general divergence. We also note how our custom SDE suffers less from the exponential μ effect, as prior to the crash the custom SDE is 8% the observed price on October 20th 2020, whereas the naive GBM is nearly twice as far off at 15% (Fig 19). Furthermore, it is apparent how the price forecast graph from the custom SDE more accurately captures the nuance of price behavior showing capability of price level to not only increase but also decrease multiple months later back towards a predicted price close towards the initial price.

Proceeding to our SDE's output for *No Available Alternative Commodity 2*, the most relevant SDE output to our ensuing trading behavior over the live-market experiment period, we note one of the most positive model results.

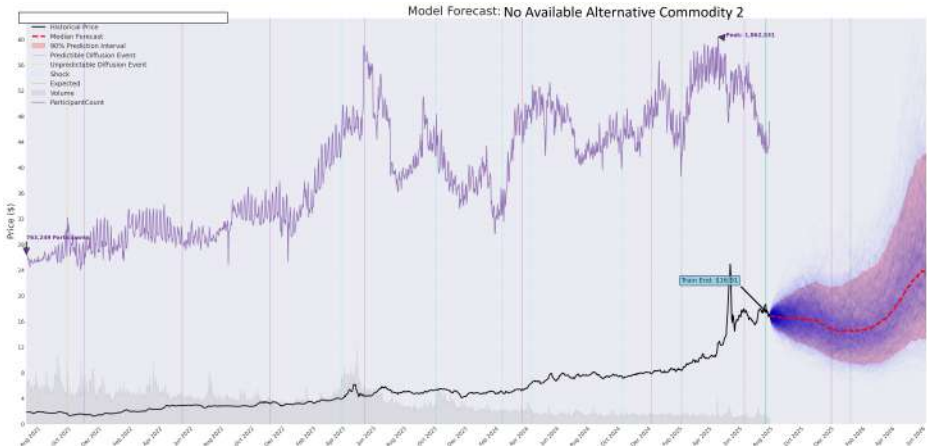


Figure 32
No Available Alternative Commodity 2 Custom SDE Price Forecast Graph.

We first highlight how our SDE quickly enforces our prior understanding of this commodity. We previously understood this commodity to be one that was already highly invested in as it is the only competitively priced commodity for its respective highest generable asset class pool, and is believed to already have a scarce supply and known to have significantly higher demand than most other commodities sharing a similar price. Thus, we previously believed that much of this commodities investment potential had already been realized and had a negative sentiment towards the relative strength of the commodity compared to other long-term commodity investments. Our beliefs towards this commodity was greatly reinforced by the price forecast immediately loosely trending downwards, and generally only holding a negative or stagnant price direction within a reasonable time period where we initially believed the model to produce determinant results for (approx 3 months).



Figure 33
No Available Alternative Commodity 2 Custom SDE Price Accuracy Graph.

The retrospective price analysis confirms our initial judgment of this specific commodities SDE output being deterministic and requiring immediate correction of our portfolio positioning. Whereas the naive GBM price predictions quickly diverged from the observed at the $1\frac{1}{2}$ month point, our model (given some grace for the divergences experienced during high price volatility periods) remains mostly accurate for the nearly 4 month period of predictions prior to the market crash event. On the date shortly before the market crash event on October 20th 2025, our SDE is .75% as in < 1% off the observed price, whereas the naive GBM was >20x more inaccurate at 20.4% off the observed price (Fig 19).

We unfortunately return to a more disappointing predictive result from our SDE for *No Available Alternative Commodity 3* as its retrospective price behavior is still somewhat surprising to us even now.

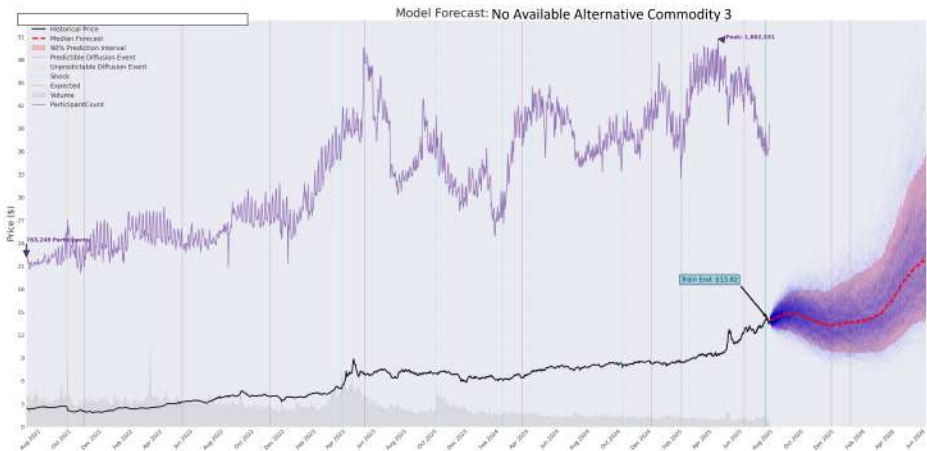


Figure 34
No Available Alternative Commodity 3 Custom SDE Price Forecast Graph.

A small win for our respective SDE relative to the naive GBM's is the accuracy of its price forecast direction and thus in regards to the goal of the SDE, differentiation of the commodity as another poor investment choice through the displayed negative price direction around the start of October 2025 relative to the naive GBM's (Fig 20) entirely positive price direction forecast.



Figure 35
No Available Alternative Commodity 3 Custom SDE Price Accuracy Graph.

Due to the shockingly poor performance of this commodity over the prediction period, analysis of the price accuracy and comparison to the naive GBM is a purely conceptual exercise as there were no wins for either our custom SDE or the naive GBM at all. For both price predictions the actually observed price quickly diverges, and September 1st 2025 the naive GBM's forecast is actually more accurate than our custom SDE's, however our custom SDE's more nuanced, largely cyclical μ , demonstrates the correction of price direction towards the observed, and thus produces significantly more accurate, albeit still inaccurate, price predictions relative to the naive GBM. Of course once again in terms of price prediction accuracy for this commodity category, our custom SDE's inaccuracy is far less than the naive GBM's, at 22.85% for our SDE compared to the naive GBM's 35.87% inaccuracy on October 20th.

We once again provide the excerpts for this respective categories covariate coefficients, although the narrative is much similar to that for the drift covariates of the *Available Alternative Commodities*.

Table 31
Significant & Skewed Drift Coefficients — No Available Alternative Commodities

Parameter	No Available Alternative Commodity 1					No Available Alternative Commodity 2					No Available Alternative Commodity 3				
	Mean	SD	HDI 3%	HDI 97%	Sig.	Mean	SD	HDI 3%	HDI 97%	Sig.	Mean	SD	HDI 3%	HDI 97%	Sig.
<i>Significant Covariates</i>															
sin(Yearly ₁)	0.00219	0.00102	0.00019	0.00407	✓	0.00247	0.00075	0.00106	0.00380	✓	0.00218	0.00064	0.00100	0.00341	✓
cos(Weekly ₁)	-0.00393	0.00093	-0.00571	-0.00226	✓	-0.00132	0.00072	-0.00268	0.00000	—	-0.00388	0.00061	-0.00506	-0.00279	✓
sin(Weekly ₁)	-0.00622	0.00094	-0.00793	-0.00438	✓	-0.00048	0.00075	-0.00186	0.00103	—	-0.00231	0.00060	-0.00343	-0.00116	✓
cos(Weekly ₂)	-0.00287	0.00098	-0.00472	-0.00107	✓	-0.00187	0.00072	-0.00318	-0.00046	✓	-0.00128	0.00062	-0.00253	-0.00023	✓
sin(Weekly ₂)	0.00246	0.00097	0.00068	0.00434	✓	0.00200	0.00073	0.00073	0.00347	✓	0.00367	0.00059	0.00256	0.00472	✓
<i>Skewed / Near-Significant Covariates</i>															
sin(Yearly ₂)	-0.00119	0.00097	-0.00309	0.00055	—	-0.00080	0.00072	-0.00219	0.00057	—	-0.00153	0.00061	-0.00260	-0.00032	✓
Pre-Event	-0.00336	0.00215	-0.00745	0.00057	—	-0.00365	0.00175	-0.00693	-0.00031	✓	-0.00156	0.00135	-0.00403	0.00108	—
<i>Jump Parameters</i>															
σ (GBM vol.)		0.02565					0.01967					0.01646			
λ_J (jumps/yr)		1.97033					1.97033					1.97033			
μ_J		0.00547					-0.00305					-0.00015			
σ_J		0.02672					0.01426					0.01439			

We once again highlight how the market truths driving the μ for commodities once again escapes the capture of our model as the majority of significance is found within the harmonic frequency covariates on the weekly scale. It is some consolation that one yearly frequency covariate remains consistently significant with a relatively large coefficient effect, but it is most disappointing how only one of these commodities achieves multiple yearly covariate significance. For these specific less liquid commodities, it is a near-certainty that any significance demonstrated through the weekly frequency coefficients is theoretically equivalent to that of unexplained drift α . However, the general higher of price predictions for the *No Available Alternative Commodities* category alongside the changing price direction in the forecasts produces some hope that a complete and more thoroughly parameterized SDE may one day properly produce truly accurate and deterministic price forecasts under the same data constraints.

Another consolation and confirmation of our categorical determinism hypothesis is demonstrated through the jump parameters for this category. For these specific illiquid and isolated commodities we typically expect a near 0 log μ_J , and for the specifically higher priced commodities of this category, *No Available Alternative Commodity 2* and *No Available Alternative Commodity 3*, we expect a slightly negative μ_J as a small proportion of investors liquidate regardless of jump sentiment to invest in other investments and investments that possibly have new price behaviors affected by jump-events, and this is a negative effect is observed proportional to the scarcity of the two *No Available Alternative Commodities* confirming our prior hypothesis on these commodities responses to jump events.

7.2.4 High-Demand Stable Commodity Prediction & Covariate Coefficients

We now proceed to our SDE's output for the *High-Demand Stable Commodities* category, one we previously expected our model to produce the least accurate and most in-determinant predictions for as the presence of derivatives for commodities in this category is blatant in the naming convention and we understood the relationship between derivatives was not parameterized in our model(Sec 6.5).

For *High-Demand Stable Commodity 1*, which has multiple derivatives with significant price differentials and significant price behavior relations, which is also the cheapest available option out of all the derivatives thus experiencing one of the strongest price behavior relations possible, we anticipated a significantly inaccurate and in-determinant price forecast from our SDE, which is exactly what we get.

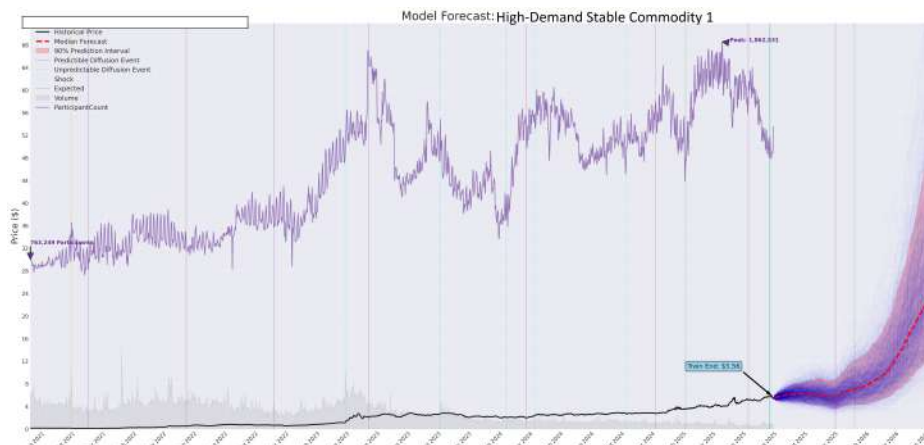


Figure 36
High-Demand Stable Commodity 1 Custom SDE Price Forecast Graph.

Firstly, the price forecast exhibits the exponential price growth behavior synonymous with an inaccurate prediction for our SDE.

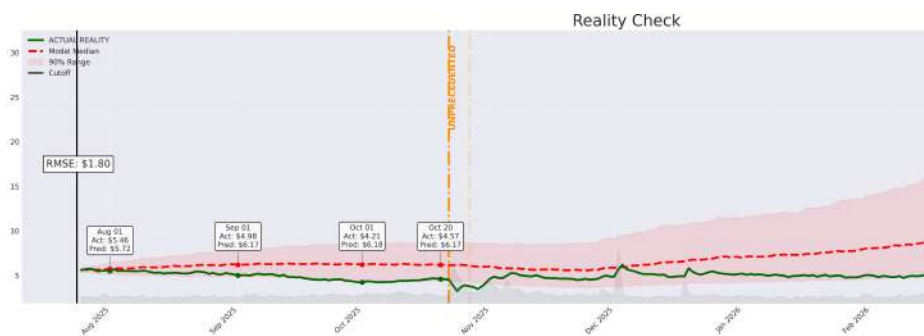


Figure 37
High-Demand Stable Commodity 1 Custom SDE Price Accuracy Graph.

Unsurprisingly the unparameterized price, supply, and demand relation between alternatives, like the naive GBM, with this commodities high previous $\mu \approx .8$ results in a significantly inaccurate price forecast by our model. Like the naive GBM it quickly diverges within less than a month. We once again retain the consolation of our SDE producing a less shitty prediction relative to the naive GBM (Fig 23), of 35% compared to the naive GBM's 46.8% on October 20th 2025, but there's really no "win" to be derived when the inaccuracy percentage is that, although it was entirely anticipated.

For the last commodity under analysis, *High-Demand Stable Commodity 2*, we initially believed the SDE to be able to provide deterministic results as this commodities price behavior relation to its derivative was nearly identical. However, this commodities derivative did spend a decent period of time in the training period as an *Actively Supplying Commodity* which will soon prove to be another problematic commodity characteristic left unparameterized.

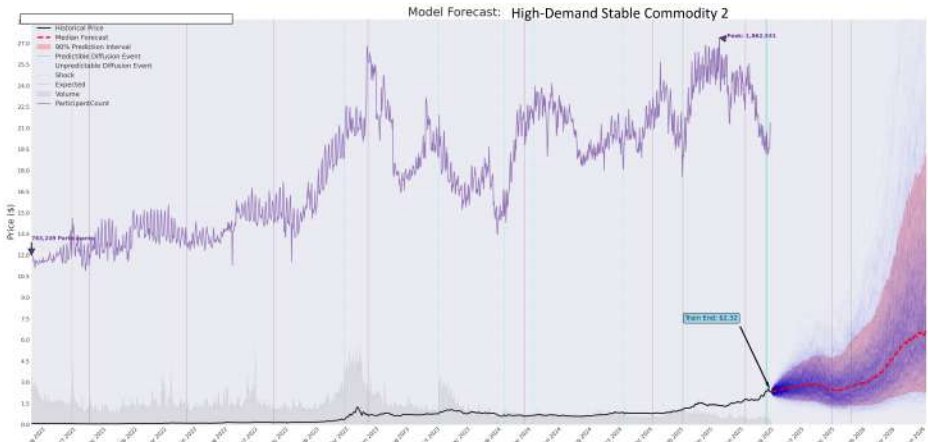


Figure 38
High-Demand Stable Commodity 2 Custom SDE Price Forecast Graph.

Although we note a positive sign in the prediction from the initial price stagnation and short negative price direction duration, as shown below this prediction was far from reality.



Figure 39
High-Demand Stable Commodity 2 Custom SDE Price Accuracy Graph.

As can be seen in the price accuracy graph, this was the first SDE output that actually misled our ensuing trading behavior as our previous confidence in this specific commodity as a strong investment led to an oversight of the signs of a poor investment displayed through the period of negative price direction, and we drank the exponential price forecast curve kool-aid for the first time and paid for it. Most surprisingly our custom SDE produced a more inaccurate price forecast than the naive GBM, our staggering 50.26% inaccuracy compared to the also staggering 47.12% inaccuracy of the naive GBM.

The covariate coefficients once again contain evidence for why our custom SDE produced a more accurate prediction for *High-Demand Stable Commodity 1* relative to the naive GBM whereas the custom SDE fell short of a naive GBM for *High-Demand Stable Commodity 2*.

Table 32
Significant & Skewed Drift Coefficients — High-Demand Stable Commodities

Parameter	High-Demand Stable Commodity 1					High-Demand Stable Commodity 2				
	Mean	SD	HDI 3%	HDI 97%	Sig.	Mean	SD	HDI 3%	HDI 97%	Sig.
<i>Significant Covariates</i>										
sin(Yearly ₁)	0.00423	0.00096	0.00234	0.00595	✓	0.00248	0.00129	-0.00015	0.00485	—
sin(Yearly ₂)	-0.00192	0.00091	-0.00368	-0.00034	✓	-0.00147	0.00126	-0.00386	0.00084	—
cos(Weekly ₁)	-0.00490	0.00087	-0.00653	-0.00331	✓	-0.00426	0.00124	-0.00674	-0.00213	✓
sin(Weekly ₁)	-0.00211	0.00087	-0.00367	-0.00050	✓	-0.00742	0.00127	-0.00979	-0.00505	✓
sin(Weekly ₂)	0.00402	0.00090	0.00226	0.00563	✓	0.00275	0.00131	0.00025	0.00519	✓
Pre-Event	-0.00618	0.00201	-0.01005	-0.00254	✓	-0.00701	0.00290	-0.01302	-0.00187	✓
<i>Skewed / Near-Significant Covariates</i>										
Anticipated Central Authority Updates	-0.02054	0.02174	-0.06196	0.02014	—	0.01567	0.02827	-0.03735	0.06795	—
Event Cooldown	-0.00303	0.00230	-0.00772	0.00093	—	0.00017	0.00319	-0.00644	0.00562	—
<i>Jump Parameters</i>										
σ (GBM vol.)			0.02401					0.03436		
λ_J (jumps/yr)			1.97033					1.96900		
μ_J			0.00864					-0.00444		
σ_J			0.01471					0.02732		

We note how for *High-Demand Stable Commodity 1* two yearly frequency coefficients are significant and exude a strong influence that oppose one another over their different periods in the yearly cycle to provide nuance to the price direction of the commodity and still revert to a baseline drift. Although it still displays significant weekly coefficients which are certainly not expected for such an asset that spent the majority if not entirety of its training data period in its respective *High-Demand Stable Commodities* category, clearly demonstrating a significant true underlying unexplained drift α , the presence of a significant negative event diffusion, and two negative near-significant binary flagged events, one being a duration diffusion and the other being the single-day price-level adjusting diffusion, serves as sign of some determinism in regards to known commodity price behavior being captured by the SDE, and that it will produce a more accurate prediction than a uninformed naive GBM would.

Moving along to the problem commodity, *High-Demand Stable Commodity 2*, there were clear signs that the SDE did not produce a deterministic or at all accurate prediction in the covariate coefficients that were ignored due to our prior beliefs on the commodities strength as a long-term investment. Furthermore, we cannot confidently disprove our hypothesis on deterministic modeling of commodities which share parallel price behavior to their derivatives, as the duration of the training period its derivative spent in the *Actively Supplying Commodities* category is believed to be at least 5 months, enough to warrant a separate regime as the price behavior would have been so significantly influenced by the increasing supply of its derivative. However in regards to the SDE producing a terrible prediction, the writing was absolutely on the wall as not a single yearly frequency coefficient is by definition statistically significant, only a single diffusion event was significant, and the single-day price-level adjustment diffusion is skewed in the positive direction! If it wasn't obvious by the price forecast curves exponential shape, the prediction was clearly in-determinant and should not have been used to reinforce our beliefs of the commodity as a strong long-term investment option.

Further analysis of jump parameters is at least consistent with our prior understanding of these respective commodities reactions to jump events. The *High-Demand Stable Commodity 1* highest generable asset class pool has historically be positively influenced by jump events, and this is somewhat evident by the $\log \mu_J \approx .008$ although a high volatility for this μ_J is also present $\sigma_J \approx .014$. For the *High-Demand Stable Commodity 2* our understanding of cheaper commodities reactions to jump events typically being negative or near zero is reinforced.

7.3 Analytical Drift & Drift Structure Reflections

7.4 Merton Jump Implementation Reflections

7.5 Redefining “Determinism” for our SDE

7.6 Most Influential and Considered Commodity Model Output

We've previously foreshadowed how *No Available Alternative Commodity 2*'s respective custom SDE prediction would prove to be one of the most influential outputs, and we now explain our initial in-depth investigation into this respective output and provide chat-log and trade-history evidence of the respective judgments and trading behavior we exhibited in response to this prediction.

As explained alongside the custom SDE analysis of *No Available Alternative Commodity 2* (around Fig 32),

despite the markets extremely positive sentiment towards this specific commodity due to its extremely strong underlying fundamentals, our belief at the time (July 2025) was that any investing potential that this commodity possessed due to underlying supply and demand fundamentals had already been realized as significant price movement was observed, but it was nearly entirely realized in the proceeding year alongside the peak market participant observations. We thus believed it wouldn't be until another significant portion of its supply was consumed that the commodity would resume the strong fundamental related growth seen in the 1-3 year time period prior to the recent growth.

This intuitive and heuristic analysis of the commodity was reinforced by our SDE producing its strongest negative prediction that opposed even the exponential μ over-influence traditionally seen in other commodities predictions.

7.6.1 Influenced Commodity Portfolio Position

Part IV

Live-Market Deployment & Crisis Response, Portfolio Results, Project Reflections & Future Development

8 Market Crash - Trading Response & Model Fragility

Around 8PM ET on October 23rd the central market authority did a “thing”.

9 Overall Project Results and Reflections

We didn't journal anything, so we tried our best to retrospectively get pricing information regarding the assets we held at the start date and the information we were able to quickly fetch about our trades made throughout the period. A lot of information from the exchanges we used had expired and were no longer fetch-able, and the amount of initial capital deployed was also very hard to measure because it was already dispersed across crypto accounts, realistic marketplace balances, tokenized marketplace balances, outstanding orders, and real assets all across multiple accounts. Meaning that both initial investment capital and final capital was spread across over dozens of marketplaces and platform balances.

It doesn't help our project analysis that our development and commit history is also regrettably fucked. Development was across IDE's and not even properly committed directly after completion. During IDE switches Author One literally had work disappear due to WSL and git jank, that was only recoverable because it was sent and documented over discord and google docs. We stress finding development dates and thus retrospective trading periods was insanely difficult and inaccurate as well.

Furthermore, as we grew as developers a lot throughout the project, we realized that we literally committed API keys and our environments and other extremely important account related data to the repository. Author One and Author Two promised each other the repository would never see the light of day and trudged forward.

This data accuracy was made much worse by the most central platforms update, where trades became reversible to protect market users from account hijackings. Trades could be reversed for one week from the central asset storage platform. Although we believe these reverses could be measured, reversals on marketplaces made fetching trade history pretty much impossible because many platforms did not properly track these events although returned platform balance.

Furthermore, nearly every model developed through was deployed discontinuously, so we denote the time period from the date we originally began mass trading, approximately July 1st 2025, to the date we effectively ceased mass trading, January 12 2026, as the total time our entire project was in deployment. We subtract the regrettably long time duration we ceased arbitrage trading due to suspicious market behavior until the arbitrage model was fixed as a result of the market crash (October 10th 2025 - December 25th 2025).

Authors would really like to stress that because of these factors, **the books and records are totally fucked**. We'll be leaning on the conservative side for estimates and records for everything.

9.1 Conservative Performance Summary

Capital Possibly Used with Models and Gains we Confidently Attribute to the Models Influence

Our most conservative and realistic bookkeeping is the information below, related to the returns we're willing to attribute to trades and knowledge we believed were related specifically to the models outlined in this project. The Unrealized + Realized Gains account for Platform Fees

Metric	Value
Deployment Status	Paused (Inactive)
Live Since	July 1, 2025
Reporting Date	January 12, 2026
Days in Operation	175 days (~5.7 months)
Arbitrage Days in Deployment	103 (~3.4 months)
SDE Days in Deployment	113 (~3.7 months)
Initial Capital Deployed	\$13,500
Current Portfolio Value	\$19,100
Unrealized + Realized Gains	\$5,600
Return to Date	+41.5%
Annualized Rate (CAGR)	+106.8%

Table 33
Conservative deployment performance estimates as of January 1-6, 2026

9.2 General Performance Summary

Just a general perspective start to finish for the entire portfolio to the end valuation from an inventory management tool (we deemed accurate) and our platform balances.

Metric	Value
Deployment Status	Paused (Inactive)
Live Since	July 1, 2025
Reporting Date	January 12, 2026
Days in Operation	175 days (~5.7 months)
Arbitrage Days in Deployment	103 (~3.4 months)
SDE Days in Deployment	113 (~3.7 months)
Initial Capital Deployed	\$15,000
Current Portfolio Value	\$24,050
Unrealized + Realized Gains	\$9,050
Return to Date	+60.3%
Annualized Rate (CAGR)	+167.7%

Table 34
General portfolio performance estimates as of January 1-6, 2026

9.3 Strategy Attribution

The returns are attributable to the following components of the analytical framework:

1. **Opportunity Identification:** The filtering pipeline identifies price discrepancies between markets with sufficient margin to cover transaction costs and market friction.
2. **Duration Prediction:** The Bayesian survival model provides probability estimates for opportunity persistence, enabling informed position sizing and timing decisions.
3. **Risk Management:** The hierarchical category structure and uncertainty quantification allow for differentiated risk assessment across asset classes.
4. **Semi-Continuous Refinement:** We tried to fix when we had time from life and school

9.4 Comparative Context

For reference, during the same August 2025 – January 2026 period:

- S&P 500 Index: Approximately +7.98%
- Risk-free rate (T-bills): ~2% annualized
- This strategy (to date): +41.5%

9.5 Considerations

9.6 Summary

10 Ongoing Development & Future Project Progression

We're working on itttttttttt very hard with EE bro!

Acknowledgments Author One would like to thank Author Two for financial support, and for entrusting Author One with his savings. Both authors would like to acknowledge frustration with marketplaces hoarding their data. If you want more trading give more data.

Conflict of interest The authors declare a minimal conflict of interest since this article may influence prospective internship opportunities.

Artificial Intelligence Below is a placeholder but we absolutely used AI to learn how to do the project. Fun fact it was not great at doing this project really at all because it's a little too absurd. Hence every line of code other than some pieces to generate graphs (which we had to fact check) were either entirely written or modified by us. Specifically almost everything to do with Arbitrage Model and the SDE's implementation are AI-free. I'm still leaving this to cover the legal basis LOL! This research utilized AI tools to assist in data analysis, manuscript drafting, and figure generation. All AI-generated content was critically reviewed and validated by the authors to ensure accuracy and alignment with the scientific integrity of the study. The use of AI adhered to ethical guidelines, ensuring transparency and compliance with academic standards. Any biases or limitations inherent to the AI tools were carefully considered in the interpretation of results. The authors affirm that the AI tools did not compromise the originality or integrity of the work.

Data availability Bro we have NOTHING like NOTHING we're sorryyy, actually u can lowkey ask for code pieces,

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A Additional tables and figures

Table A1
Weibull AFT Model Development Comparison: Old Model (Phase 2) vs. New Model (Phase 4)

Parameter	Old Model				New Model			
	Mean	SD	HDI 3%	HDI 97%	Mean	SD	HDI 3%	HDI 97%
<i>Intercepts</i>								
Global Intercept ($\beta_{0,global}$)	2.505	0.555	1.457	3.576	2.188	0.373	1.507	2.888
Offset: Variation A	0.192	0.558	-0.904	1.220	0.074	0.389	-0.658	0.808
Offset: Variation B	-0.019	0.555	-1.109	1.008	0.492	0.376	-0.199	1.204
Offset: Variation C	-0.020	0.616	-1.243	1.076	0.409	0.381	-0.281	1.127
<i>Global Covariates</i>								
Initial Exchange Price (P_T)	0.388	0.571	-0.668	1.480	0.100	0.130	-0.152	0.342
Arbitrage Price Ratio (APR_{anchor})	0.154	0.562	-0.947	1.156	0.038	0.118	-0.163	0.274
Benchmark Ratio Momentum	-0.130	0.566	-1.170	0.919	-0.007	0.120	-0.229	0.220
Price Volatility (σ_u)	0.014	0.569	-1.069	1.034	0.152	0.132	-0.101	0.390
Exchange Capacity (L_{cap})	0.103	0.578	-1.056	1.131	—	—	—	—
Ratio to Benchmark (P_T/P_B)	-0.078	0.552	-1.156	0.940	—	—	—	—
Benchmark Volatility	0.059	0.565	-1.003	1.114	—	—	—	—
Congestion Metric	—	—	—	—	0.240	0.119	0.013	0.456
Benchmark Ratio Deviation	—	—	—	—	-0.099	0.117	-0.325	0.114
<i>Offsets: Variation A</i>								
Initial Exchange Price	0.292	0.612	-0.878	1.391	-0.021	0.145	-0.297	0.252
Arbitrage Price Ratio	0.068	0.573	-1.042	1.117	0.035	0.127	-0.208	0.262
Benchmark Ratio Momentum	-0.048	0.577	-1.135	0.998	0.043	0.127	-0.196	0.278
Price Volatility	0.239	0.629	-0.891	1.479	0.017	0.145	-0.250	0.294
Exchange Capacity	0.247	0.584	-0.824	1.354	—	—	—	—
Ratio to Benchmark	-0.101	0.560	-1.170	0.948	—	—	—	—
Benchmark Volatility	0.064	0.575	-1.041	1.094	—	—	—	—
Congestion Metric	—	—	—	—	0.035	0.132	-0.208	0.292
Benchmark Ratio Deviation	—	—	—	—	-0.009	0.122	-0.229	0.228
<i>Offsets: Variation B</i>								
Initial Exchange Price	-0.196	0.573	-1.276	0.871	-0.022	0.132	-0.264	0.226
Arbitrage Price Ratio	0.008	0.565	-1.034	1.091	-0.011	0.120	-0.225	0.220
Benchmark Ratio Momentum	0.134	0.570	-0.899	1.198	-0.021	0.124	-0.257	0.206
Price Volatility	-0.177	0.572	-1.237	0.884	0.023	0.133	-0.223	0.270
Exchange Capacity	0.121	0.579	-0.908	1.274	—	—	—	—
Ratio to Benchmark	0.171	0.552	-0.923	1.175	—	—	—	—
Benchmark Volatility	-0.056	0.570	-1.183	0.962	—	—	—	—
Congestion Metric	—	—	—	—	0.078	0.120	-0.150	0.301
Benchmark Ratio Deviation	—	—	—	—	-0.081	0.122	-0.301	0.152
<i>Offsets: Variation C</i>								
Initial Exchange Price	-0.021	0.601	-1.159	1.098	0.059	0.135	-0.193	0.312
Arbitrage Price Ratio	-0.051	0.569	-1.143	1.004	-0.014	0.121	-0.249	0.204
Benchmark Ratio Momentum	-0.069	0.580	-1.126	1.002	-0.023	0.123	-0.244	0.214
Price Volatility	-0.078	0.570	-1.163	0.967	-0.011	0.135	-0.262	0.237
Exchange Capacity	-0.272	0.685	-1.526	1.048	—	—	—	—
Ratio to Benchmark	-0.045	0.562	-1.125	0.976	—	—	—	—
Benchmark Volatility	0.007	0.580	-1.078	1.104	—	—	—	—
Congestion Metric	—	—	—	—	-0.080	0.127	-0.318	0.155
Benchmark Ratio Deviation	—	—	—	—	0.069	0.121	-0.161	0.298
<i>Family Specific</i>								
Weibull Shape (α)	1.039	0.040	0.962	1.113	1.122	0.032	1.060	1.181

Table A2
GBM Analytical Drift Covariate Posterior Estimates by Commodity Category

Parameter	Available Alt. Commodity 1				Available Alt. Commodity 2			
	Mean	SD	HDI 3%	HDI 97%	Mean	SD	HDI 3%	HDI 97%
<i>Category: Available Alternative Commodities</i>								
α	0.00034	0.01004	-0.01844	0.01906	0.00043	0.00944	-0.01774	0.01729
log(ParticipantCount)	0.00008	0.00075	-0.00134	0.00150	0.00041	0.00071	-0.00083	0.00179
TimeIndex	0.00000	0.00000	-0.00000	0.00000	0.00000	0.00000	-0.00000	0.00000
cos(Yearly ₁)	0.00178	0.00106	-0.00013	0.00379	0.00081	0.00100	-0.00117	0.00259
sin(Yearly ₁)	0.00337	0.00107	0.00118	0.00524	0.00302	0.00094	0.00113	0.00463
cos(Yearly ₂)	-0.00271	0.00107	-0.00482	-0.00073	0.00028	0.00099	-0.00151	0.00211
sin(Yearly ₂)	0.00118	0.00121	-0.00117	0.00332	0.00084	0.00110	-0.00108	0.00296
cos(Yearly ₃)	-0.00124	0.00102	-0.00318	0.00068	0.00044	0.00092	-0.00126	0.00216
sin(Yearly ₃)	-0.00002	0.00104	-0.00209	0.00176	-0.00065	0.00098	-0.00236	0.00121
cos(Weekly ₁)	-0.00282	0.00101	-0.00466	-0.00092	-0.00325	0.00093	-0.00503	-0.00154
sin(Weekly ₁)	-0.00020	0.00101	-0.00227	0.00157	-0.00226	0.00091	-0.00392	-0.00053
cos(Weekly ₂)	0.00006	0.00097	-0.00169	0.00198	0.00107	0.00095	-0.00078	0.00275
sin(Weekly ₂)	0.00430	0.00103	0.00222	0.00613	0.00433	0.00091	0.00277	0.00617
Pre-Event	-0.00732	0.00226	-0.01138	-0.00305	-0.00442	0.00211	-0.00859	-0.00069
Event Active	0.00168	0.00293	-0.00361	0.00714	-0.00265	0.00278	-0.00777	0.00236
Post-Event	0.00509	0.00300	-0.00055	0.01089	0.00322	0.00277	-0.00185	0.00840
Event Cooldown	0.00204	0.00263	-0.00299	0.00669	-0.00309	0.00240	-0.00730	0.00171
Unpredictable Event Active	-0.00286	0.00291	-0.00843	0.00257	-0.00440	0.00266	-0.00952	0.00053
Anticipated Central Authority Updates	-0.00740	0.02269	-0.05109	0.03151	-0.00202	0.02317	-0.04216	0.04578
σ	0.02737	0.00049	0.02648	0.02832	0.02529	0.00045	0.02448	0.02616

Parameter	No Avail. Alt. Commodity 1				No Avail. Alt. Commodity 2				No Avail. Alt. Commodity 3			
	Mean	SD	HDI 3%	HDI 97%	Mean	SD	HDI 3%	HDI 97%	Mean	SD	HDI 3%	HDI 97%
<i>Category: No Available Alternative Commodities</i>												
α	-0.00072	0.01037	-0.01979	0.01905	0.00239	0.00988	-0.01525	0.02115	0.00303	0.00933	-0.01450	0.02060
log(ParticipantCount)	0.00019	0.00077	-0.00121	0.00169	-0.00008	0.00073	-0.00140	0.00130	-0.00009	0.00069	-0.00147	0.00114
TimeIndex	0.00000	0.00000	-0.00000	0.00000	0.00000	0.00000	-0.00000	0.00000	-0.00000	0.00000	-0.00000	0.00000
cos(Yearly ₁)	0.00147	0.00104	-0.00046	0.00334	0.00076	0.00077	-0.00065	0.00222	-0.00008	0.00066	-0.00141	0.00107
sin(Yearly ₁)	0.00219	0.00102	0.00019	0.00407	0.00247	0.00075	0.00106	0.00380	0.00218	0.00064	0.00100	0.00341
cos(Yearly ₂)	-0.00007	0.00107	-0.00209	0.00188	-0.00145	0.00081	-0.00293	0.00005	-0.00018	0.00067	-0.00145	0.00102
sin(Yearly ₂)	0.00204	0.00113	-0.00010	0.00416	-0.00051	0.00087	-0.00223	0.00104	-0.00001	0.00072	-0.00129	0.00135
cos(Yearly ₃)	0.00002	0.00093	-0.00172	0.00174	0.00055	0.00070	-0.00068	0.00199	0.00027	0.00062	-0.00086	0.00140
sin(Yearly ₃)	-0.00119	0.00097	-0.00309	0.00055	-0.00080	0.00072	-0.00219	0.00057	-0.00153	0.00061	-0.00260	-0.00032
cos(Weekly ₁)	-0.00393	0.00093	-0.00571	-0.00226	-0.00132	0.00072	-0.00268	0.00000	-0.00388	0.00061	-0.00506	-0.00279
sin(Weekly ₁)	-0.00622	0.00094	-0.00793	-0.00438	-0.00048	0.00075	-0.00186	0.00103	-0.00231	0.00060	-0.00343	-0.00116
cos(Weekly ₂)	-0.00287	0.00098	-0.00472	-0.00107	-0.00187	0.00072	-0.00318	-0.00046	-0.00128	0.00062	-0.00253	-0.00023
sin(Weekly ₂)	0.00246	0.00097	0.00068	0.00434	0.00200	0.00073	0.00073	0.00347	0.00367	0.00059	0.00256	0.00472
Pre-Event	-0.00336	0.00215	-0.00745	0.00057	-0.00265	0.00175	-0.00693	-0.00031	-0.00156	0.00135	-0.00403	0.00108
Event Active	-0.00003	0.00273	-0.00503	0.00512	-0.00195	0.00219	-0.00649	0.00170	-0.00064	0.00181	-0.00417	0.00269
Post-Event	-0.00144	0.00270	-0.00652	0.00357	-0.00051	0.00217	-0.00470	0.00358	-0.00212	0.00189	-0.00595	0.00125
Event Cooldown	-0.00056	0.00247	-0.00519	0.00421	-0.00056	0.00199	-0.00420	0.00339	-0.00053	0.00150	-0.00351	0.00219
Unpredictable Event Active	-0.00144	0.00284	-0.00671	0.00402	0.00109	0.00205	-0.00290	0.00470	-0.00047	0.00170	-0.00337	0.00310
Anticipated Central Authority Updates	-0.00331	0.02327	-0.04545	0.04174	-0.00930	0.01899	-0.04480	0.02663	-0.00365	0.01591	-0.03220	0.02607
σ	0.02565	0.00050	0.02472	0.02656	0.01967	0.00036	0.01892	0.02028	0.01646	0.00030	0.01590	0.01705

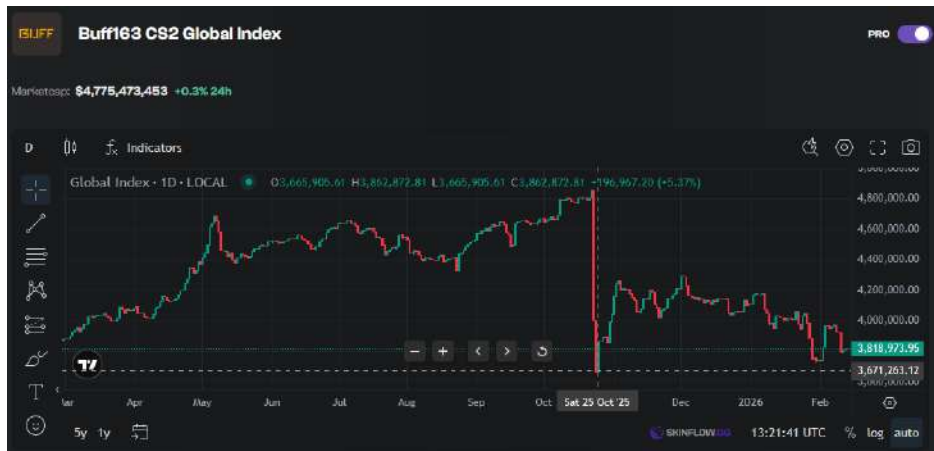


Figure 40

One websites calculated global market cap graph (some sites approximations differ)

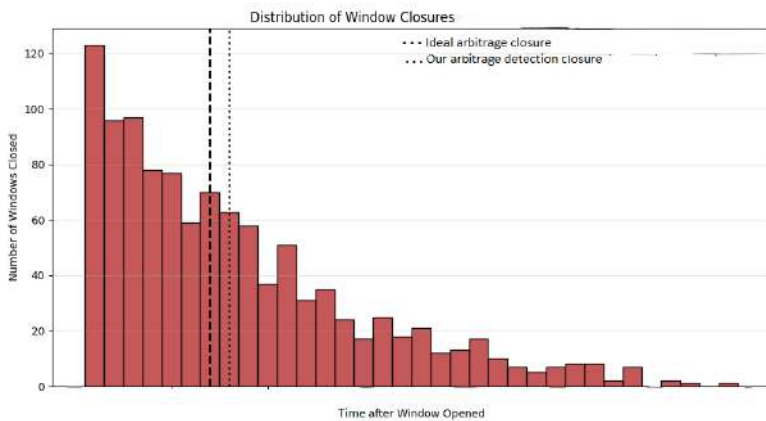


Figure A1

The “our model” line was for the old model on all windows defined, so the final wasn’t getting MOGGED that hard

Parameter	High-Demand Stable Commodity 1				High-Demand Stable Commodity 2			
	Mean	SD	HDI 3%	HDI 97%	Mean	SD	HDI 3%	HDI 97%
<i>Category: High-Demand Stable Commodities</i>								
α	0.00046	0.00982	-0.01746	0.01811	0.00007	0.00989	-0.01904	0.01751
log(ParticipantCount)	0.00032	0.00074	-0.00110	0.00158	0.00026	0.00074	-0.00100	0.00172
TimeIndex	-0.00000	0.00000	-0.00001	0.00000	-0.00000	0.00000	-0.00001	0.00000
cos(Yearly ₁)	0.00115	0.00093	-0.00056	0.00286	0.00093	0.00135	-0.00140	0.00361
sin(Yearly ₁)	0.00423	0.00096	0.00234	0.00595	0.00248	0.00129	-0.00015	0.00485
cos(Yearly ₂)	-0.00026	0.00094	-0.00198	0.00155	-0.00146	0.00137	-0.00384	0.00123
sin(Yearly ₂)	-0.00033	0.00108	-0.00235	0.00171	0.00172	0.00152	-0.00102	0.00470
cos(Yearly ₃)	0.00081	0.00086	-0.00070	0.00250	-0.00067	0.00122	-0.00296	0.00156
sin(Yearly ₃)	-0.00192	0.00091	-0.00368	-0.00034	-0.00147	0.00126	-0.00386	0.00084
cos(Weekly ₁)	-0.00490	0.00087	-0.00653	-0.00331	-0.00426	0.00124	-0.00674	-0.00213
sin(Weekly ₁)	-0.00211	0.00087	-0.00367	-0.00050	-0.00742	0.00127	-0.00979	-0.00505
cos(Weekly ₂)	-0.00017	0.00090	-0.00189	0.00148	-0.00089	0.00128	-0.00336	0.00142
sin(Weekly ₂)	0.00402	0.00090	0.00226	0.00563	0.00275	0.00131	0.00025	0.00519
Pre-Event	-0.00618	0.00201	-0.01005	-0.00254	-0.00701	0.00290	-0.01302	-0.00187
Event Active	0.00176	0.00268	-0.00336	0.00639	-0.00319	0.00365	-0.00997	0.00359
Post-Event	-0.00092	0.00270	-0.00612	0.00390	0.00332	0.00363	-0.00360	0.00990
Event Cooldown	-0.00303	0.00230	-0.00772	0.00093	0.00017	0.00319	-0.00644	0.00562
Unpredictable Event Active	0.00097	0.00254	-0.00378	0.00554	-0.00199	0.00368	-0.00912	0.00444
Anticipated Central Authority Updates	-0.02054	0.02174	-0.06196	0.02014	0.01567	0.02827	-0.03735	0.06795
σ	0.02401	0.00045	0.02310	0.02479	0.03436	0.00063	0.03315	0.03549

Note: Posterior estimates produced by the Arviz Python library for our finished and behaviour influencing custom SDE (2025/07/24). Parameters correspond to the log-return drift equation $\mu(t) = \alpha + \beta \cdot X(t)$. Estimates reported as posterior mean with standard deviation (SD) and 94% HDI. Fourier terms capture seasonal periodicity at yearly and weekly frequencies. Pre-Event, Event Active, Post-Event, and Event Cooldown are binary indicators for the four phases of predictable drift events; Unpredictable Event Active flags the active window of unpredictable diffusion events; and Anticipated Central Authority Updates is a single-day binary indicator for infrequent dates where the central market authority was anticipated to release information or updates with known market sentiment implications. Unlike previous μ diffusion flags, this indicator captures a single point-in-time drift adjustment (*only one such date falls in the training window*). (see Table 22).

Table A3
Merton Jump Component Parameters by Commodity

Commodity	Jump Frequency λ_j (jumps/year)	Mean Jump Size μ_j	Jump Size Volatility σ_j
<i>Available Alternative Commodities</i>			
Available Alt. Commodity 1 (Clutch)	1.97033	-0.00347	0.01896
Available Alt. Commodity 2 (Revolver)	1.97033	0.00951	0.03460
<i>No Available Alternative Commodities</i>			
No Avail. Alt. Commodity 1 (Falchion)	1.97033	0.00547	0.02672
No Avail. Alt. Commodity 2 (Glove)	1.97033	-0.00305	0.01426
No Avail. Alt. Commodity 3 (Breakout)	1.97033	-0.00015	0.01439
<i>High-Demand Stable Commodities</i>			
High-Demand Stable Commodity 1 (Chroma 3)	1.97033	0.00864	0.01471
High-Demand Stable Commodity 2 (Prisma)	1.96900	-0.00444	0.02732

Note: Jump parameters are estimated empirically from residuals on declared jump dates (shock and operation start dates). λ_j is the annualised Poisson intensity; μ_j and σ_j are the mean and standard deviation of the log-return residual on jump days, defining the Normal jump size distribution $J(t) \sim \mathcal{N}(\mu_j, \sigma_j^2)$. See Table 23 for jump date classifications.

Price Inference Uncertainty Error Proof